

COMP 3311 Group 20

Supplementary Note

Group Project Activity 2

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1. Introduction

The Library Management System is designed to provide students, staff, authors, and librarians with efficient access to and management of library services.

2. Technical Requirements

The project is built with the programming platform IntelliJ IDEA Community Edition in Java Language and uses Maven to manage (integrated in IntelliJ). The detailed dependency requirements are:

- JDK 21 or above
- Opencsv 4.6
- Javafx-controls 21
- Javafx-fxml 21
- Controlsfx 11.2.1
- Ikonli-javafx 12.3.1
- Openedf 1.3.39
- Jbcrypt 0.3 or above

The dependencies for testing are :

- Testfx-junit5 4.0.16-alpha
- Javafx-swing 21
- Junit-jupiter 5.10.2
- Mockito-core 4.6.1
- Mockito-junit-jupiter 4.6.1
- Kotlin-test 2.2.0
- Kotlin-stdlib-jdk8 2.2.0
- Junit-jupiter 5.10.2
- Junit-jupiter-engine 5.10.2
- Junit-jupiter-api 5.10.2

3. Data Storage

3.1 Overview

Data is stored inside the user's PC. The structure tree is shown as follows


```

YourProjectRoot/
├── book/
│   ├── AllBooks.txt
│   └── bookcontent/
│       ├── BookTitle1_Author1.txt
│       ├── BookTitle2_Author2.txt
│       └── ...
├── data/
│   ├── user_data.txt
│   ├── borrows.txt
│   └── notifications.txt
└── src/
    └── ...

```

Data are stored on two main files --- the book file and data file.

Book file saves the book details and content.

Data file saves the users' data, borrowing records, and notification records, respectively.

3.2 Detailed Explanation of each file

AllBooks.txt : Contains every book information exist in the system

Item Name	Description	Data Type	Example
bookTitle	Title of the Book	String	demobook
authorUsername	Username of the author	String	Jason
bookAbstract	The summary of the boo	String	This is the summary of the book
noOfTimesBorrowed	The total number of time that this has been borrowed	int	10
bookStatus	Current status of the book. Can be "APPROVED", "REJECTED""PENDING"and "DELETED" Manipulate in book operations	Enum BookStatus	APPROVED

bookcontent/ : A file containing all the actual book content.

Item Name	Description	Data Type	Example
Book.txt	A renamed, immutable book content text file chosen by the author upon publishing book. Renaming simply combine the txt file name, underscore and the author name.	Txt file	Book_Jason.txt

user_data.txt : This text file stores the user information.

Item Name	Description	Data Type	Example
username	Username of the user	String	cherrylai
password	Password of the user account, encrypted before storage	String	\$2a\$10\$FyVHQcxAvH/MHs0LSUf
fullName	Full Name of the user	String	Lai Cheuk Ying
status	Active status of the user, must be either “Activated” or “Deactivated”	Enum Status	Activated
role	Role of the user, must be either “Student”, “Author” or “Librarian”	Enum Role	Librarian

borrows.txt : This text file stores the borrowing records.

Item Name	Description	Data Type	Example
bookTitle	Title of the Book	String	Java Programming
authorUsername	Username of the Author	String	johnDoe
borrowerUsername	Username of the Borrower	String	alice123
duration	Borrowing Duration	Duration	P2D
borrowedDateTime	Borrowing Time	LocalDateTime	2023-10-01T10:15:30

notifications.txt : This file stores the notification that sends to the users.

Item Name	Description	Data Type	Example
receiverUsername	Username of the receiver	String	cherrylai
message	Message that the receiver receives	String	Your book “BookName” has been Approved!

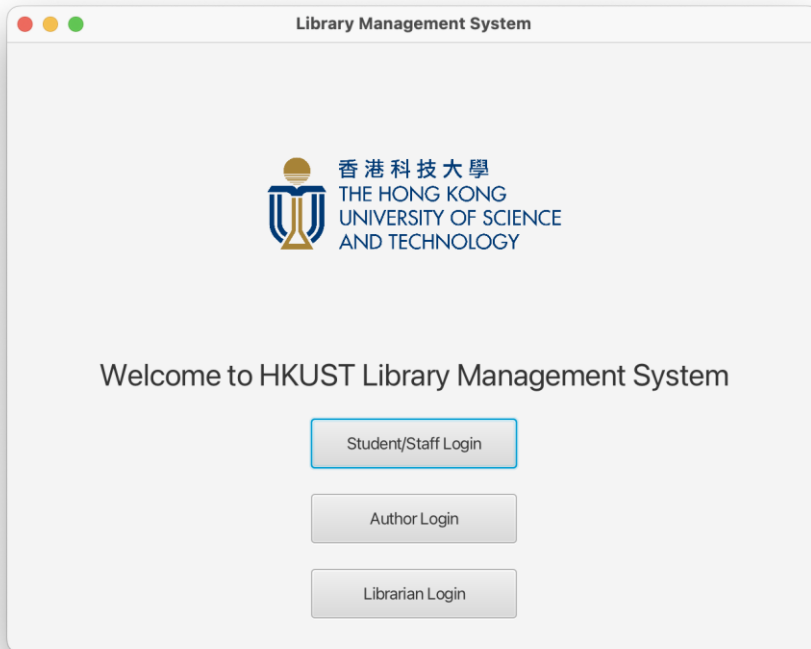
4. User Guide

To facilitate better use of the program, the following section provides a basic user guide for running

the Library Management System using IntelliJ IDEA.

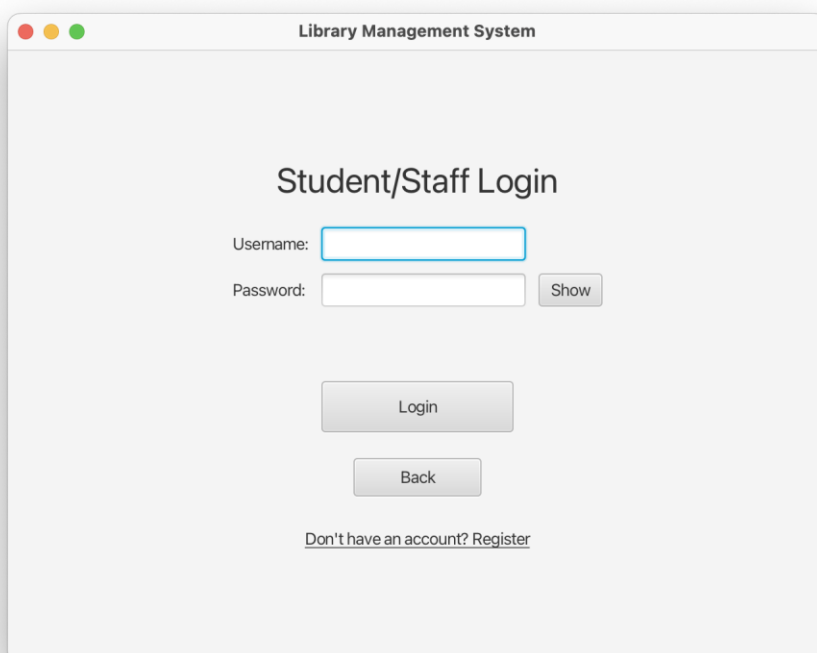
4.1 Login

When you enter the Library Management System, you will see the Home Page.



Select your role to log in to the corresponding dashboard.

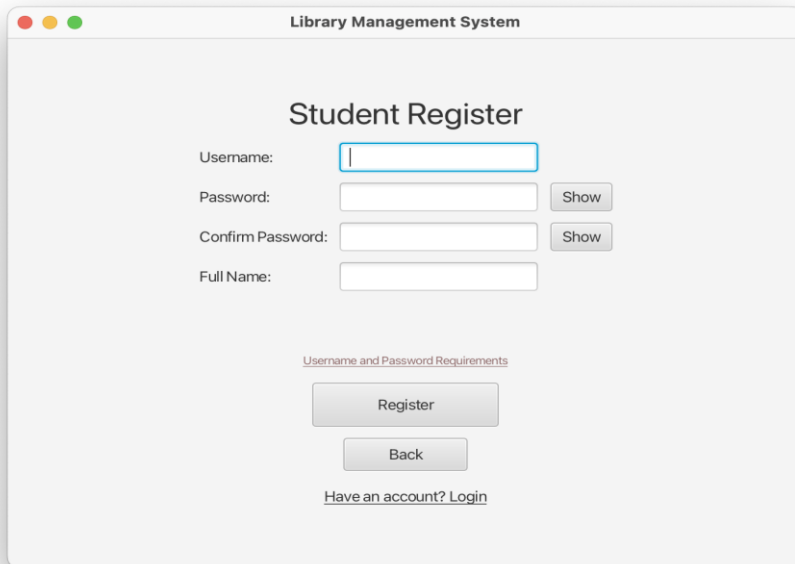
For example, select “Student/Staff Login” to log in as a “Student/Staff”.



4.2 Register

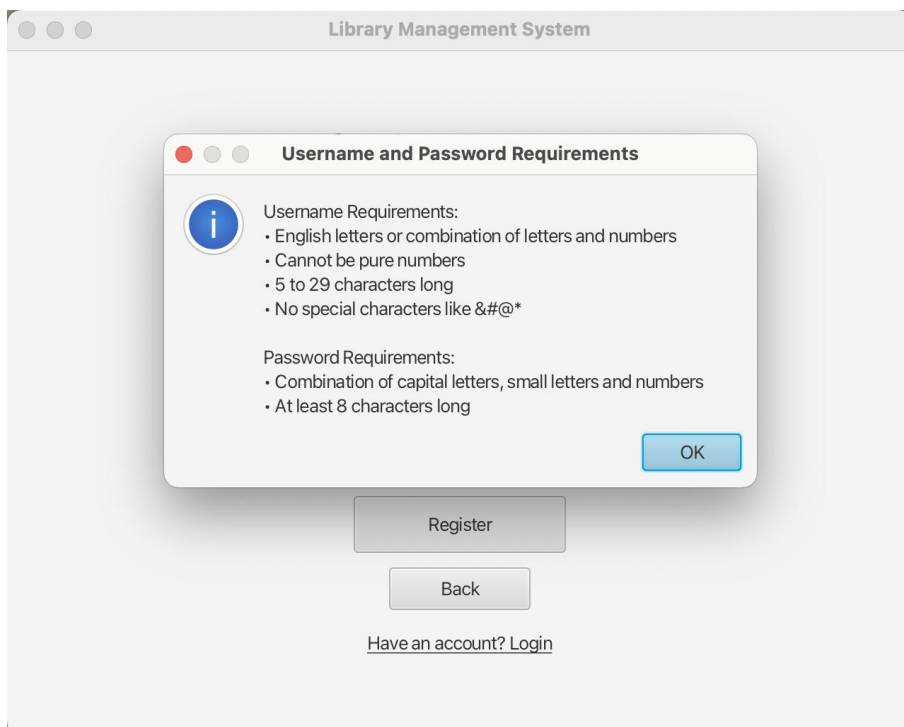
If you do not have an account, you can click the hyperlink “Don’t have an account? Register” in Login Page to access the Register Page.

For example, the following is the Register Page for Student/Staff.



The screenshot shows a web browser window titled "Library Management System". Inside, there is a form titled "Student Register". The form contains four input fields: "Username:", "Password:", "Confirm Password:", and "Full Name:". The "Password:" and "Confirm Password:" fields have a "Show" button next to them. Below the input fields, there is a link "Username and Password Requirements" in red text. At the bottom of the form, there are two buttons: "Register" and "Back". At the very bottom, there is a link "Have an account? Login".

To learn more about username and password requirements, you can click the hyperlink “Username and Password Requirements”. The requirements will be displayed in a popup window.



The screenshot shows a web browser window titled "Library Management System" with a popup window titled "Username and Password Requirements". The popup window contains an information icon (i) and the following text:

Username Requirements:

- English letters or combination of letters and numbers
- Cannot be pure numbers
- 5 to 29 characters long
- No special characters like &#@*

Password Requirements:

- Combination of capital letters, small letters and numbers
- At least 8 characters long

At the bottom right of the popup window is an "OK" button. In the background, the "Student Register" form is visible, showing the "Register" and "Back" buttons and the "Have an account? Login" link.

Enter the username, password, and full name to create an account.

Library Management System

Student Register

Username:

Password:

Confirm Password:

Full Name:

[Username and Password Requirements](#)

[Have an account? Login](#)

Click the “Show” button next to the password field to view password.

Library Management System

Student Register

Username:

Password:

Confirm Password:

Full Name:

[Username and Password Requirements](#)

[Have an account? Login](#)

Click the “Register” button to create an account. A success message will appear when the account has been successfully created.

The screenshot shows a web browser window titled "Library Management System". The main heading is "Student Register". Below the heading, there are four input fields: "Username:" with the value "amanda", "Password:" with masked characters and a "Show" button, "Confirm Password:" with masked characters and a "Show" button, and "Full Name:" with the value "amanda yeung". Below these fields, a red message states "Registration success! You can return to login page." followed by a small link "Username and Password Requirements". At the bottom, there are two buttons: "Register" (highlighted with a blue border) and "Back". A link "Have an account? Login" is positioned below the "Back" button.

You can return to the Login page by clicking the hyperlink “Have an account? Login”.

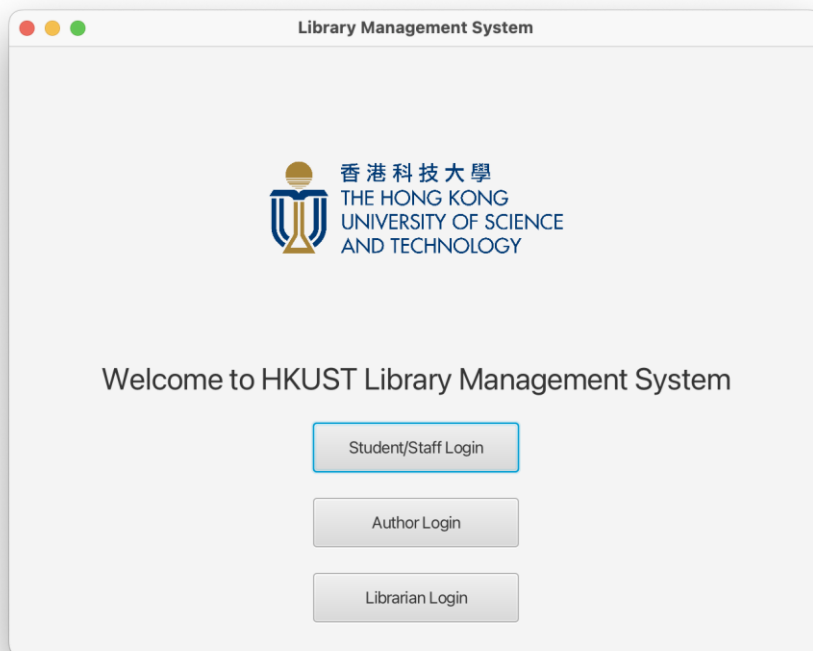
The screenshot shows a web browser window titled "Library Management System". The main heading is "Student/Staff Login". Below the heading, there are two input fields: "Username:" and "Password:", both with "Show" buttons. Below these fields, there are two buttons: "Login" and "Back". A link "Don't have an account? Register" is positioned below the "Back" button.

4.3 Student/Staff Dashboard

This section introduces how to use the Student/Staff Dashboard.

4.3.1 Login

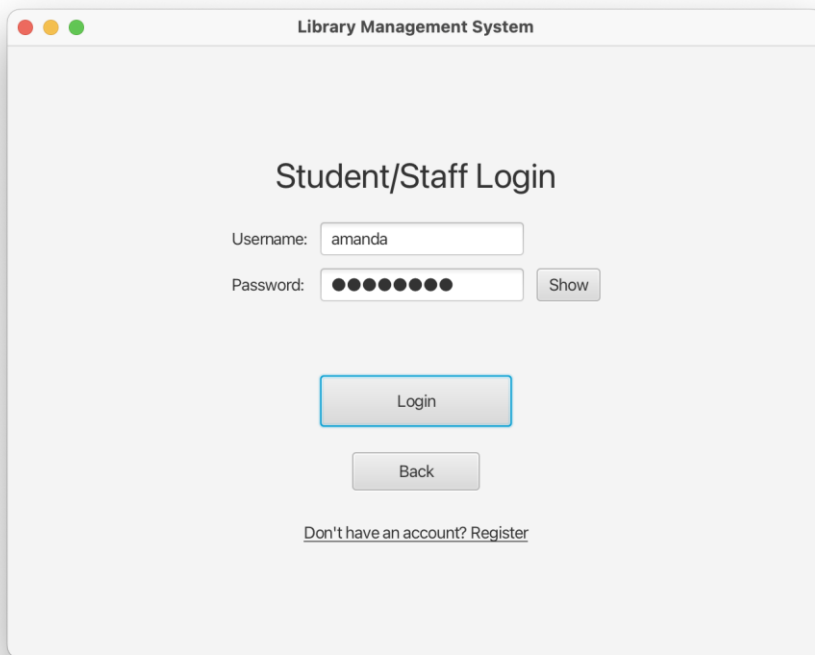
When you enter the Library Management System, you will see the Home Page.



Select "Student/Staff Login" to log in as a "Student/Staff".

A screenshot of the "Student/Staff Login" page within the "Library Management System" browser window. The page has a title "Student/Staff Login". It contains a "Username:" label followed by a text input field, and a "Password:" label followed by a text input field and a "Show" button. Below these fields are "Login" and "Back" buttons. At the bottom, there is a link: "Don't have an account? Register".

Enter your username and password.



Library Management System

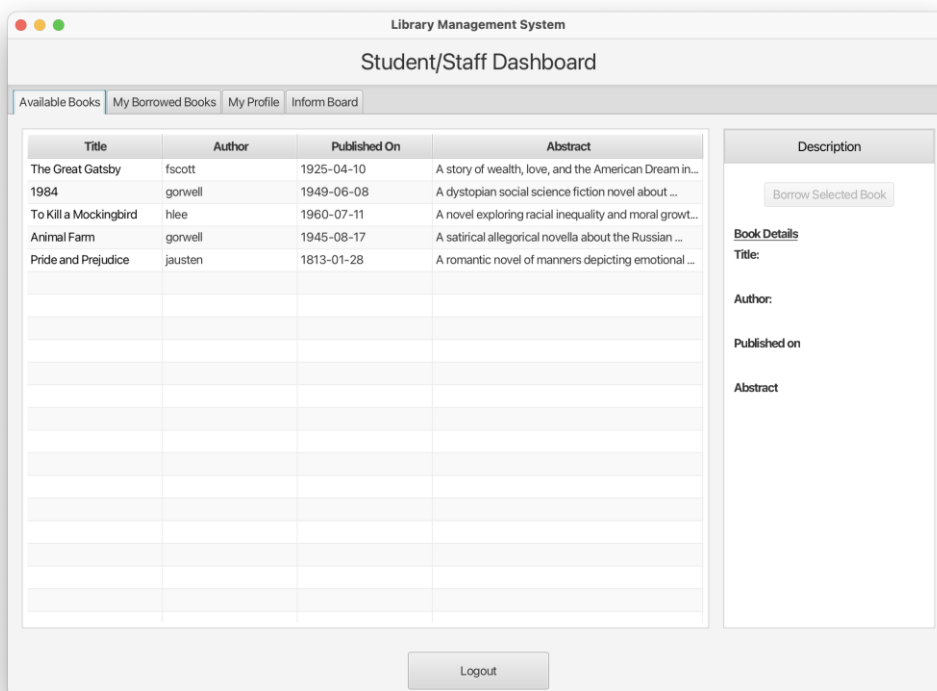
Student/Staff Login

Username:

Password:

[Don't have an account? Register](#)

If you log in successfully, you should see the Available Books tab.



Library Management System

Student/Staff Dashboard

Available Books | My Borrowed Books | My Profile | Inform Board

Title	Author	Published On	Abstract
The Great Gatsby	fscott	1925-04-10	A story of wealth, love, and the American Dream in...
1984	gorwell	1949-06-08	A dystopian social science fiction novel about ...
To Kill a Mockingbird	hlee	1960-07-11	A novel exploring racial inequality and moral growth...
Animal Farm	gorwell	1945-08-17	A satirical allegorical novella about the Russian ...
Pride and Prejudice	jausten	1813-01-28	A romantic novel of manners depicting emotional ...

Description

Book Details

Title:

Author:

Published on

Abstract

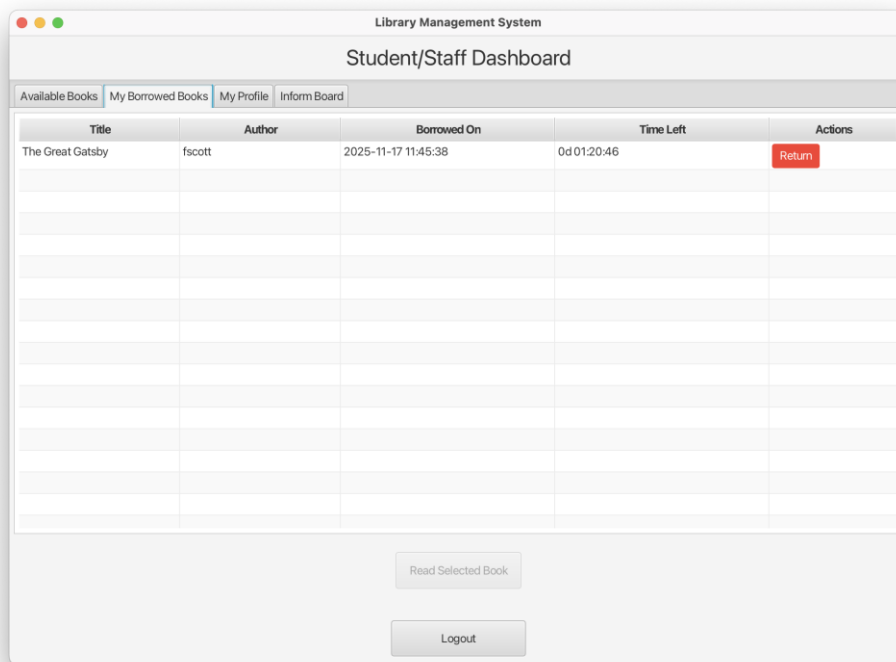
4.3.2 Available Books

In the “Available Books” tab, you will find a list of available books for you to borrow. When you select a book, the corresponding book details will be shown in the description box, in case the details are too long to be displayed in the table columns.

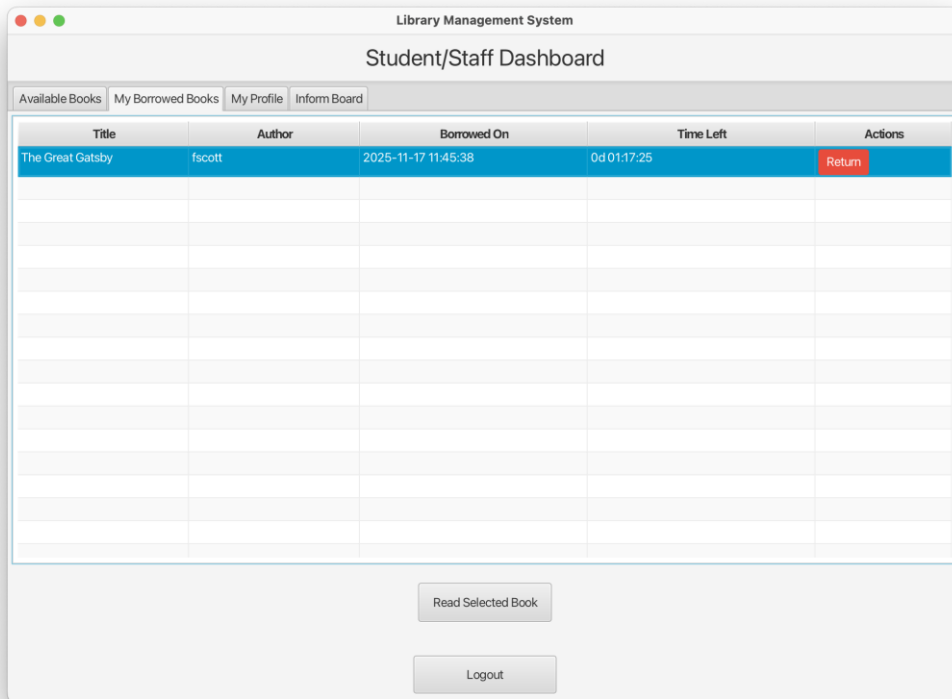
Once the book is borrowed, it will disappear from the list of available books.

4.3.3 My Borrowed Books

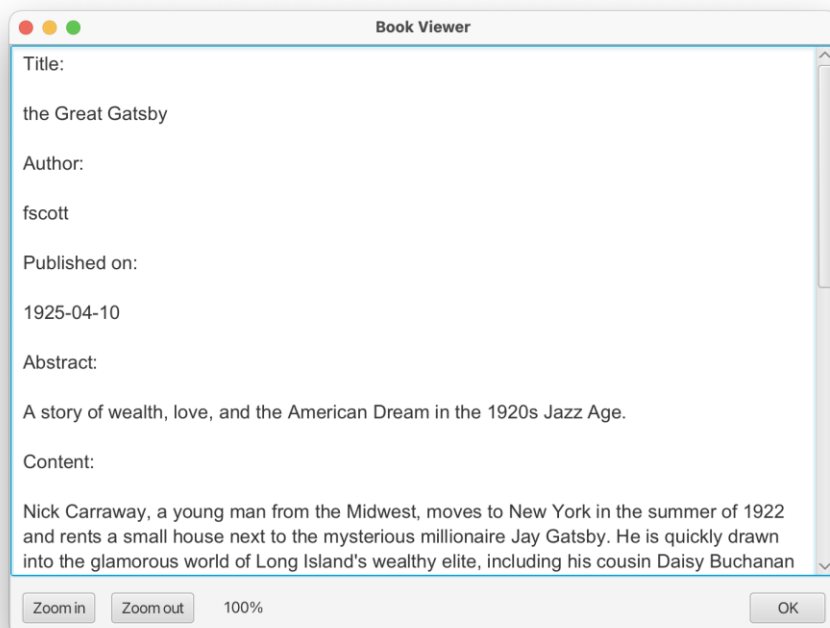
In the “My Borrowed Books” tab, you will find a list of books that you have borrowed. The borrowing details, including borrowing date and time, and time left, will be shown.



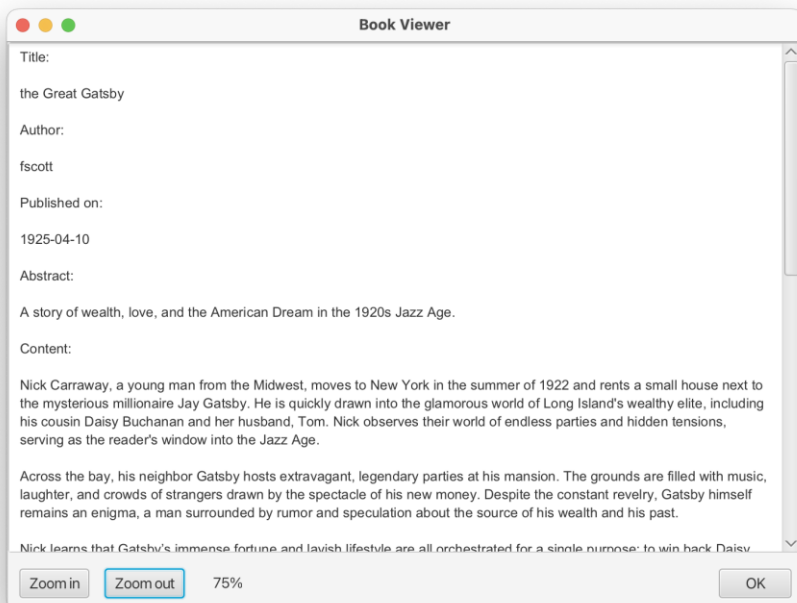
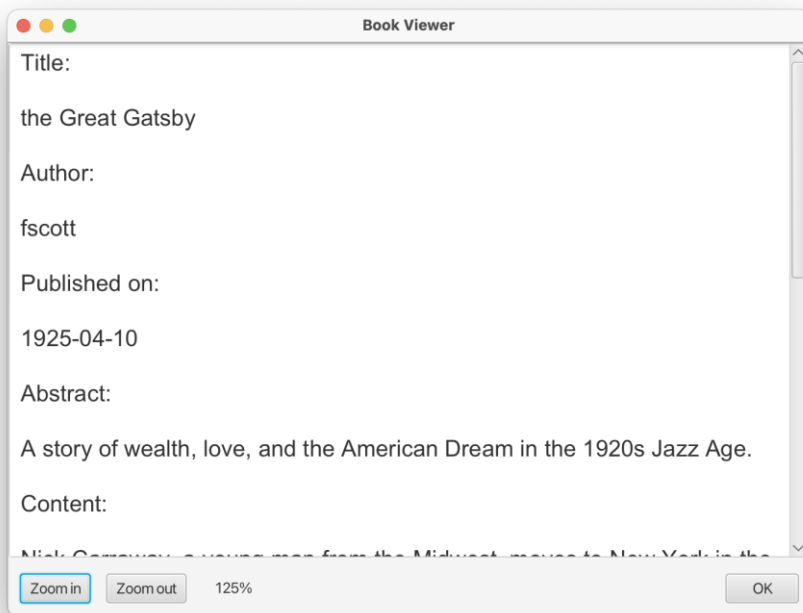
Once you have selected a book, you can press the “Read Selected Book” button to read the content of the book.



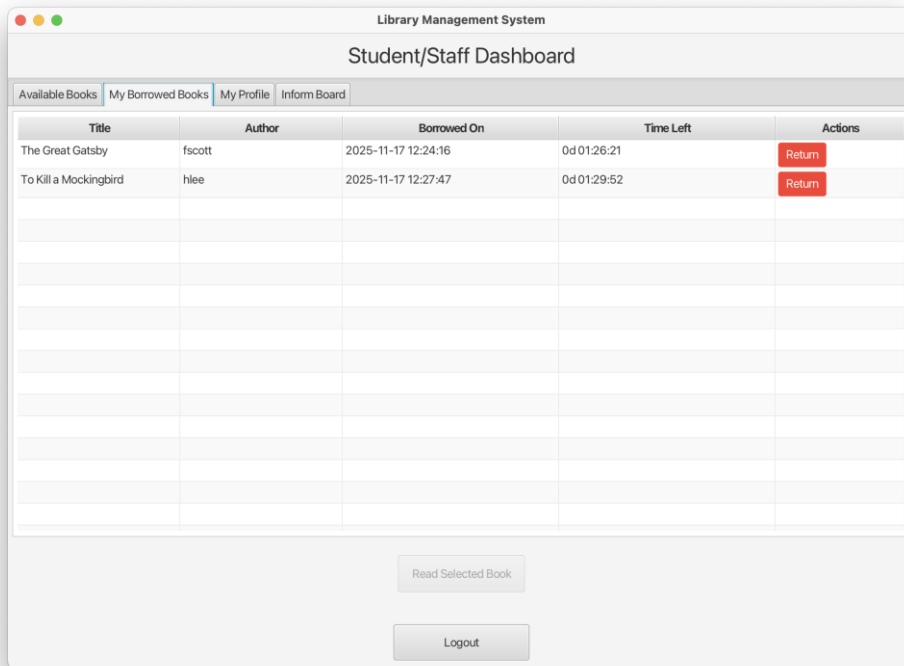
The following screen will appear.



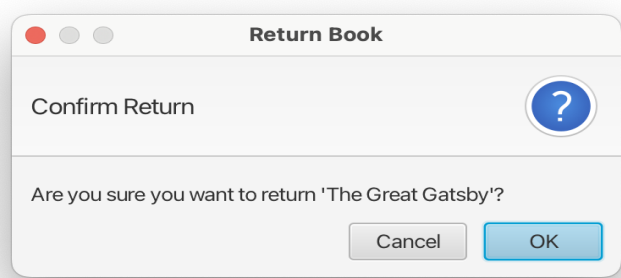
You can zoom in/out text, and can scroll up/down to read the content.



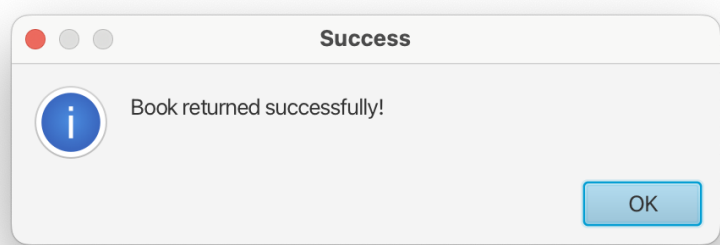
You can press the “Return” button to return the corresponding book.



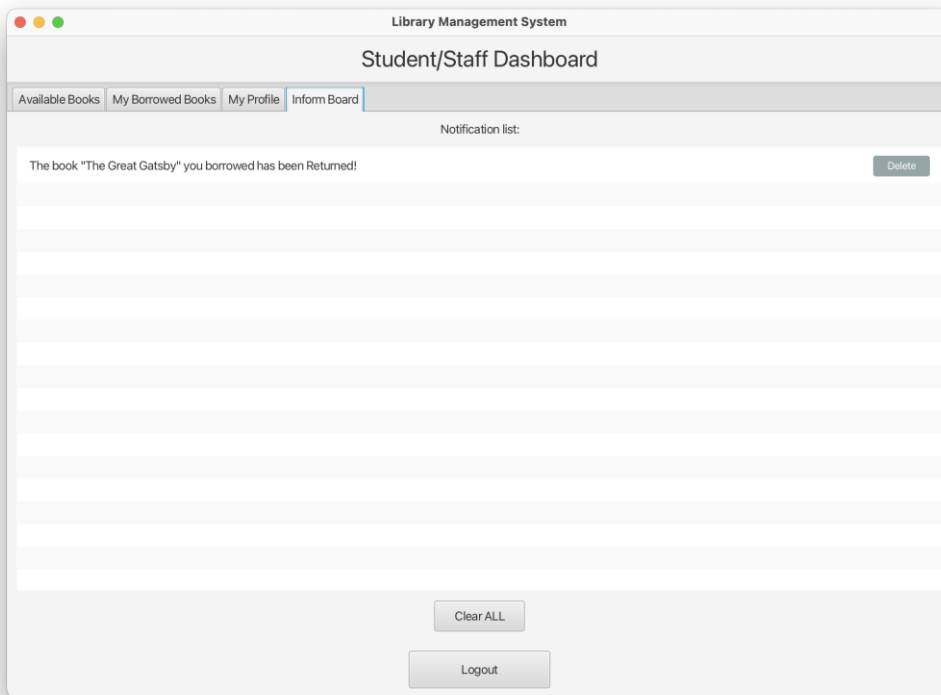
A window will appear to confirm your return action.



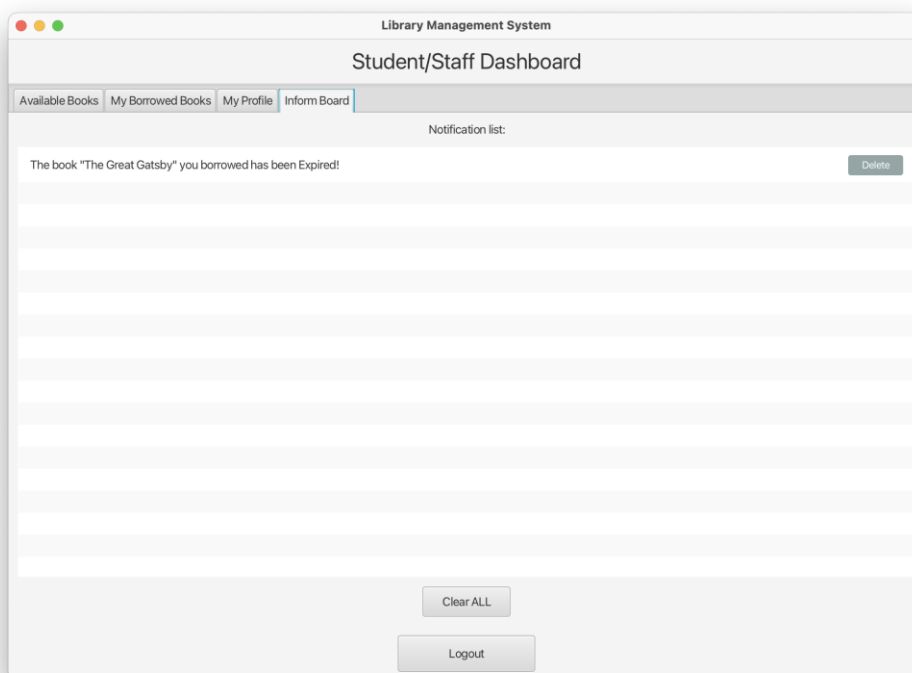
A message will appear to indicate that the book is successfully returned.



Once a book is successfully returned, a notification can be found in “Inform Board” tab.



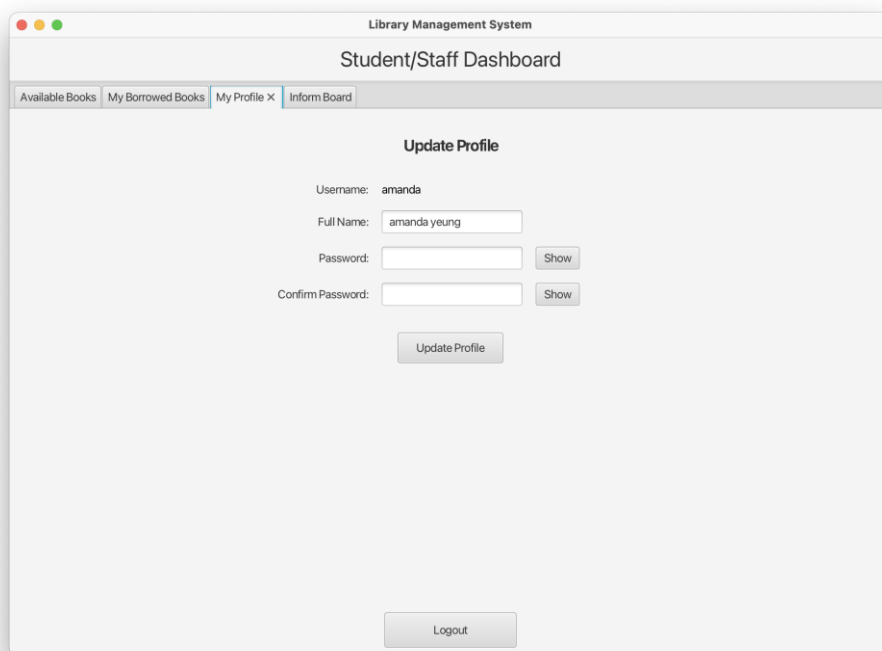
Once the book expires, it will be automatically returned. A notification of expiration can be found in “Inform Board” tab.



Once the book is returned, it will disappear from the list of borrowed books. It can be found in the list of available books.

4.3.4 Update Profile

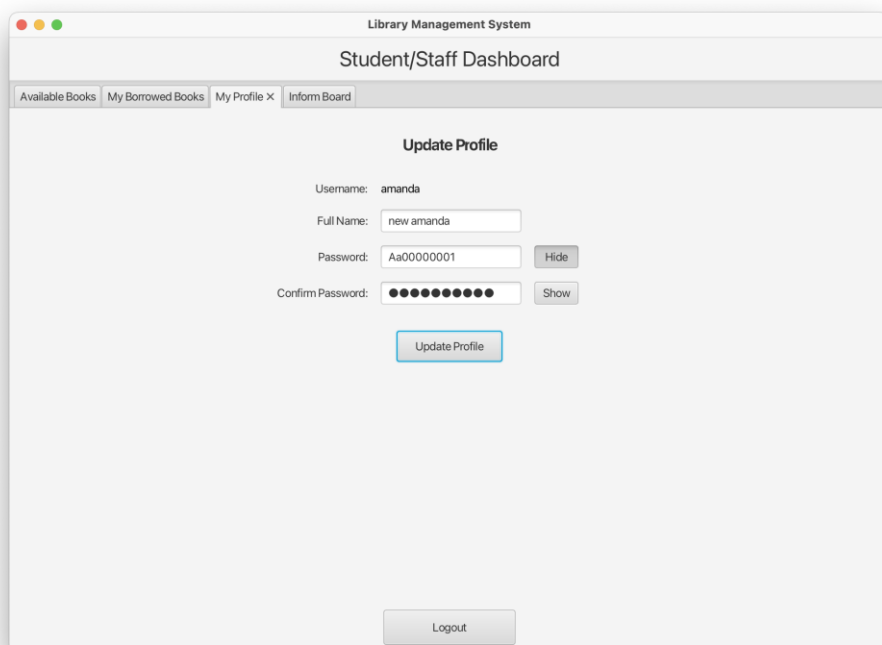
In the “My Profile” tab, you can update your profile.



The screenshot shows a web application window titled "Library Management System" with a subtitle "Student/Staff Dashboard". The navigation bar includes "Available Books", "My Borrowed Books", "My Profile X", and "Inform Board". The "My Profile X" tab is active, displaying the "Update Profile" form. The form contains the following fields and buttons:

- Username:** amanda
- Full Name:** amanda yeung
- Password:** [text input] with a "Show" button to its right.
- Confirm Password:** [text input] with a "Show" button to its right.
- Update Profile:** A button at the bottom of the form.
- Logout:** A button at the bottom right of the dashboard.

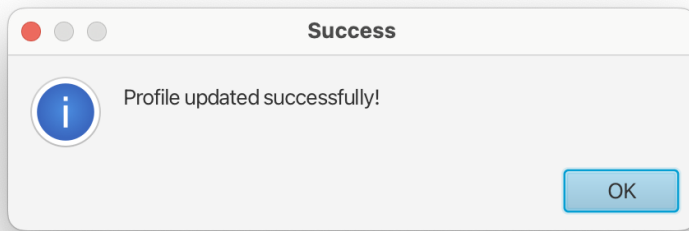
Press “Update Profile” button after entering the new password and/or the new full name. Again, you can press the “Hide/Show” button to view your password.



This screenshot shows the same "Update Profile" form as the previous one, but with updated information:

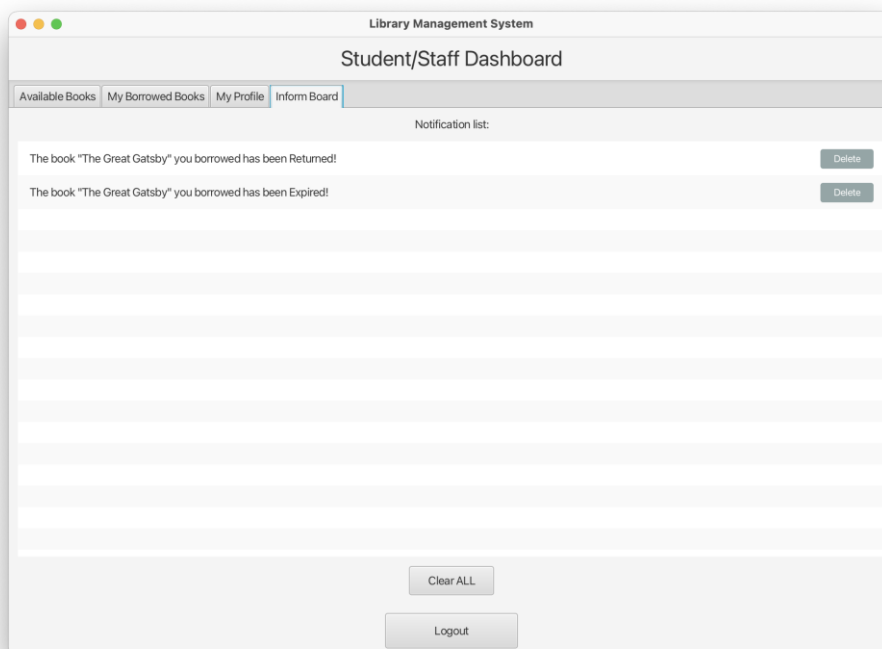
- Username:** amanda
- Full Name:** new amanda
- Password:** Aa00000001, with a "Hide" button to its right.
- Confirm Password:** [masked with dots], with a "Show" button to its right.
- Update Profile:** A button at the bottom of the form.
- Logout:** A button at the bottom right of the dashboard.

A message will appear to indicate that the profile is successfully updated.

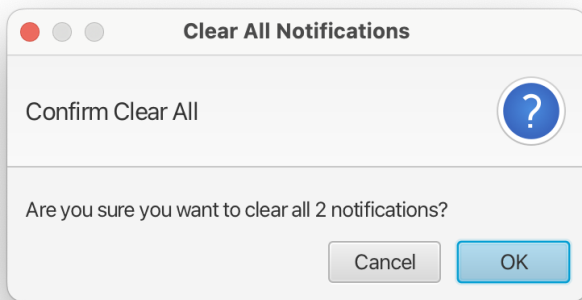


4.3.5 Inform Board

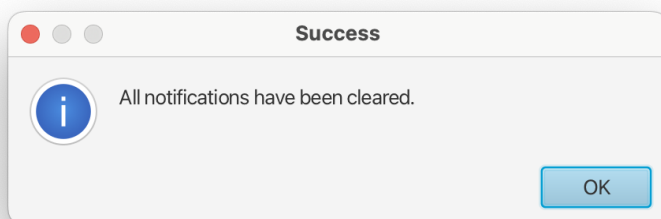
All the notifications belonging to this user will be shown on the inform board.



You can click the “Clear” button to delete the corresponding notification. You can also click the “Clear ALL” button to delete all the notifications, but double confirmation is needed to ensure the user does not accidentally misclick.

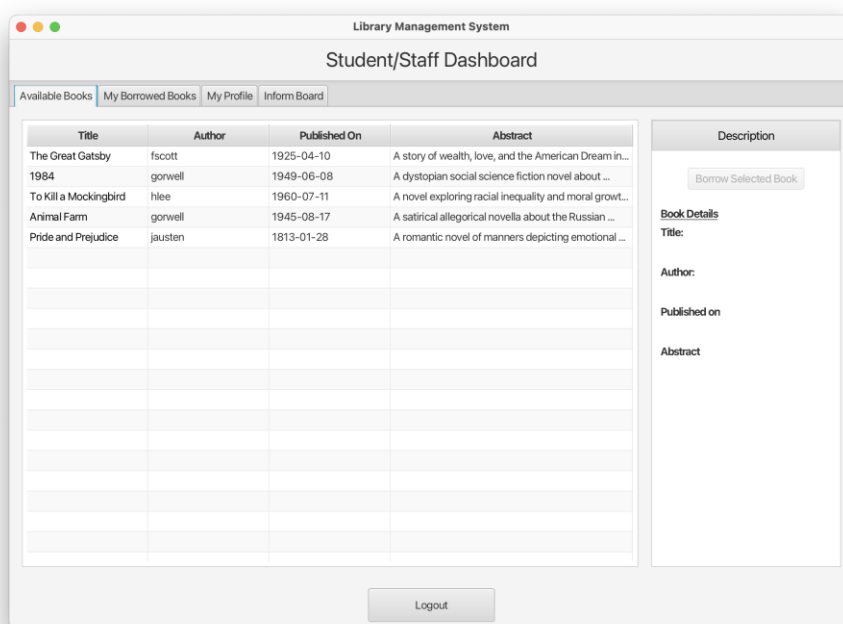


A message will appear to indicate that all notifications have been successfully cleared.

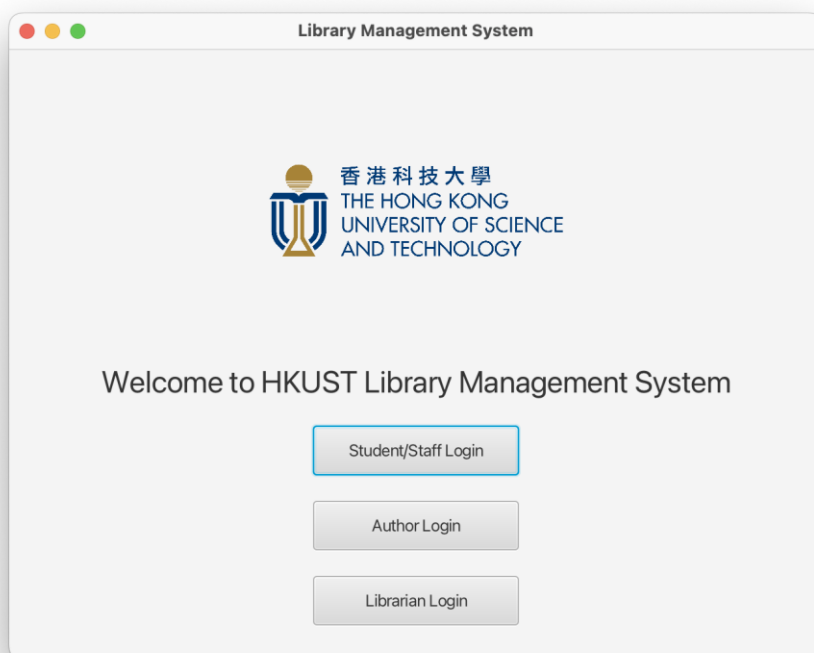


4.3.6 Logout

Press the “Logout” button to log out of the system.



You will see the Home Page if you log out of the system successfully.

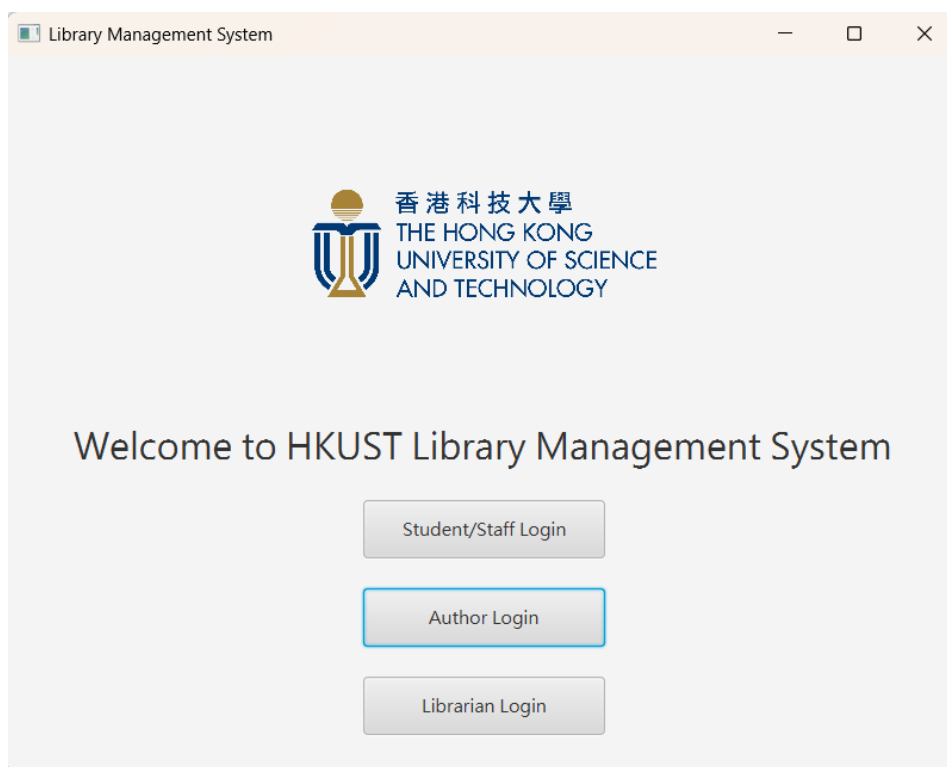


4.4 Author Dashboard

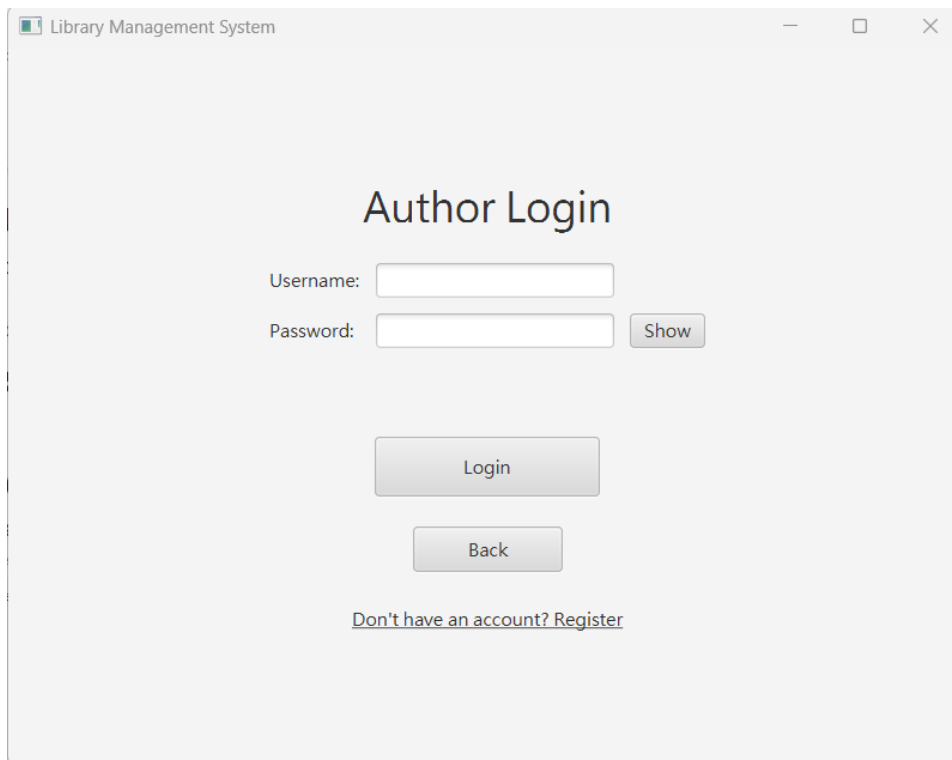
This section introduces how to use the author dashboard.

4.4.1 Login

When you run the Library Management System, you will see the Home Page.

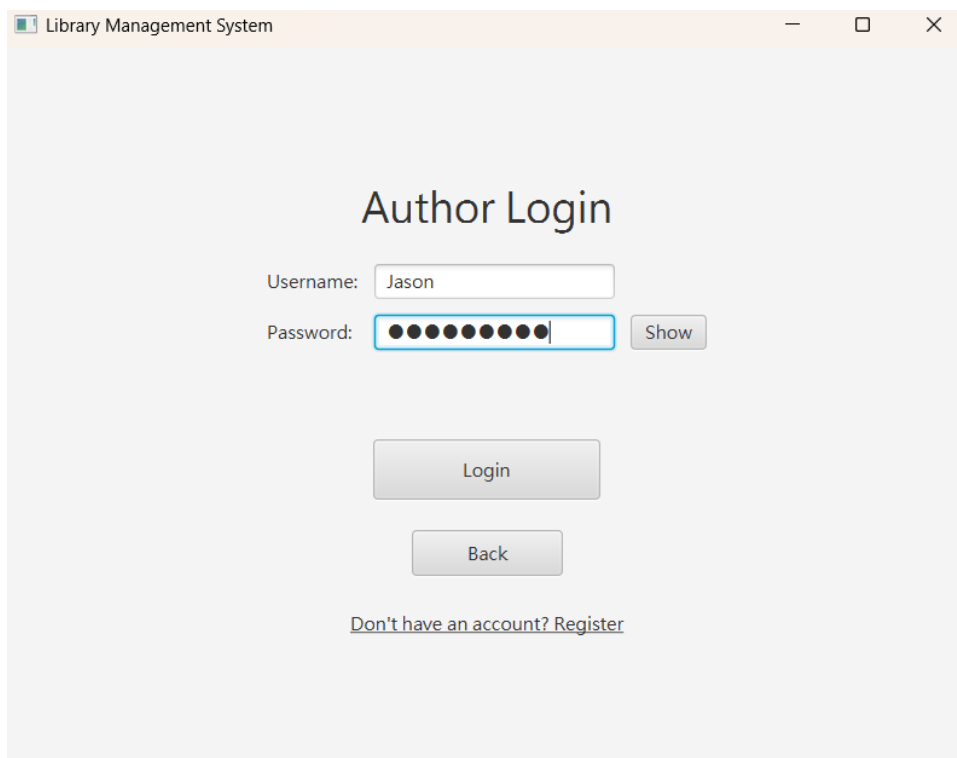


Select “Author Login” in order to access the Author Dashboard.



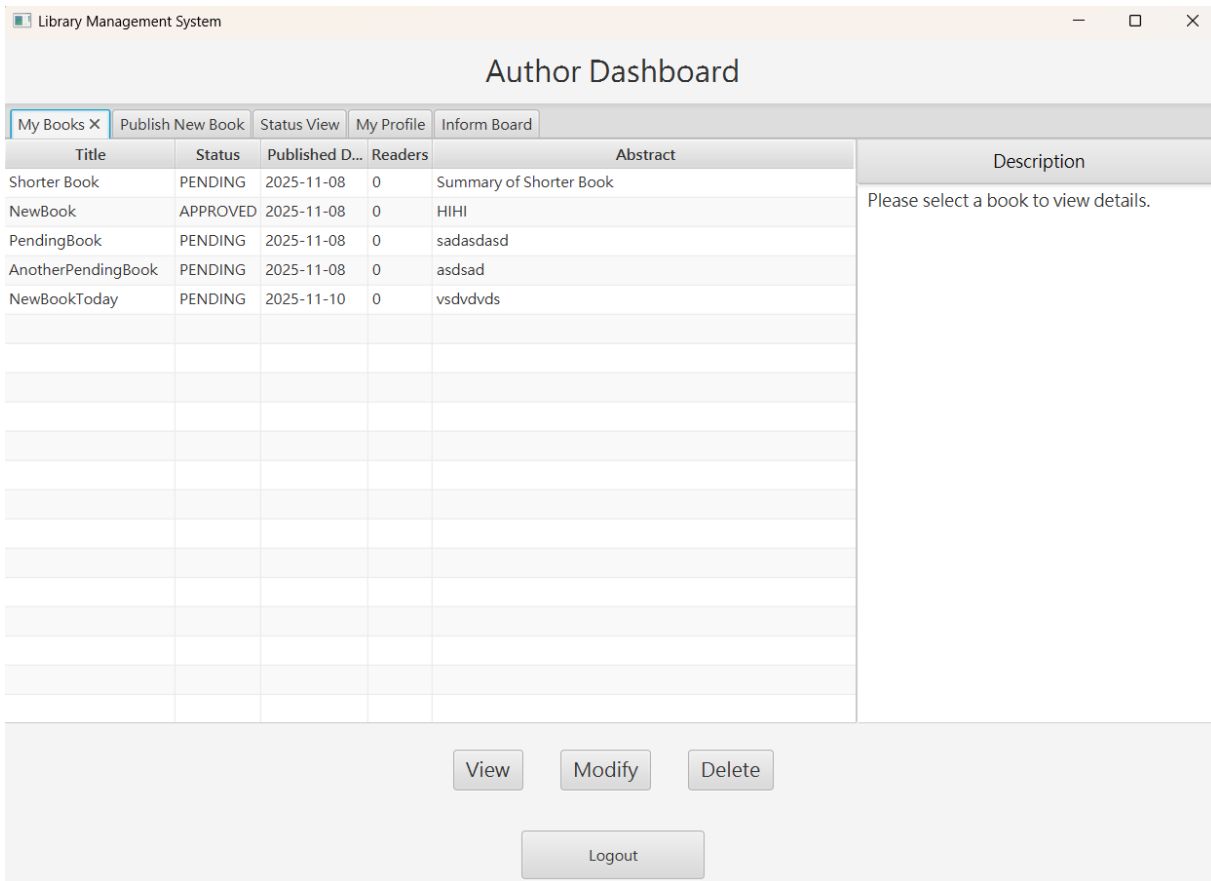
The screenshot shows a web browser window titled "Library Management System". The main heading is "Author Login". Below the heading, there are two input fields: "Username:" and "Password:". To the right of the "Password:" field is a "Show" button. Below the input fields are two buttons: "Login" and "Back". At the bottom, there is a link that says "Don't have an account? Register".

Fill in the fields with your registered information.



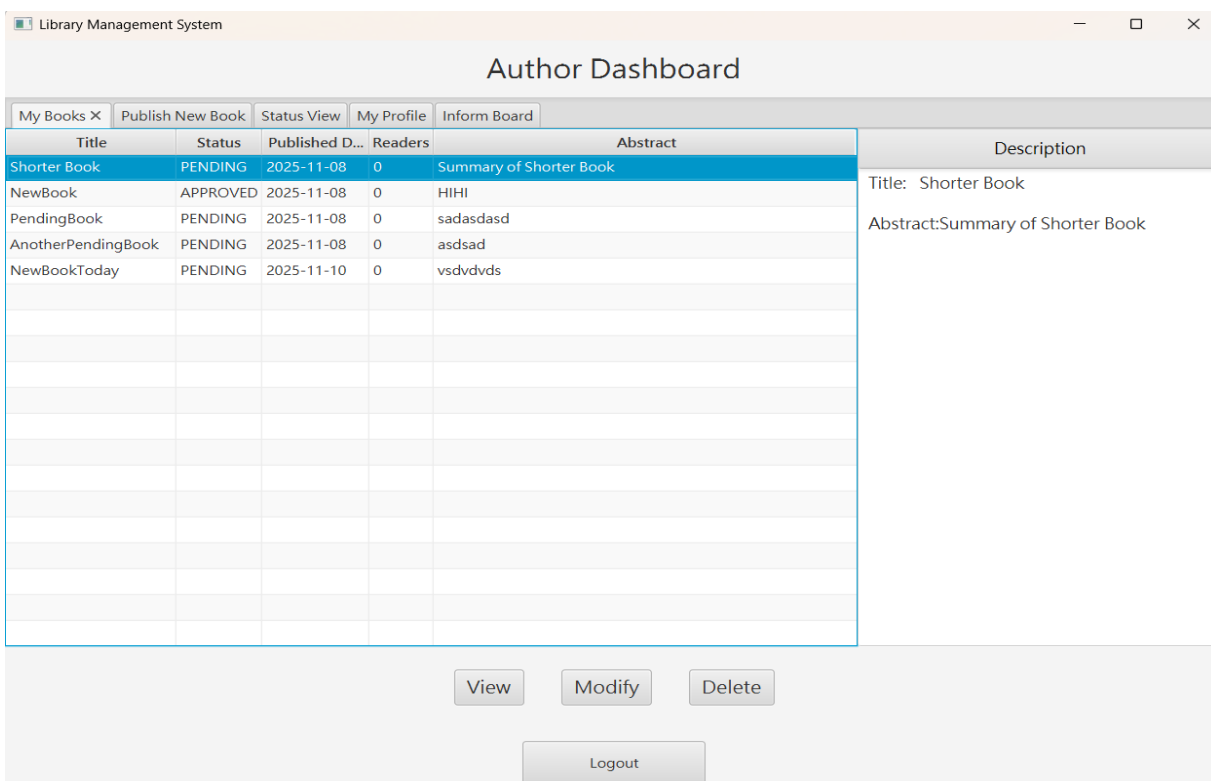
The screenshot shows the same "Author Login" page, but now the "Username:" field is filled with the text "Jason". The "Password:" field is filled with ten black dots, indicating a password is entered. The "Show" button is still present to the right of the password field. The "Login" and "Back" buttons remain below the fields. The "Don't have an account? Register" link is at the bottom.

After that, click the Login Button and you should be able to see the My Books Tab.



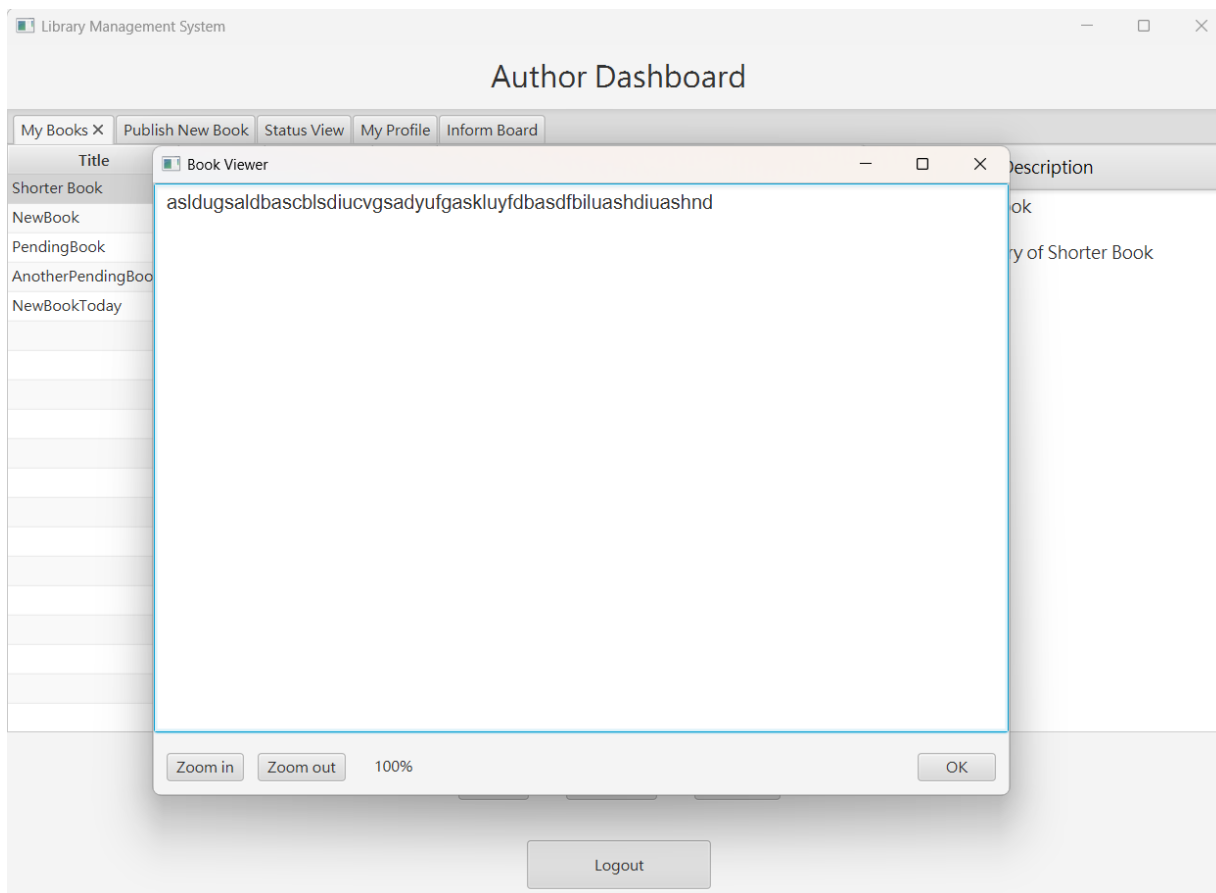
4.4.2 My Books

Select a specific row and the description box on the right side will show the Title and Abstract of the book, in case they are too long and cannot be displayed by the table

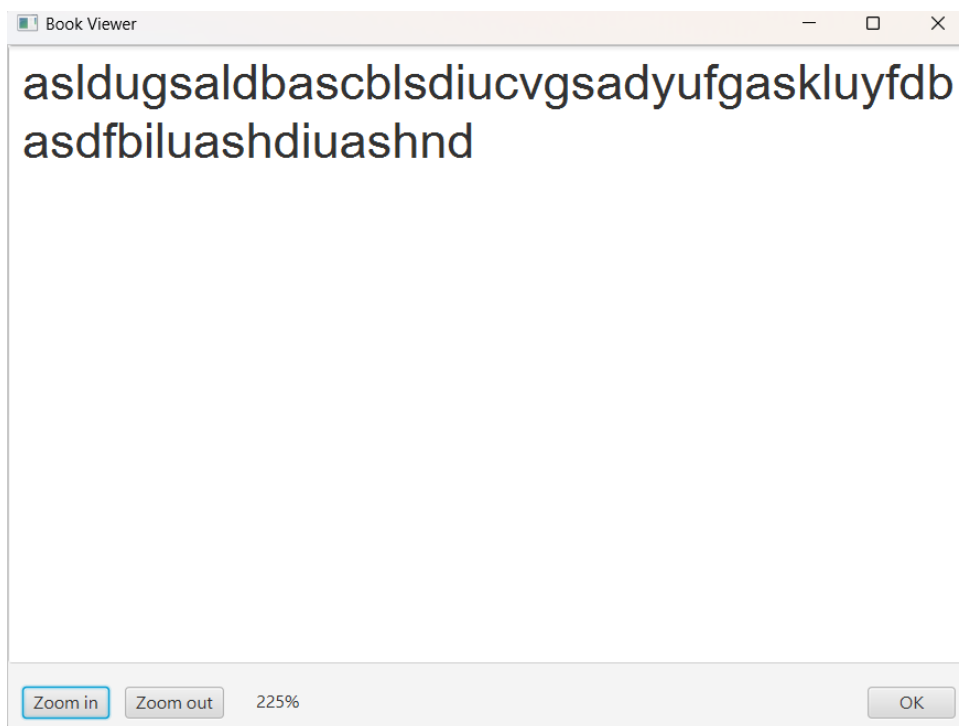


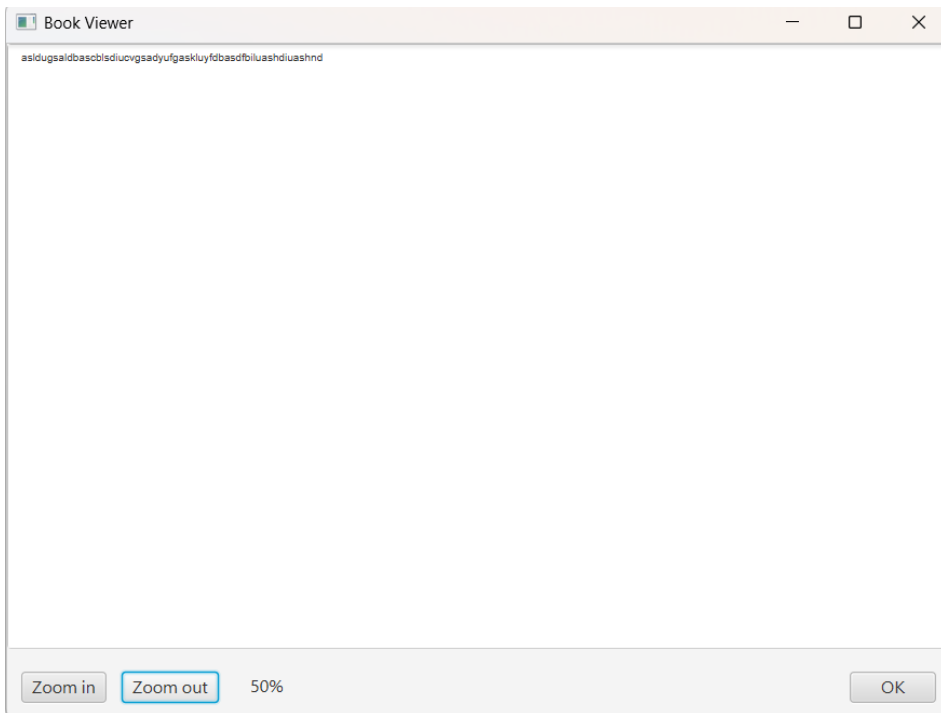
After selecting a specific book, you can read, edit and delete the book.

For reading the book, press the “View” Button.



A pop-up window jumps out and you can scroll up and down to view the whole book.

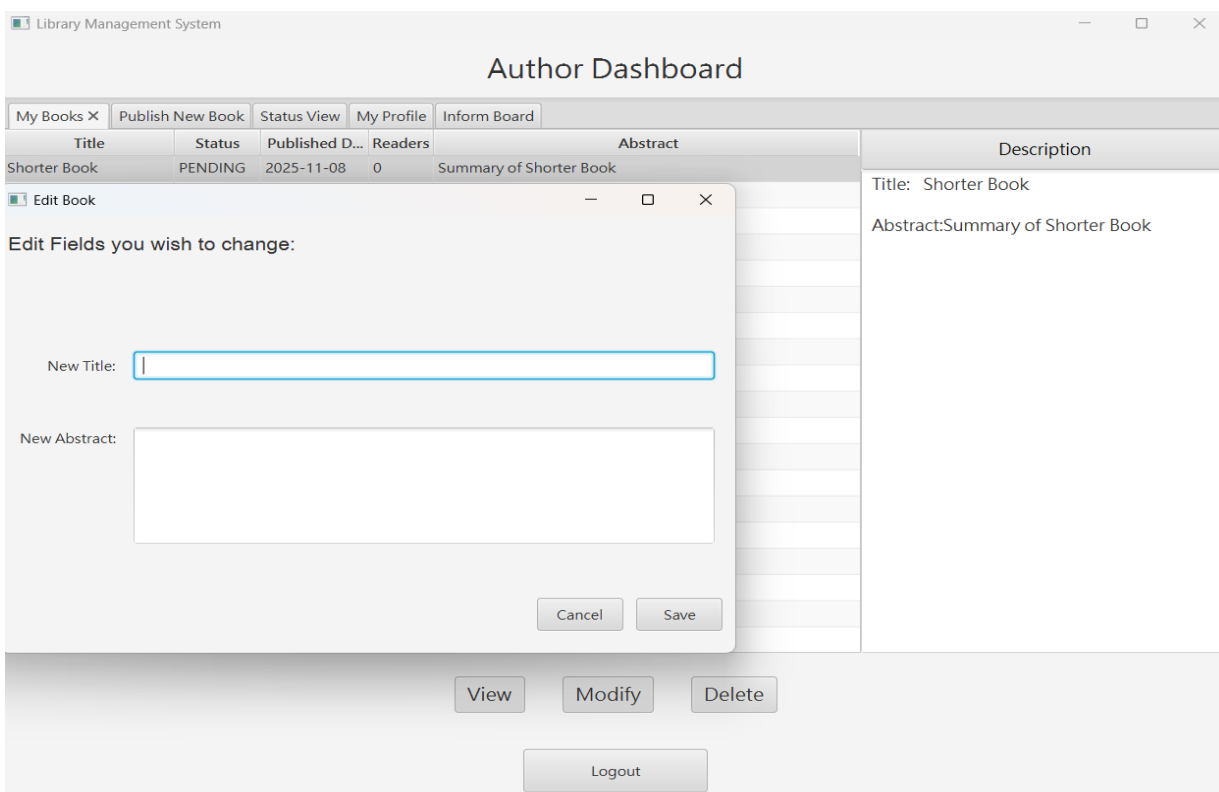




You can press Zoom in or Zoom out for adjusting comfortable font size. The enlarge scale is shown at the bottom. If you want to close the view window, simply click on the OK button.

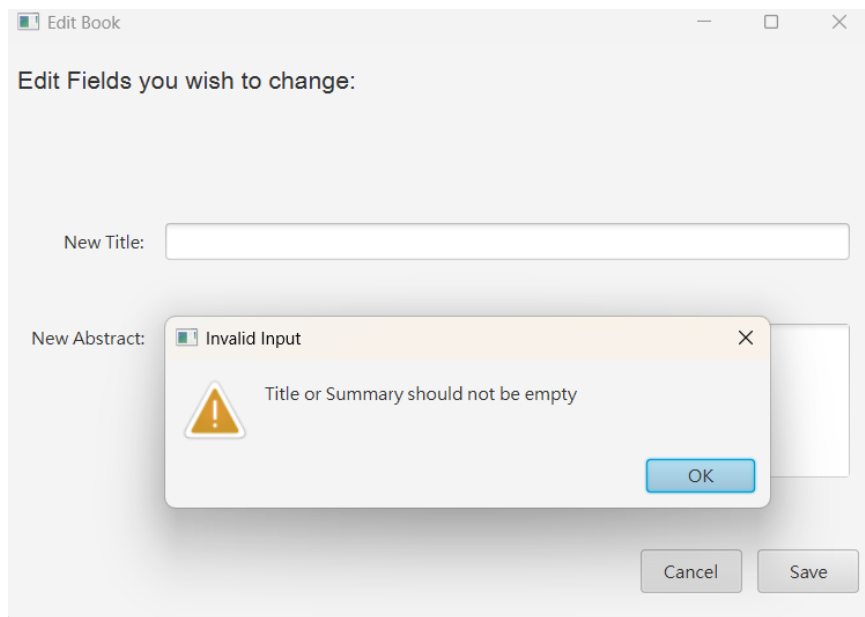
For editing the book, press the “Modify” Button.

Note that you can only modify a book that is in “Pending” Status, or “Approved” Status but with 0 readers currently.

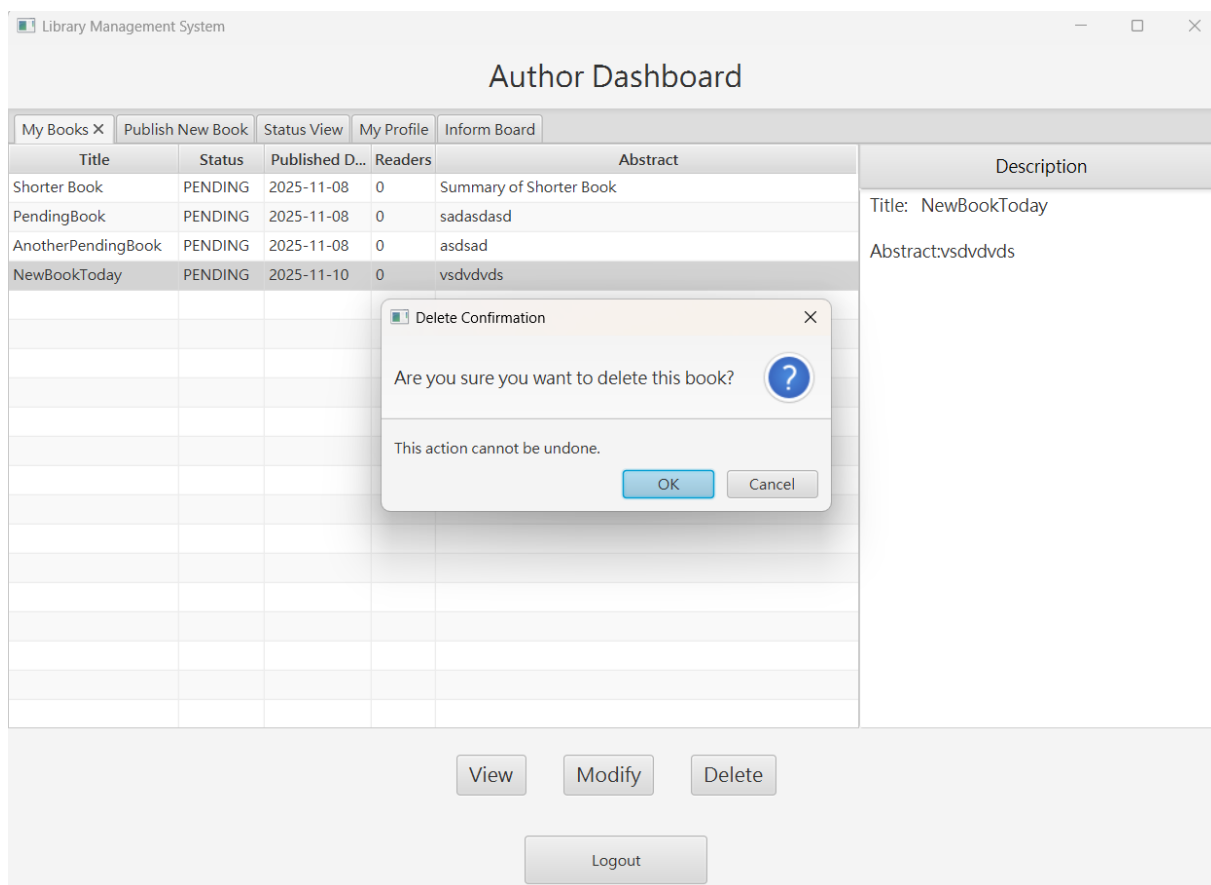


Enter the new title and new abstract and click the “Save” button to modify the book. Note that you

cannot successfully modify the book if any of the fields are empty.



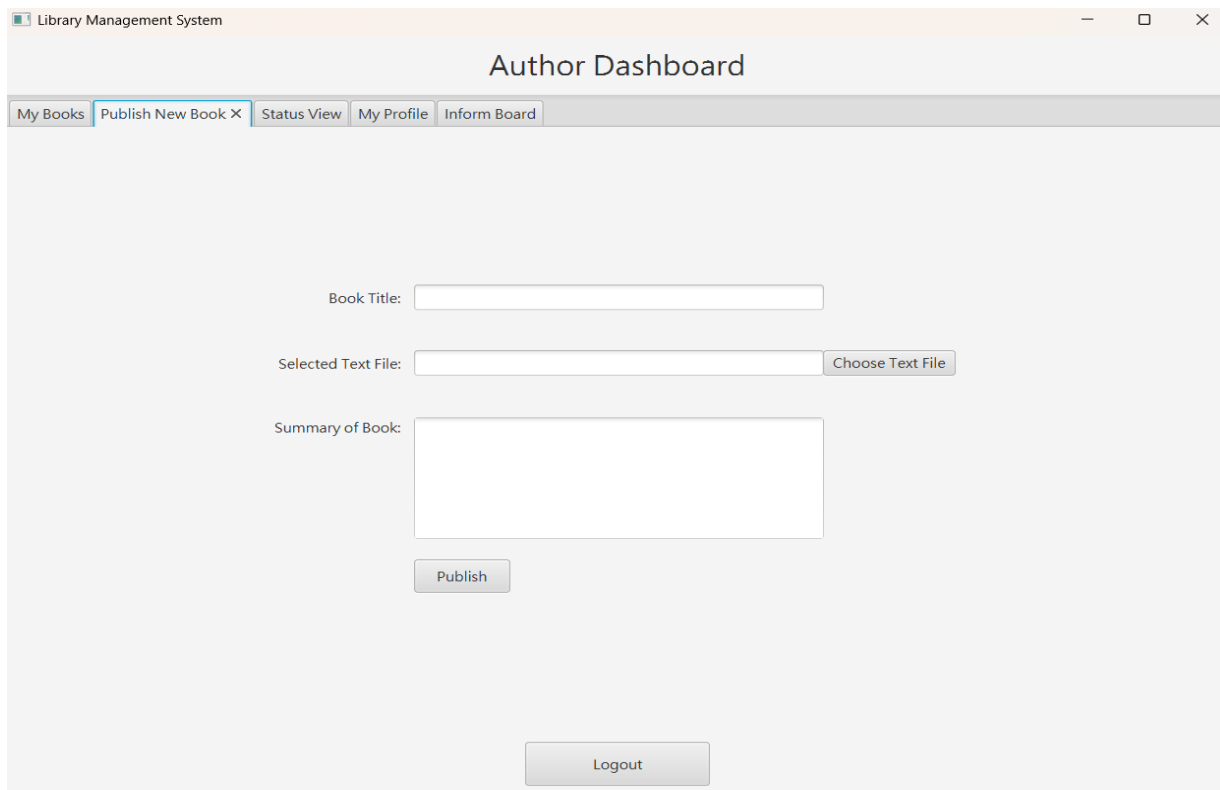
For delete the book. Similar as editing the book, you can only delete a book that is in “Pending” Status, or “Approved” Status but with 0 readers currently.



After pressing “OK”, the book will be permanently deleted. You can see the book selected from the right side of the description box to make sure you are deleting the correct book.

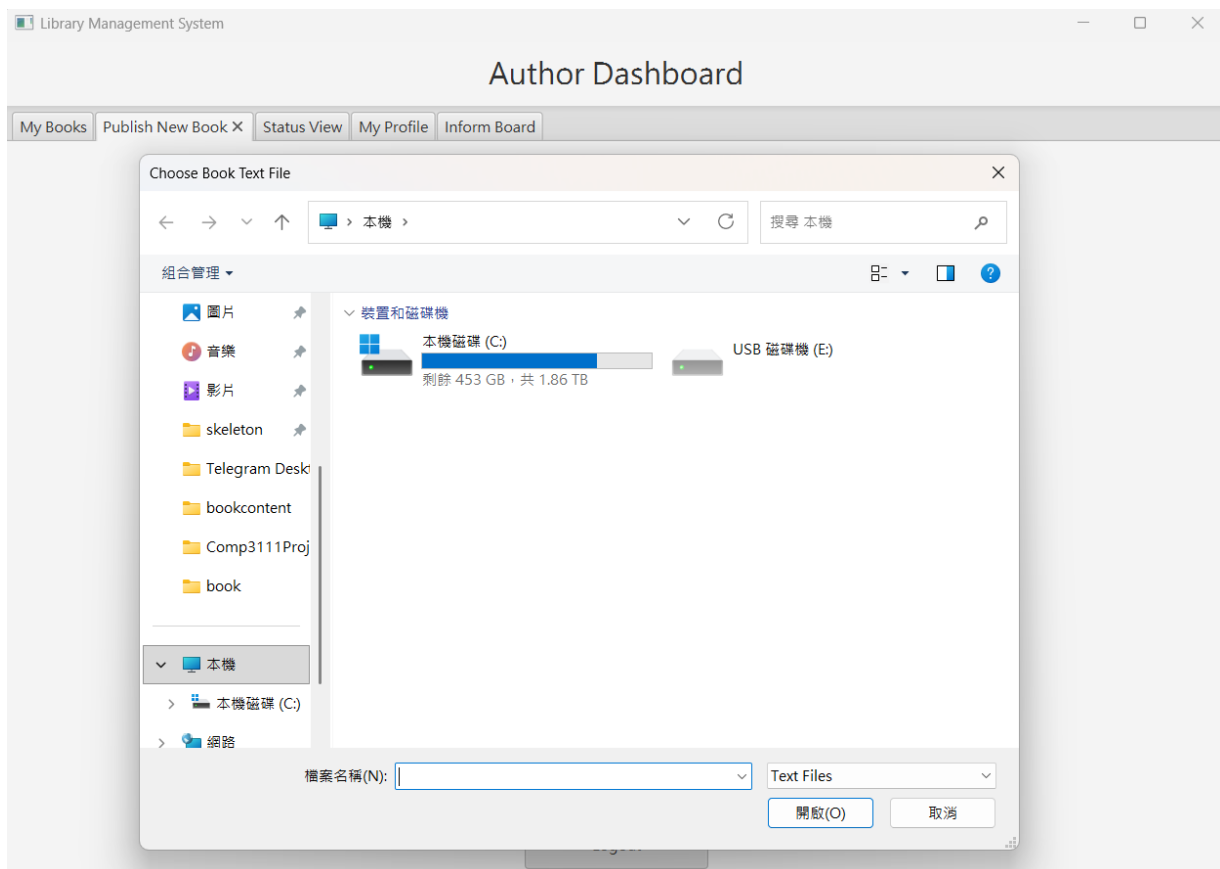
4.4.3 Publish New Book

You can publish new book in the “Publish New Book” Tab.

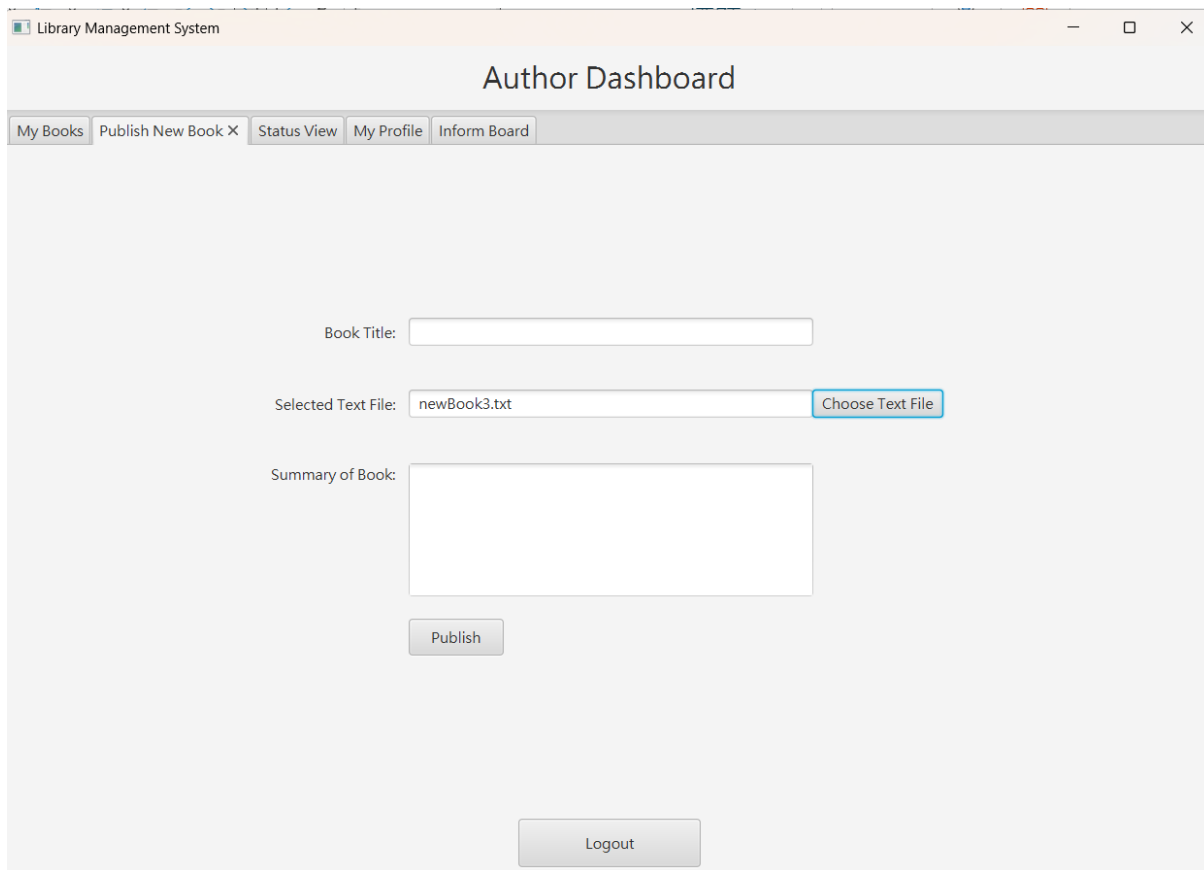


The screenshot shows the 'Author Dashboard' window of the 'Library Management System'. The dashboard has a navigation bar with tabs: 'My Books', 'Publish New Book X', 'Status View', 'My Profile', and 'Inform Board'. The 'Publish New Book X' tab is active. Below the navigation bar, there are three input fields: 'Book Title:' with a text box, 'Selected Text File:' with a text box and a 'Choose Text File' button, and 'Summary of Book:' with a larger text box. A 'Publish' button is located below the summary box. At the bottom center of the dashboard is a 'Logout' button.

You need to manually input the title and abstract of the book. For choosing the txt file of the book, you need to click on the “Choose Text File” button.



After clicking, the OS default file explorer will pop up. You can find your text file from here.

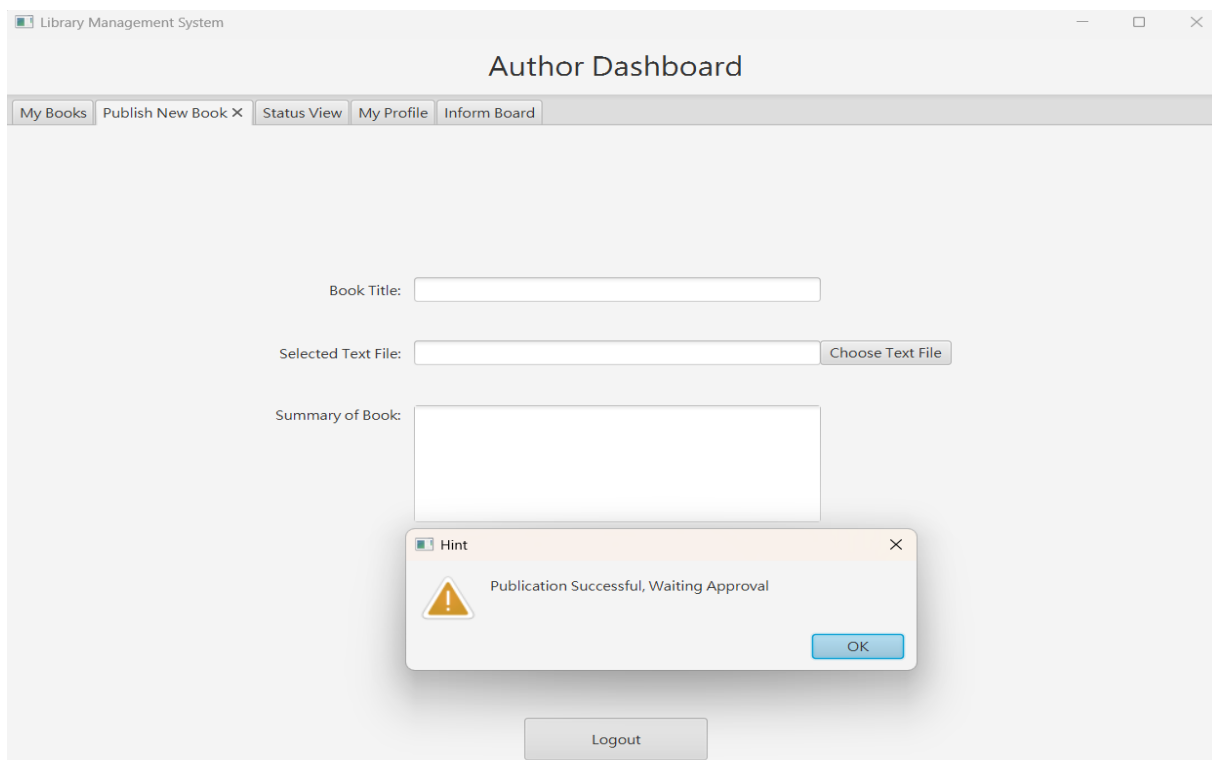


The screenshot shows the 'Author Dashboard' window of the 'Library Management System'. The window has a title bar with standard OS controls. Below the title bar is a navigation bar with tabs: 'My Books', 'Publish New Book X', 'Status View', 'My Profile', and 'Inform Board'. The main content area contains the following elements:

- A 'Book Title:' label followed by a text input field.
- A 'Selected Text File:' label followed by a text input field containing 'newBook3.txt' and a 'Choose Text File' button.
- A 'Summary of Book:' label followed by a large text area.
- A 'Publish' button located below the summary text area.
- A 'Logout' button at the bottom center of the dashboard.

After selecting the desired book filename will be displayed in the above field.

You can click “Publish” button to confirm the publication.

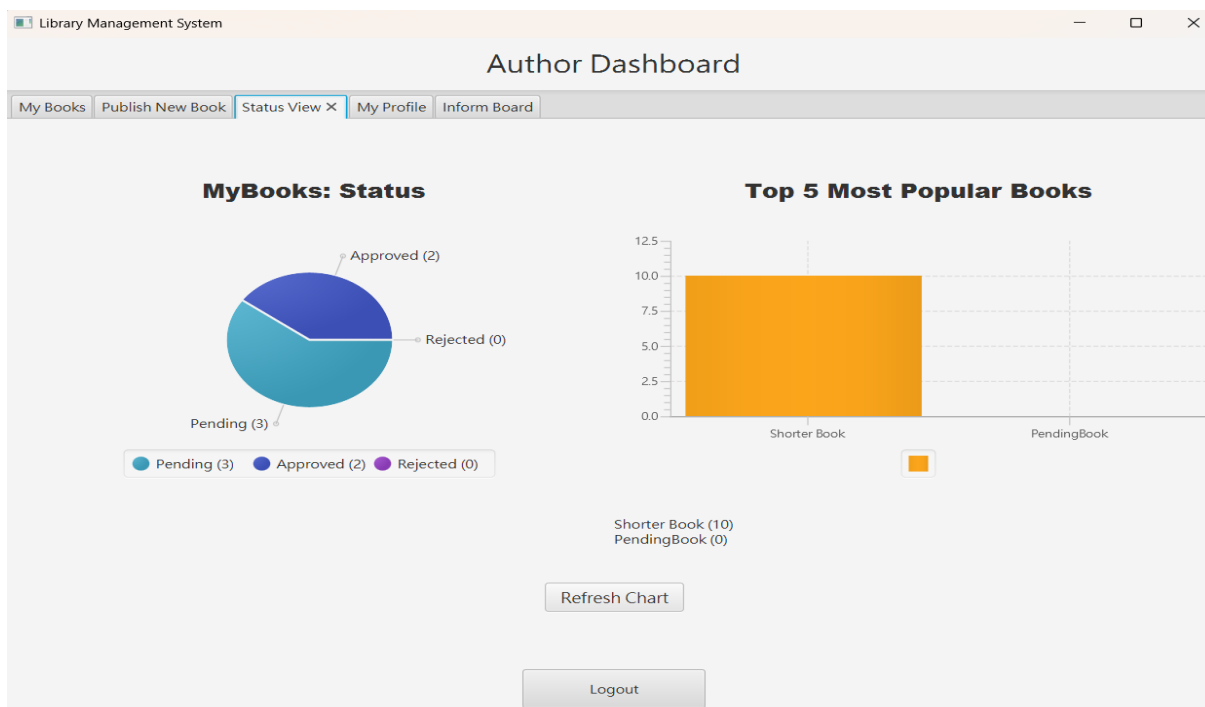


This screenshot shows the same 'Author Dashboard' as the previous one, but with a 'Hint' dialog box open. The dialog box has a title bar with a close button and contains a yellow warning icon, the text 'Publication Successful, Waiting Approval', and an 'OK' button. The background dashboard elements are visible but slightly dimmed.

A pop up hint will be shown if the publication is successful.

4.4.4 Status View

You can view your status in the "Status View" Tab.



The left Pie Chart shows the current status of all your published book. On the other hand, the right Bar Chart shows the top 5 Approved book that being borrowed the most. A exact number of list is shown under the graph for easier understanding purpose.

After clicking "Refresh" button, both graphs is refreshed.



4.4.5 Manage Profile

You can update your profile in the “My Profile Tab”

The screenshot shows the 'Author Dashboard' with the 'My Profile' tab selected. The profile information is displayed as follows: Username: Jason, Full Name: Jason, Password: [input field], Confirmation Password: [input field]. There are 'Show' buttons next to the password fields. An 'Update Profile' button is located below the password fields. A 'Logout' button is at the bottom right.

Username: Jason

Full Name:

Password:

Confirmation Password:

Enter the new password and/or the new fullname and press “Update Profile”

Library Management System

Author Dashboard

My Books Publish New Book Status View My Profile X Inform Board

Username: Jason

Full Name:

Password:

Confirmation Password:

A pop-up window is shown to indicate profile updated successfully.

Library Management System

Author Dashboard

My Books Publish New Book Status View My Profile X Inform Board

Username: Jason

Full Name:

Password:

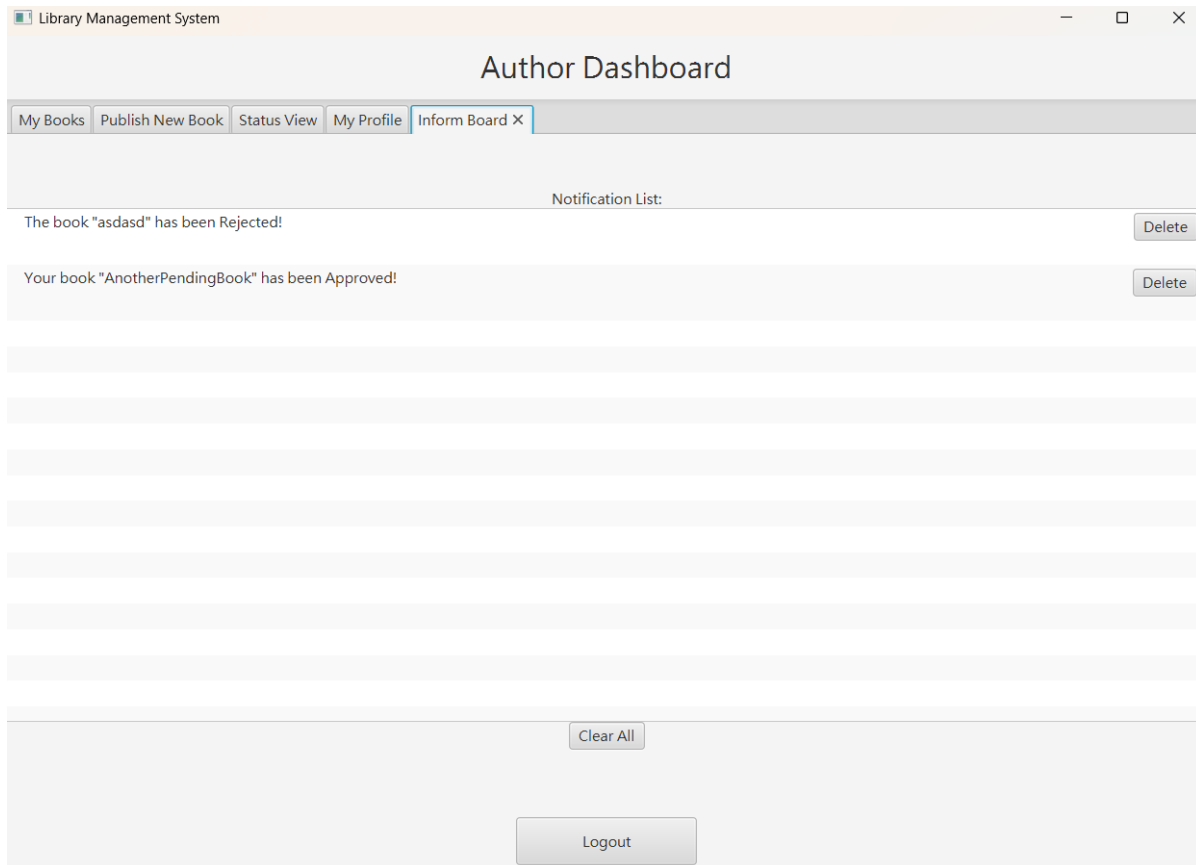
Confirmation Password:

Hint

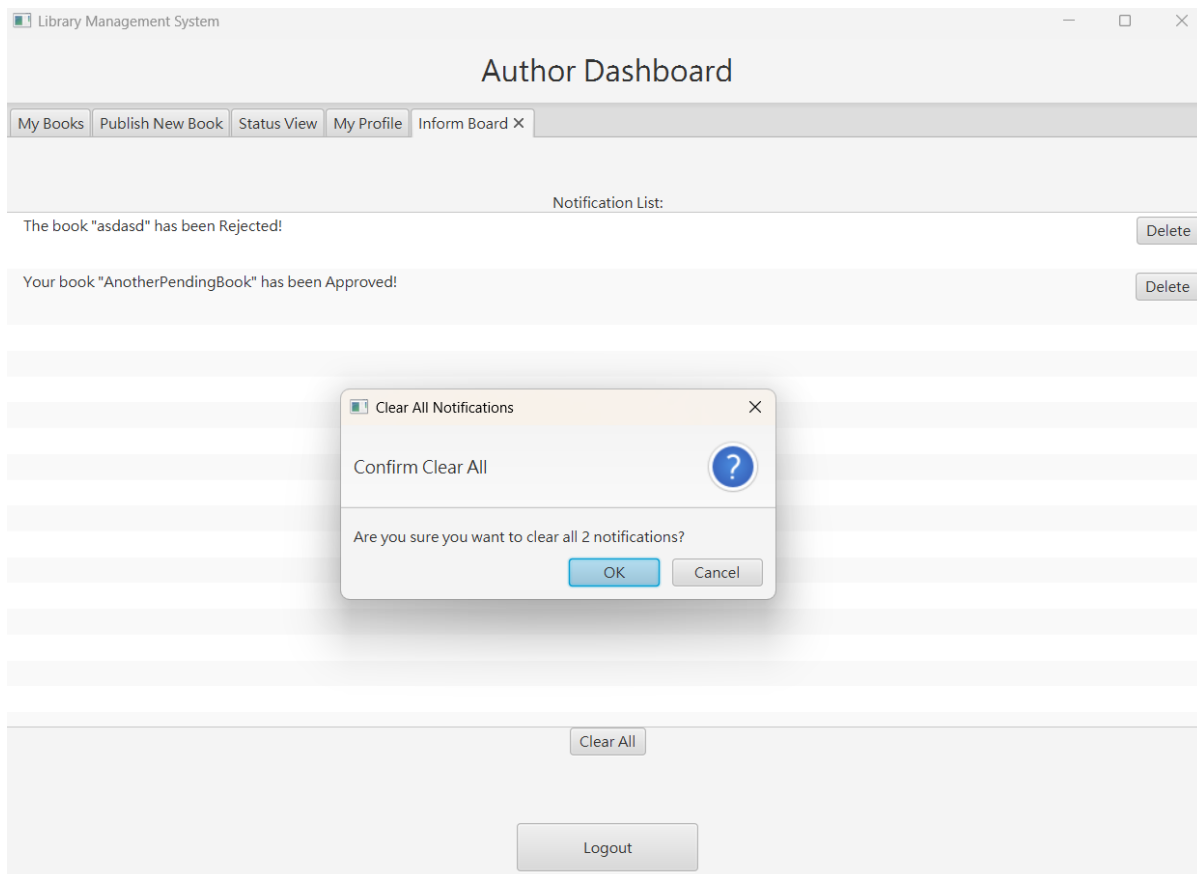
 Update profile success!

4.4.6 Inform Board

All the notification belongs to this user will be shown on the inform board.



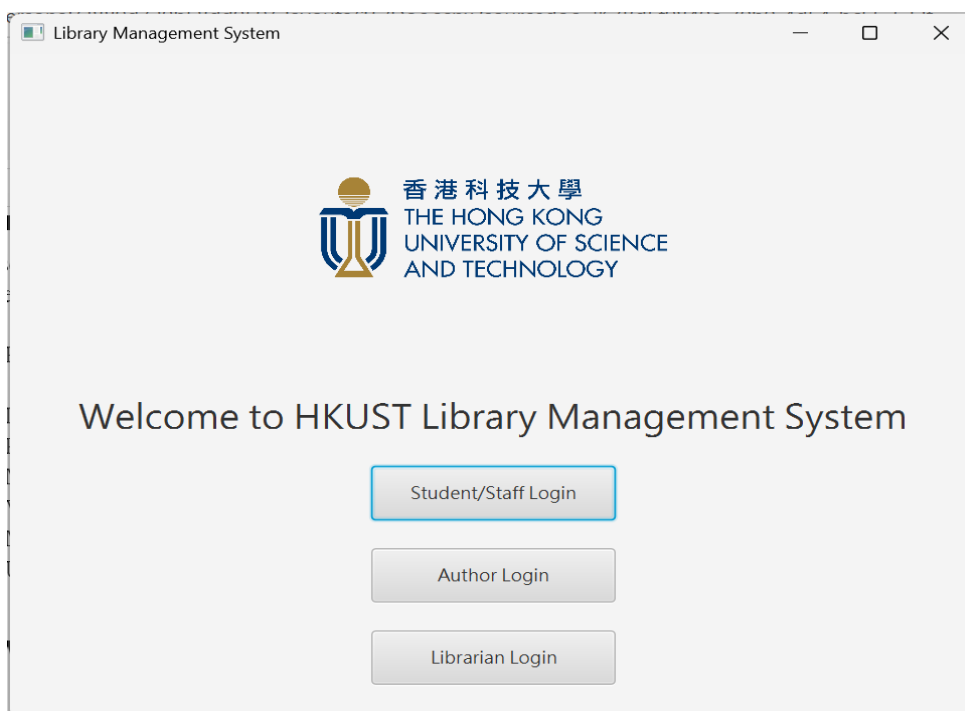
You can either delete the notification by the right “Delete” Button or “Clear All” button to delete all the notification. Note that in “Clear All” button, double confirmation is needed to ensure the user do not accidentally misclicked.



4.4.7 Logout

Press the “Logout” button exists at the bottom of the system.

You will see the Home Page if you log out of the system successfully.

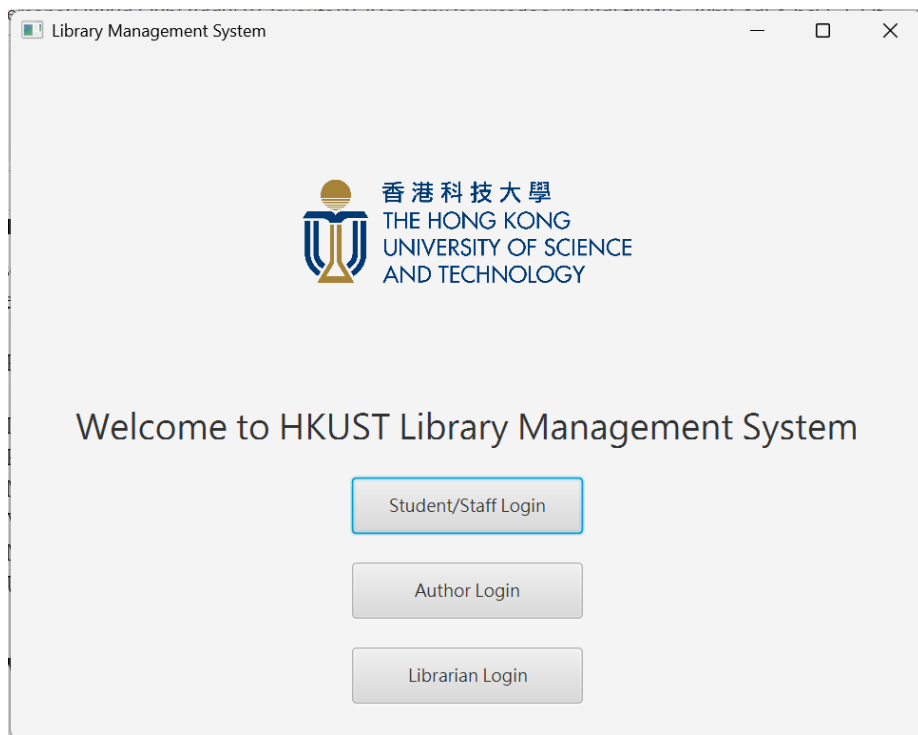


4.5 Librarian Dashboard

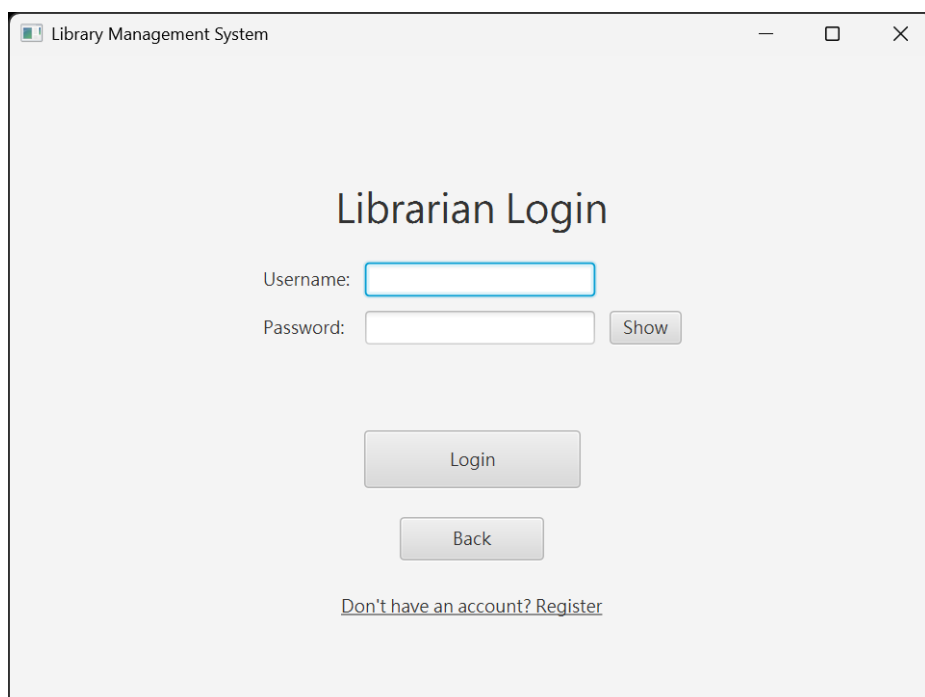
This section introduces how to use the librarian dashboard.

4.5.1 Login

When you enter the Library Management System, you will see the Home Page.



Press “Librarian Login” to log in as a “Librarian.”



Eter your username and password to log in to the system as a “Librarian.”

Library Management System

Librarian Login

Username:

Password:

[Don't have an account? Register](#)

If you log in successfully, you should see the Pending Approval tab. In the Pending Approval tab, you will find a list of pending approvals.

Library Management System

Librarian Dashboard

[Pending Approvals](#) [Manage Users](#) [My Profile](#) [Borrowed Books](#) [Published Books](#)

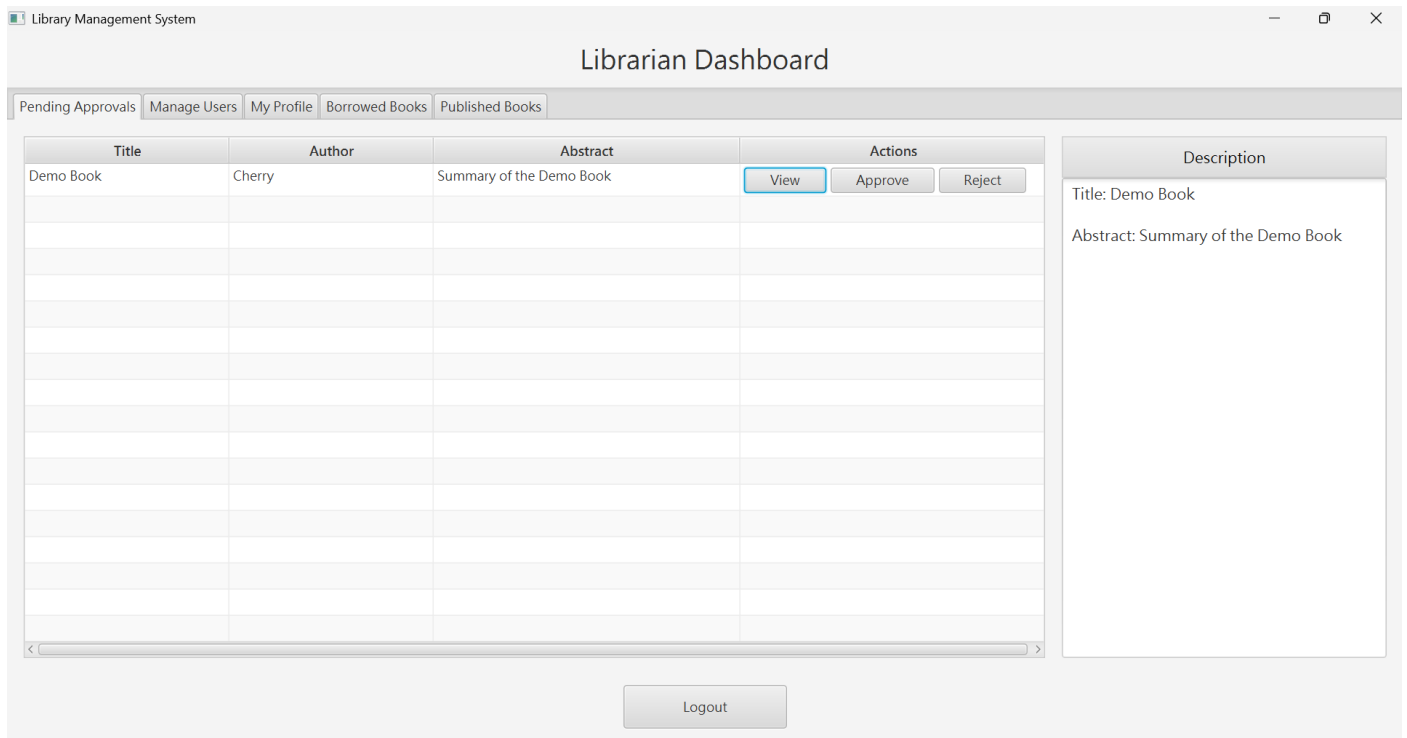
Title	Author	Abstract	Actions
Demo Book	Cherry	Summary of the Demo Book	View Approve Reject

Description

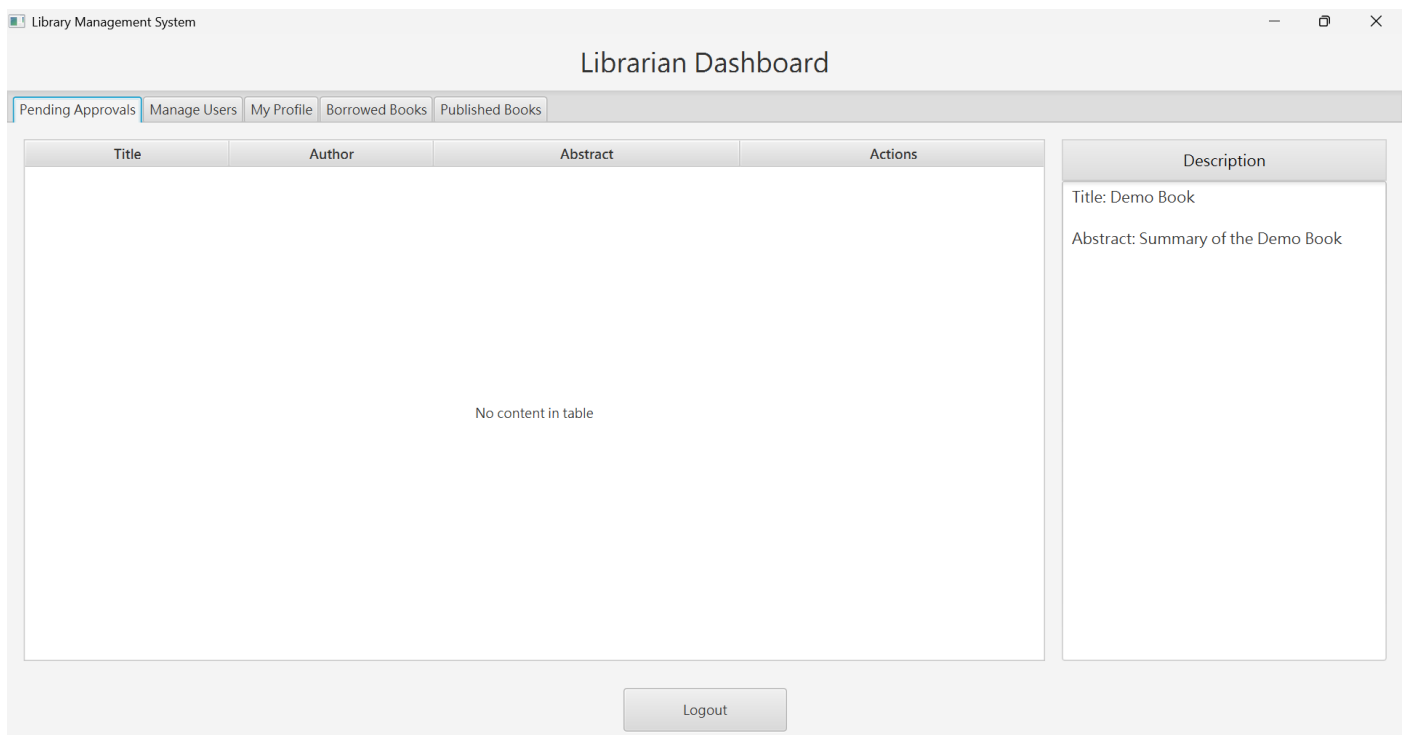
Please select a book to view details.

4.5.2 Pending Approval

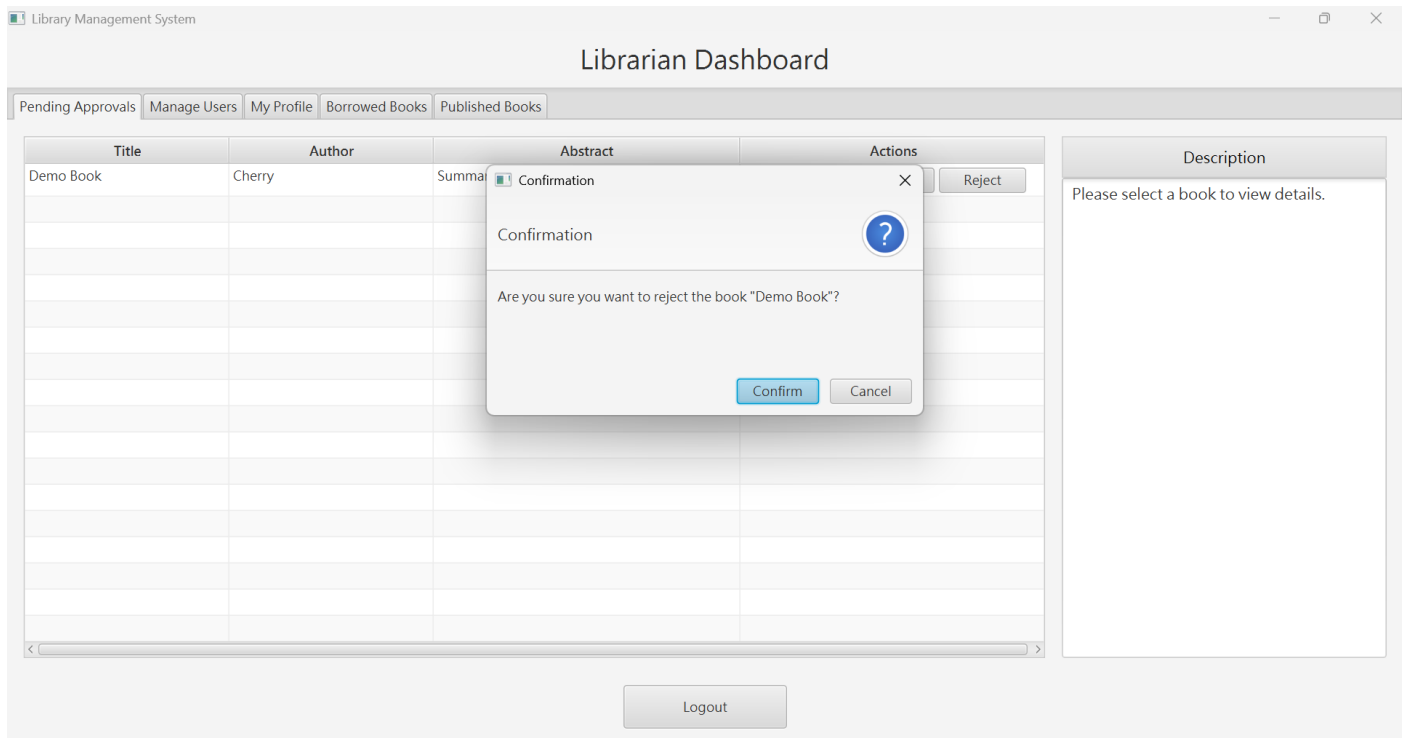
Click the “View” button in the “Actions” column to see the title and summary of the book. The description of the book will be displayed in the description box on the right-hand side.



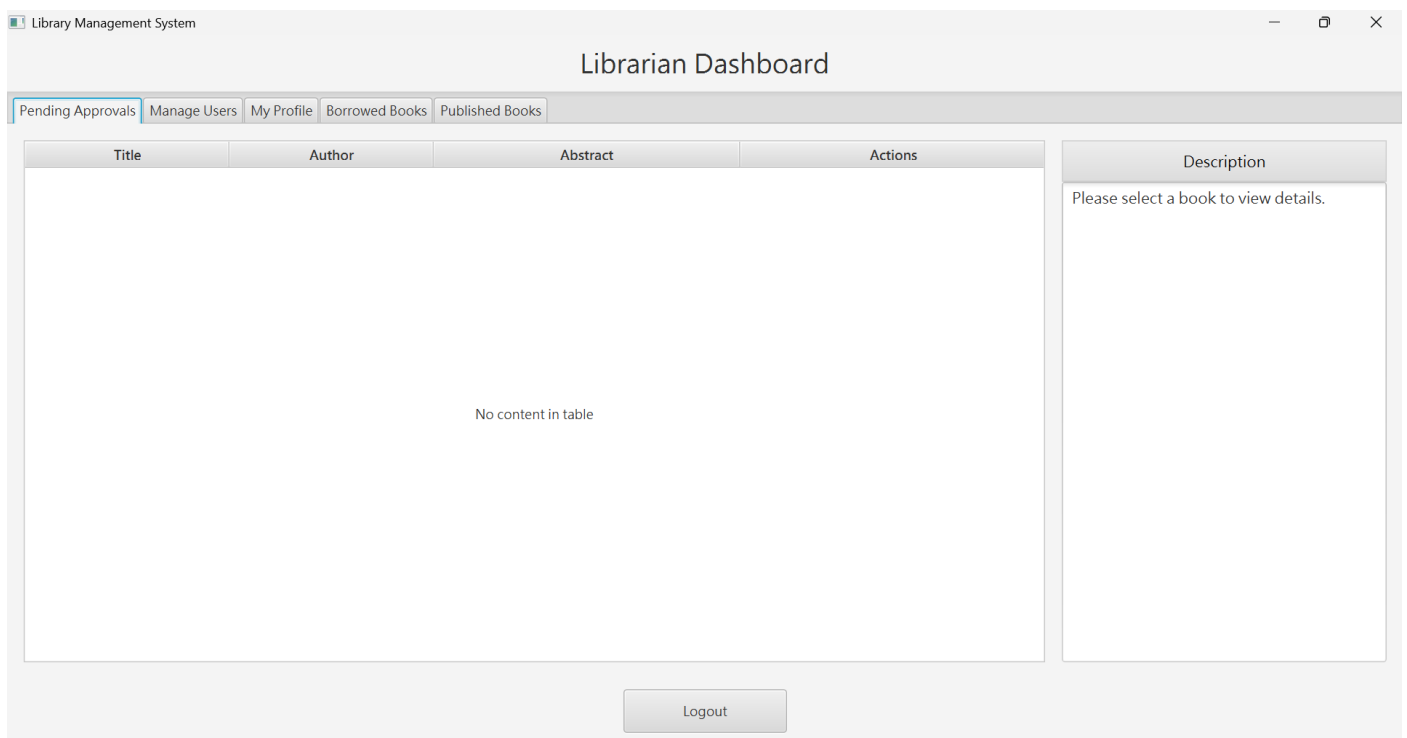
Click the “Approve” button to approve the pending approval. The pending approval will be removed if the book is published successfully. A notification will be sent to the author.



Click the “Reject” button to reject the pending approval. A confirmation window will pop up.

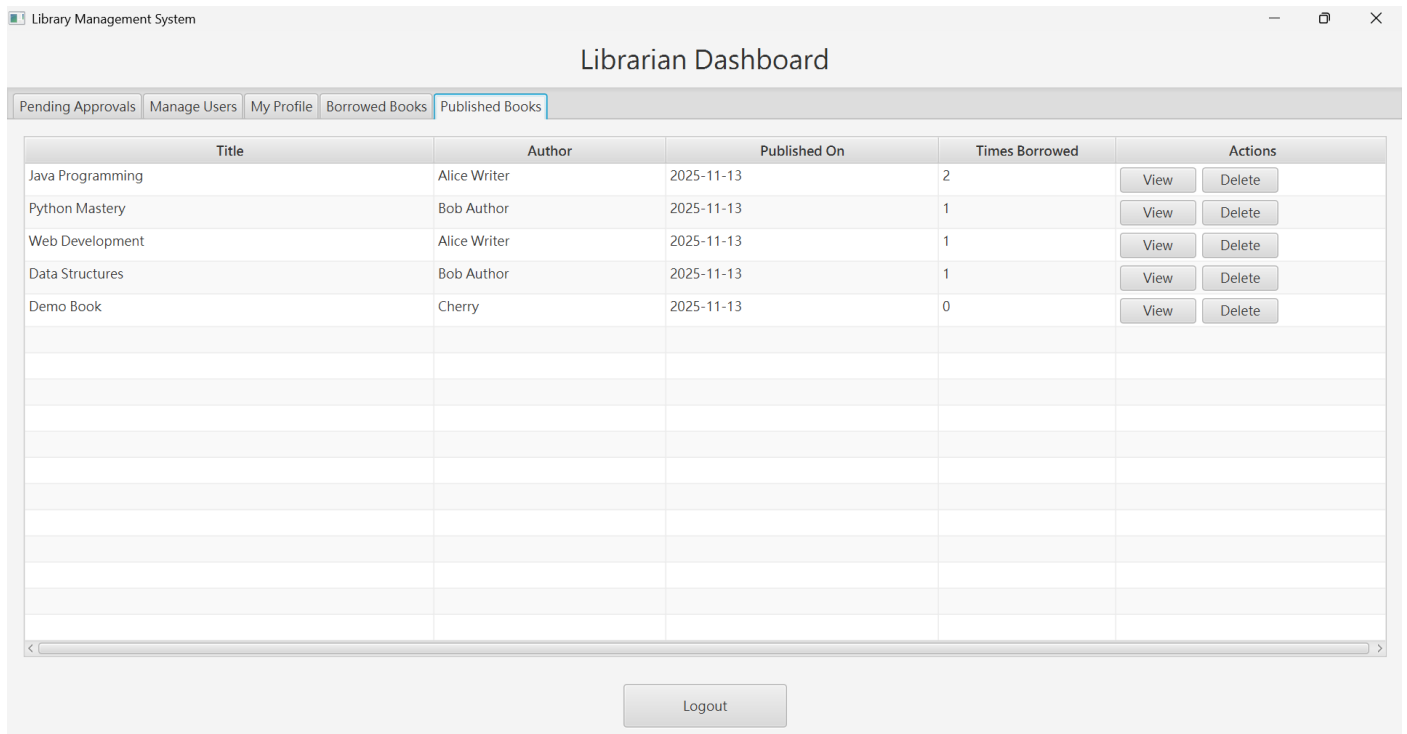


If you click “Cancel,” the pending approval will not be rejected. If you click “OK,” the pending approval will be rejected and removed from the list. A notification will be sent to the author.

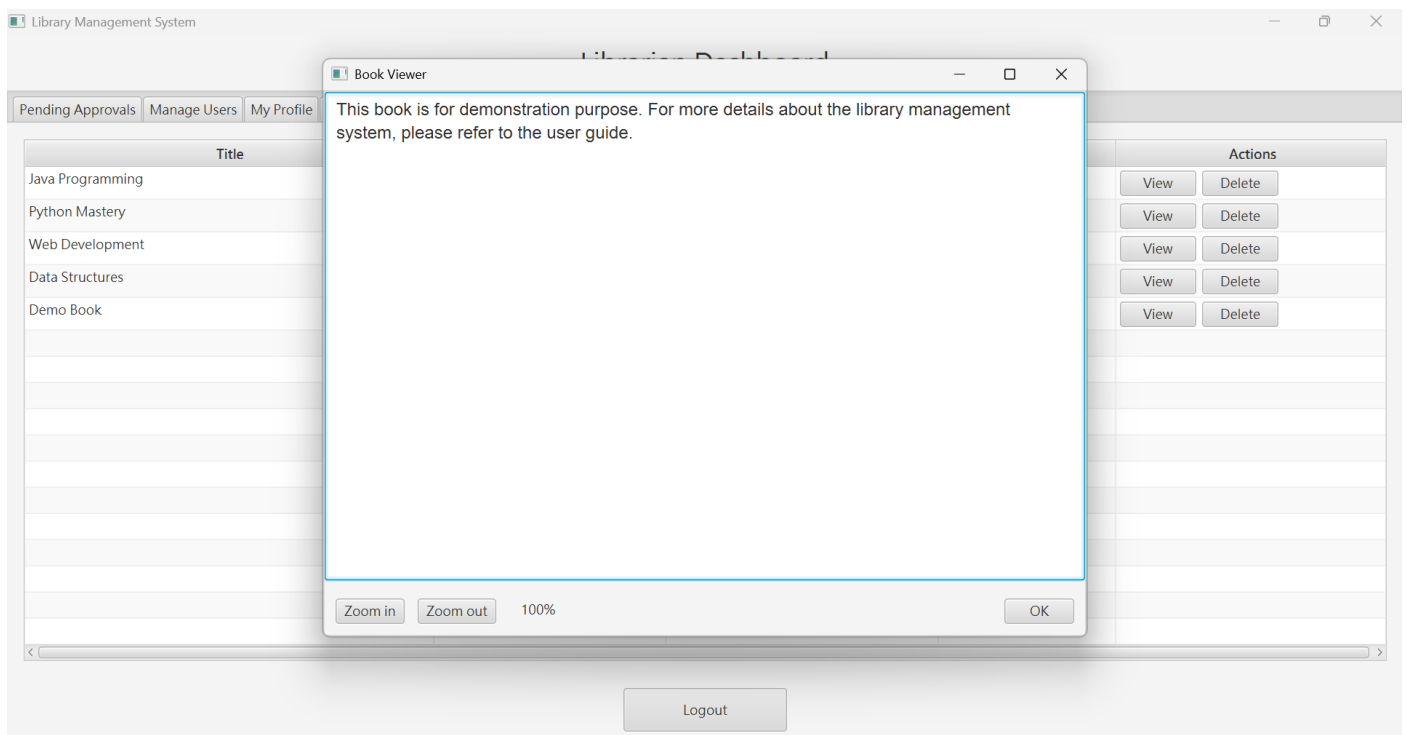


4.5.3 Manage Published Book

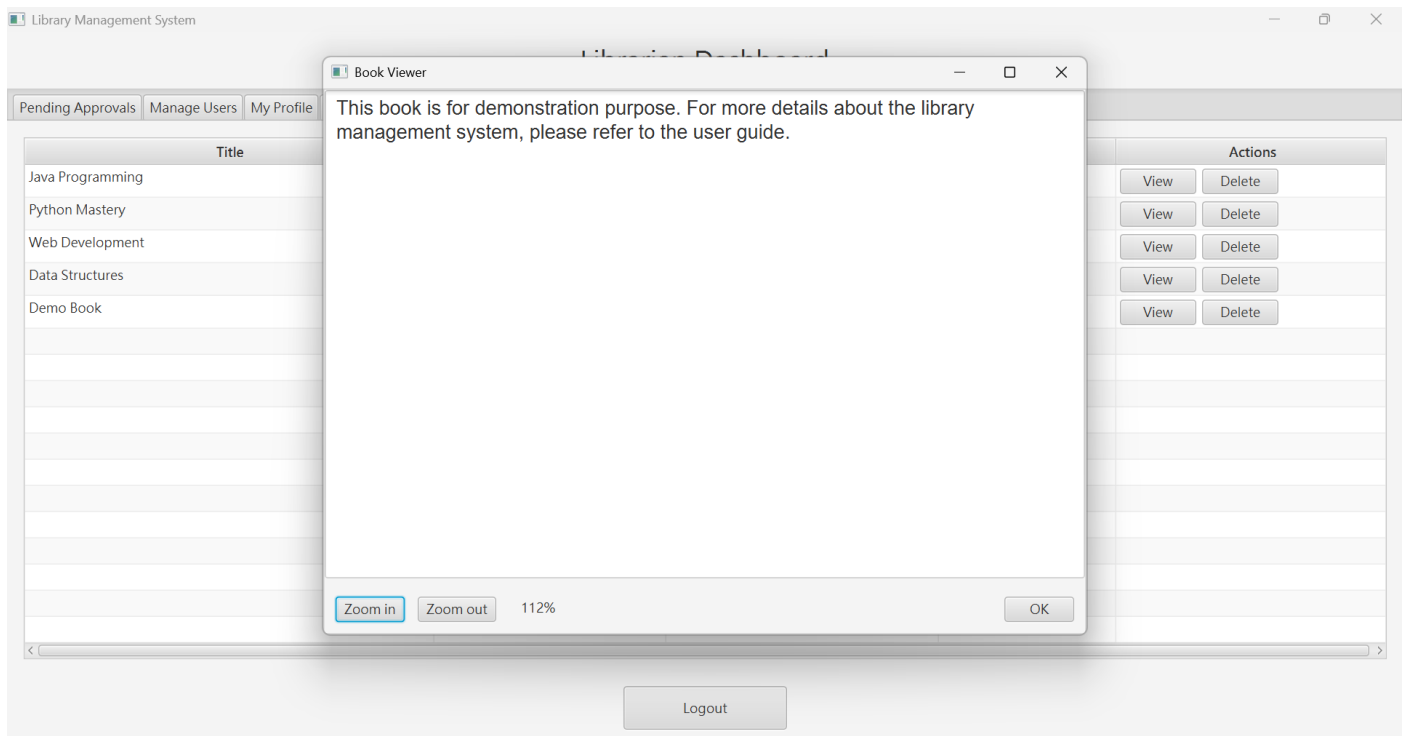
Select the “Published Books” tab to view the list of published books. These books are available for students and staff to borrow. The title, author, published date, and number of times borrowed are listed.



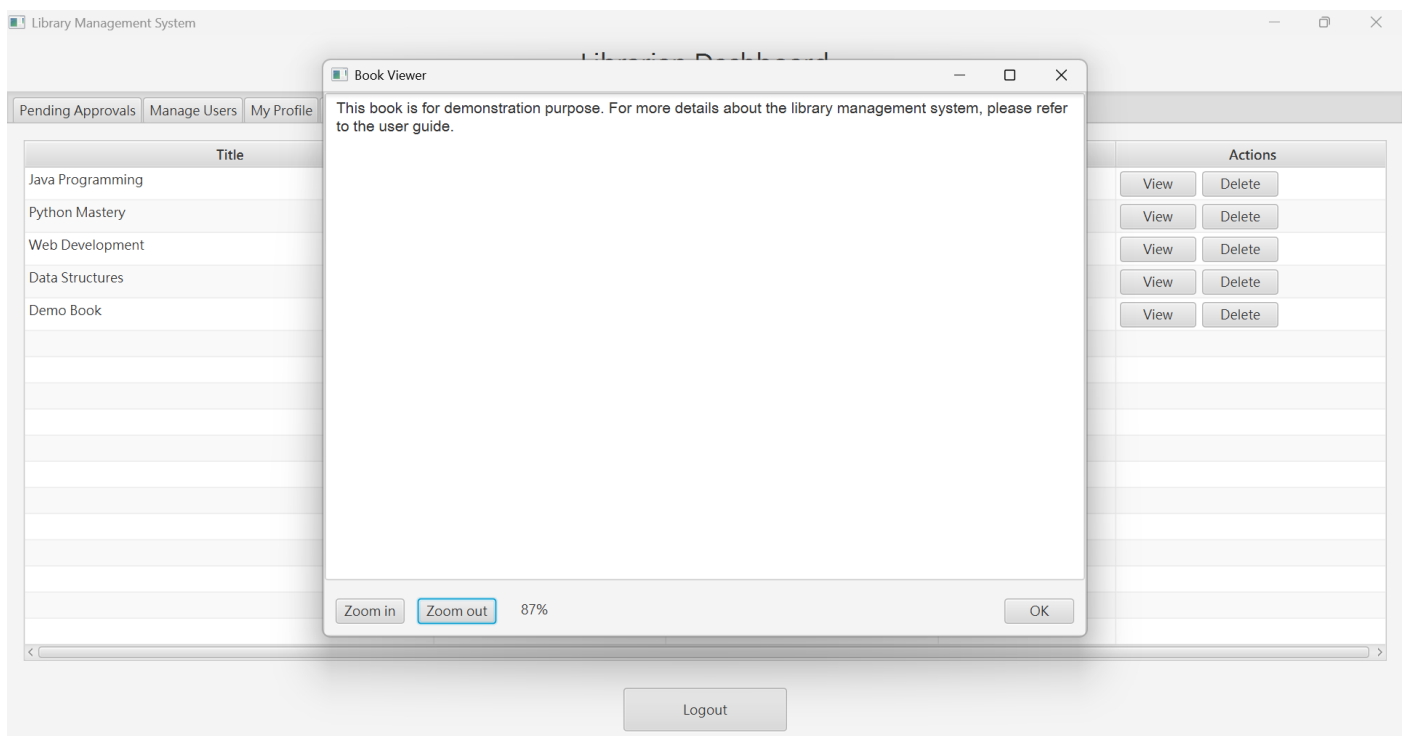
Press the “View” button in the action column to see the content of the book. A popup window will appear for reading.



Press the “Zoom In” button in the popup window to enlarge the text. The zoom scale will increase.



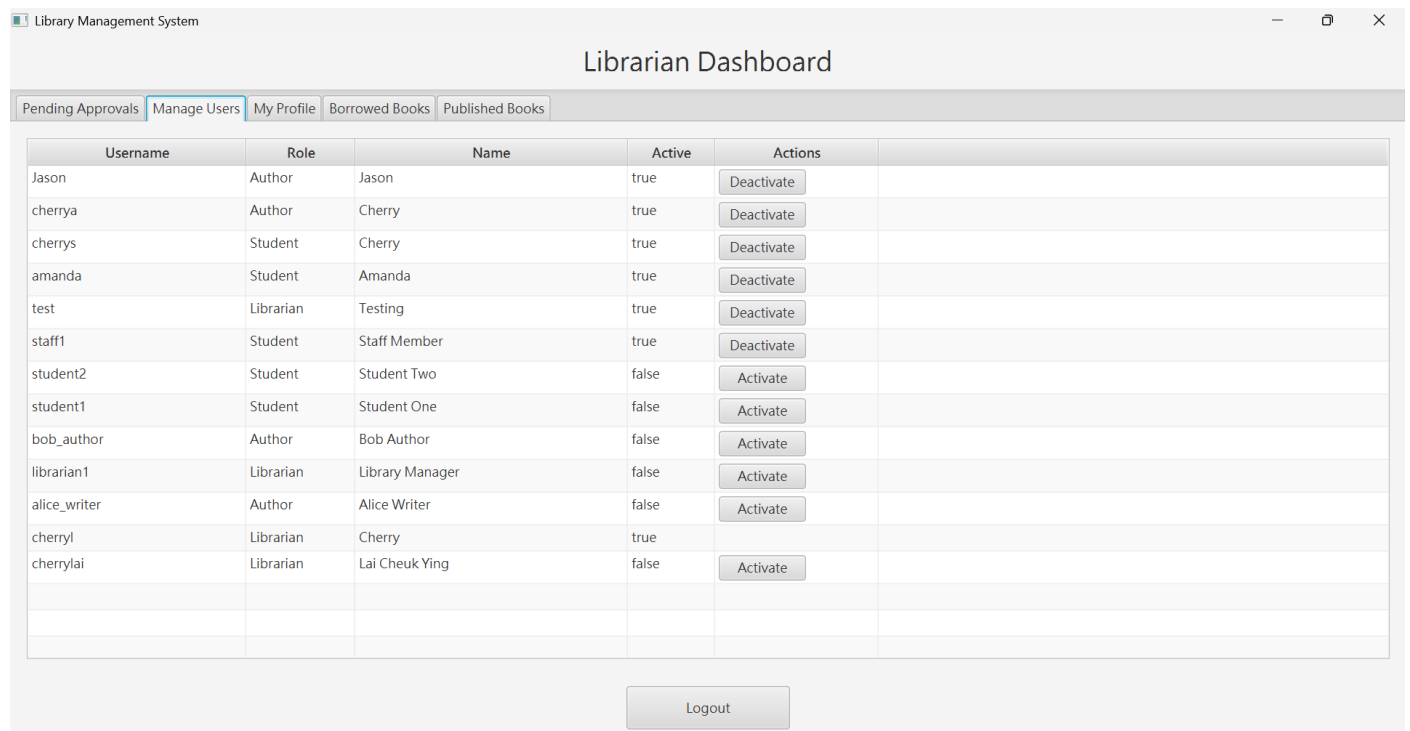
Press the “Zoom Out” button in the popup window to reduce the text. The zoom scale will decrease.



Press the “OK” button in the popup window to close it. You will return to the “Published Books” tab.

4.5.5 Manage User

Select the “Manage Users” tab to manage user statuses.



Library Management System

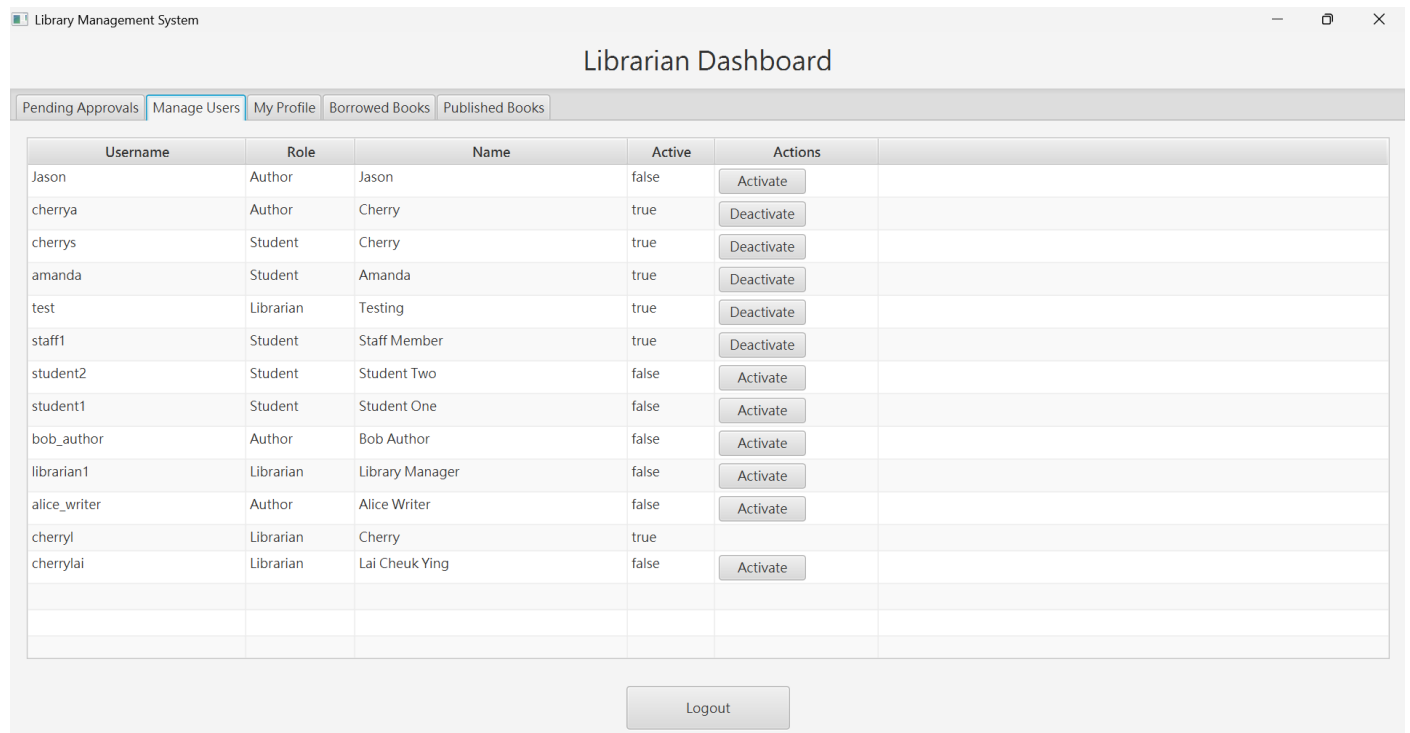
Librarian Dashboard

Pending Approvals Manage Users My Profile Borrowed Books Published Books

Username	Role	Name	Active	Actions
Jason	Author	Jason	true	Deactivate
cherrya	Author	Cherry	true	Deactivate
cherrys	Student	Cherry	true	Deactivate
amanda	Student	Amanda	true	Deactivate
test	Librarian	Testing	true	Deactivate
staff1	Student	Staff Member	true	Deactivate
student2	Student	Student Two	false	Activate
student1	Student	Student One	false	Activate
bob_author	Author	Bob Author	false	Activate
librarian1	Librarian	Library Manager	false	Activate
alice_writer	Author	Alice Writer	false	Activate
cherryl	Librarian	Cherry	true	
cherrylai	Librarian	Lai Cheuk Ying	false	Activate

Logout

For “active” users, press the “Deactivate” button to deactivate the user. Deactivated users are not allowed to log in to the system. (In this example, the first user is deactivated.)



Library Management System

Librarian Dashboard

Pending Approvals Manage Users My Profile Borrowed Books Published Books

Username	Role	Name	Active	Actions
Jason	Author	Jason	false	Activate
cherrya	Author	Cherry	true	Deactivate
cherrys	Student	Cherry	true	Deactivate
amanda	Student	Amanda	true	Deactivate
test	Librarian	Testing	true	Deactivate
staff1	Student	Staff Member	true	Deactivate
student2	Student	Student Two	false	Activate
student1	Student	Student One	false	Activate
bob_author	Author	Bob Author	false	Activate
librarian1	Librarian	Library Manager	false	Activate
alice_writer	Author	Alice Writer	false	Activate
cherryl	Librarian	Cherry	true	
cherrylai	Librarian	Lai Cheuk Ying	false	Activate

Logout

For “inactive” users, press the “Activate” button to activate the user. Activated users are allowed to log in to the system and access the services. (In this example, the first user is activated.)

Library Management System

Librarian Dashboard

Pending Approvals
Manage Users
My Profile
Borrowed Books
Published Books

Username	Role	Name	Active	Actions
Jason	Author	Jason	true	Deactivate
cherrya	Author	Cherry	true	Deactivate
cherrys	Student	Cherry	true	Deactivate
amanda	Student	Amanda	true	Deactivate
test	Librarian	Testing	true	Deactivate
staff1	Student	Staff Member	true	Deactivate
student2	Student	Student Two	false	Activate
student1	Student	Student One	false	Activate
bob_author	Author	Bob Author	false	Activate
librarian1	Librarian	Library Manager	false	Activate
alice_writer	Author	Alice Writer	false	Activate
cherryl	Librarian	Cherry	true	
cherrylai	Librarian	Lai Cheuk Ying	false	Activate

Logout

The librarian cannot modify the status of themselves.

4.5.6 Update Profile

Select the “Update Profile” tab to update your full name and password.

Library Management System

Librarian Dashboard

Pending Approvals
Manage Users
My Profile
Borrowed Books
Published Books

Update Profile

Username: cherryl
Full Name:
Password:
Confirm Password:

Enter the new full name and password.

Library Management System

Librarian Dashboard

Pending Approvals Manage Users My Profile Borrowed Books Published Books

Update Profile

Username: cherryl

Full Name:

Password: Show

Confirm Password: Show

A success alert box will appear to indicate that the update was successful.

Library Management System

Librarian Dashboard

Pending Approvals Manage Users My Profile Borrowed Books Published Books

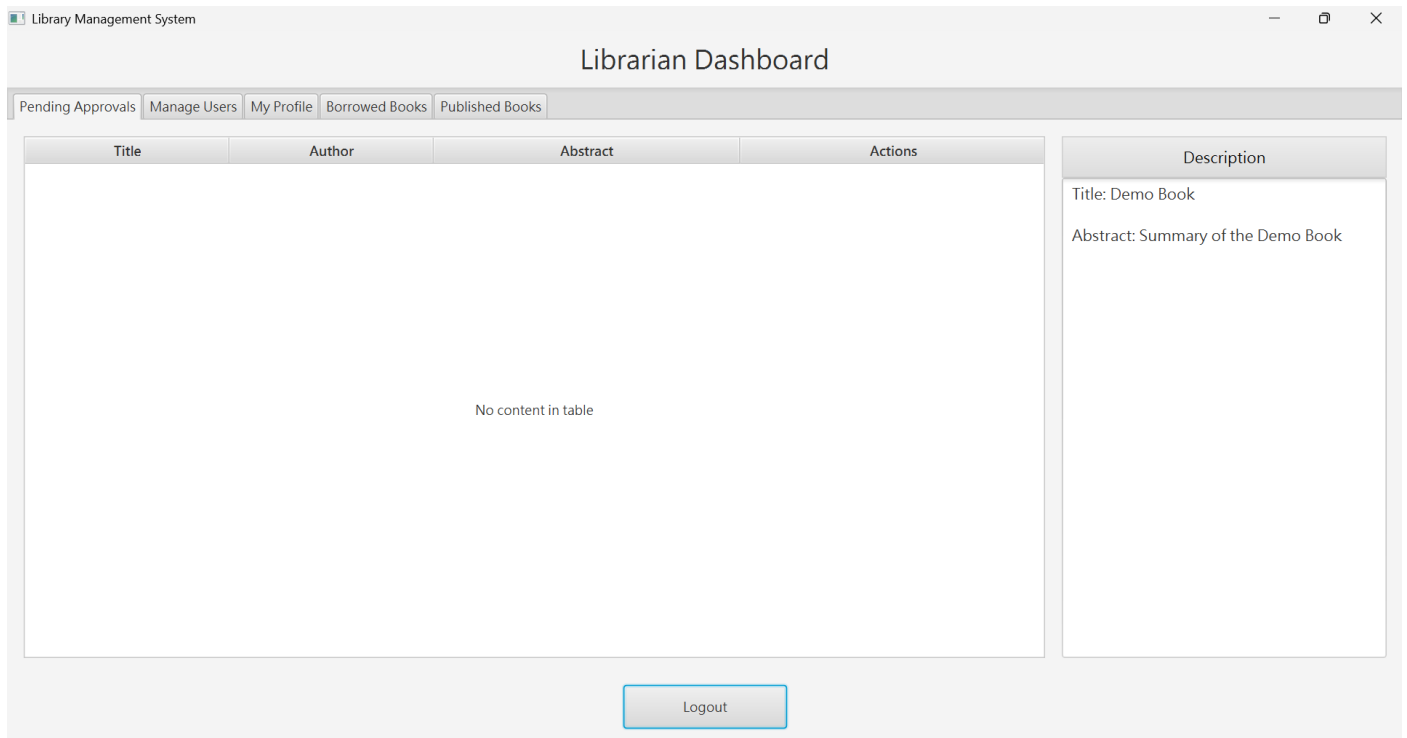
Update Profile

 Success

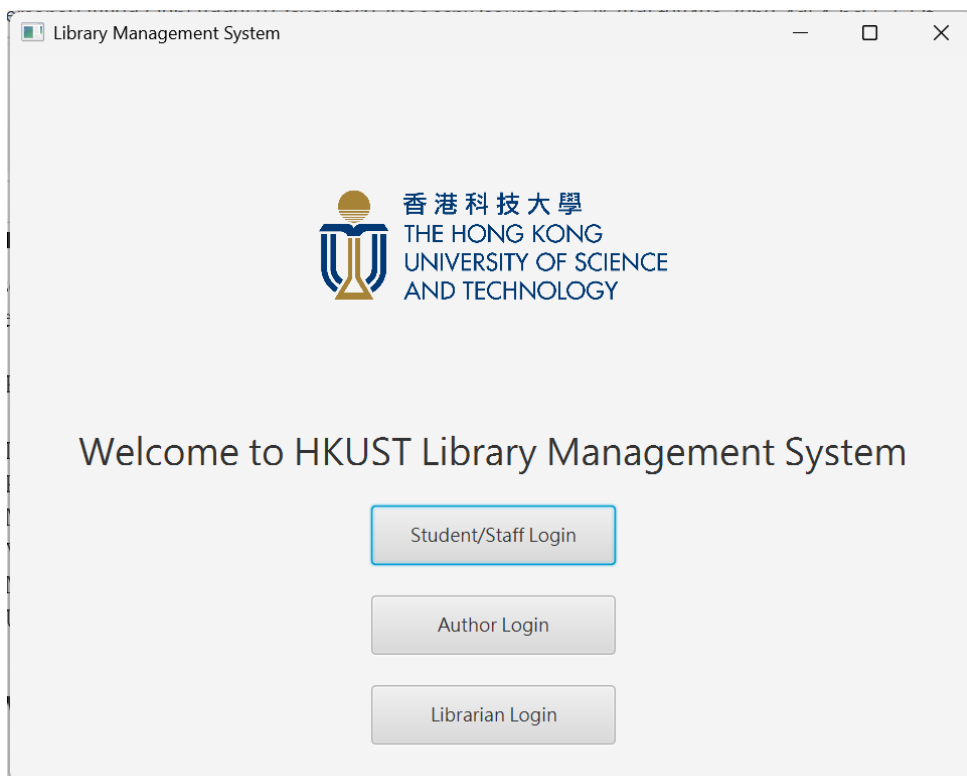
 Profile updated successfully!

4.5.7 Logout

Press the “Logout” button to log out of the system.



You will see the Home Page if you log out of the system successfully.



5. System Features

5.1 Login Authentication

The login system implements a multi-step authentication process to verify a user's identity and system access rights before granting entry to the dashboard. This process sequentially checks for the

username's existence in the system database, the correctness of the corresponding password, the account's activated status, and the user's assigned role for appropriate access rights.

a. Checking: Username's existence

The screenshot shows a web browser window titled "Library Management System". The page has a header "Student/Staff Login". Below the header, there are two input fields: "Username:" with the value "NonExistingAccount" and "Password:" with a masked password (represented by 10 dots). To the right of the password field is a "Show" button. Below the input fields, there is a red error message: "Incorrect username or password.". Below the error message, there are two buttons: "Login" and "Back". At the bottom of the page, there is a link: "Don't have an account? Register".

b. Checking: Password's correctness

The screenshot shows a web browser window titled "Library Management System". The page has a header "Student/Staff Login". Below the header, there are two input fields: "Username:" with the value "ExistingAccount" and "Password:" with a masked password (represented by 10 dots). To the right of the password field is a "Show" button. Below the input fields, there is a red error message: "Incorrect username or password.". Below the error message, there are two buttons: "Login" and "Back". At the bottom of the page, there is a link: "Don't have an account? Register".

c. Checking: Account's activated status

The screenshot shows a web browser window titled "Library Management System". The page has a header "Student/Staff Login". Below the header, there are two input fields: "Username:" with the value "ExistingAccount" and "Password:" with masked characters. To the right of the password field is a "Show" button. Below the input fields, a red error message reads: "Your account has been deactivated. Please contact the librarian to reactivate it." Below the error message are two buttons: "Login" (highlighted with a blue border) and "Back". At the bottom, there is a link: "Don't have an account? Register".

d. Checking: User's assigned role for appropriate access rights

The screenshot shows a web browser window titled "Library Management System". The page has a header "Student/Staff Login". Below the header, there are two input fields: "Username:" with the value "Author" and "Password:" with masked characters. To the right of the password field is a "Show" button. Below the input fields, a red error message reads: "You don't have access rights to this dashboard." Below the error message are two buttons: "Login" (highlighted with a blue border) and "Back". At the bottom, there is a link: "Don't have an account? Register".

5.2 Registration Validation

The registration system enforces a set of validation rules to ensure the integrity of all new users' accounts. This process checks for empty input in all mandatory fields, validates the length and

character composition of the username, and verifies its uniqueness within the system. It also enforces password length requirements, validates password strength, and confirms that the password and confirmation password match exactly. Finally, the system validates the length of the full name and, upon successful completion of all checks, displays a successful registration message.

a. Checking: Empty input

The screenshot shows a web browser window titled "Library Management System". Inside, there is a form titled "Student Register". The form has four input fields: "Username:", "Password:", "Confirm Password:", and "Full Name:". All four fields are empty. To the right of the "Password:" and "Confirm Password:" fields are "Show" buttons. Below the input fields, there is a red error message: "Please enter information." Below the error message is a link: "Username and Password Requirements". At the bottom of the form are three buttons: "Register" (highlighted with a blue border), "Back", and a link: "Have an account? Login".

b. Checking: Username's length

The screenshot shows the same "Student Register" form in the "Library Management System" browser window. The "Username:" field now contains the letter "a". The "Password:", "Confirm Password:", and "Full Name:" fields are filled with masked characters (dots). The "Full Name:" field contains the text "student". A red error message is displayed below the input fields: "Username must be 4 to 30 characters long." The "Register" button remains highlighted with a blue border. The "Back" button and the "Have an account? Login" link are also visible.

c. Checking: Username's character composition

Library Management System

Student Register

Username:

Password: [Show](#)

Confirm Password: [Show](#)

Full Name:

Invalid username.

[Username and Password Requirements](#)

[Register](#)

[Back](#)

[Have an account? Login](#)

d. Checking: Uniqueness of username

Library Management System

Student Register

Username:

Password: [Show](#)

Confirm Password: [Show](#)

Full Name:

Username already exists! Please try again.

[Username and Password Requirements](#)

[Register](#)

[Back](#)

[Have an account? Login](#)

e. Checking: Password's length

Library Management System

Student Register

Username:

Password:

Confirm Password:

Full Name:

Password must be at least 8 characters long.

[Username and Password Requirements](#)

[Have an account? Login](#)

f. Checking: Validation of password

Library Management System

Student Register

Username:

Password:

Confirm Password:

Full Name:

Invalid password.

[Username and Password Requirements](#)

[Have an account? Login](#)

g. Checking: Matching of password and confirmation password

Library Management System

Student Register

Username:

Password:

Confirm Password:

Full Name:

New password and confirmation password do not match.

[Username and Password Requirements](#)

[Have an account? Login](#)

h. Checking: Full name's length

Library Management System

Student Register

Username:

Password:

Confirm Password:

Full Name:

Full name must be 2 to 50 characters long.

[Username and Password Requirements](#)

[Have an account? Login](#)

i. Display: Successful Registration

Library Management System

Student Register

Username:

Password:

Confirm Password:

Full Name:

Registration success! You can return to login page.

[Username and Password Requirements](#)

[Have an account? Login](#)

5.3 Username & Password Requirements

During the registration process, to learn more about username and password requirements, they are provided by clicking the hyperlink “Username and Password Requirements”. The requirements will be displayed in a pop-up window.

Library Management System

Student Register

Username:

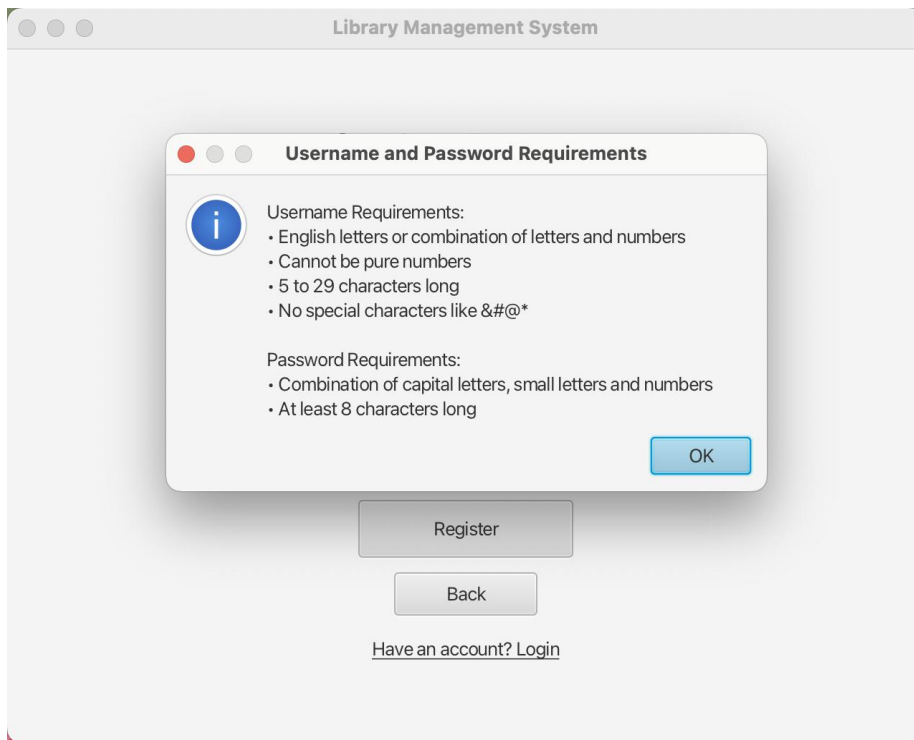
Password:

Confirm Password:

Full Name:

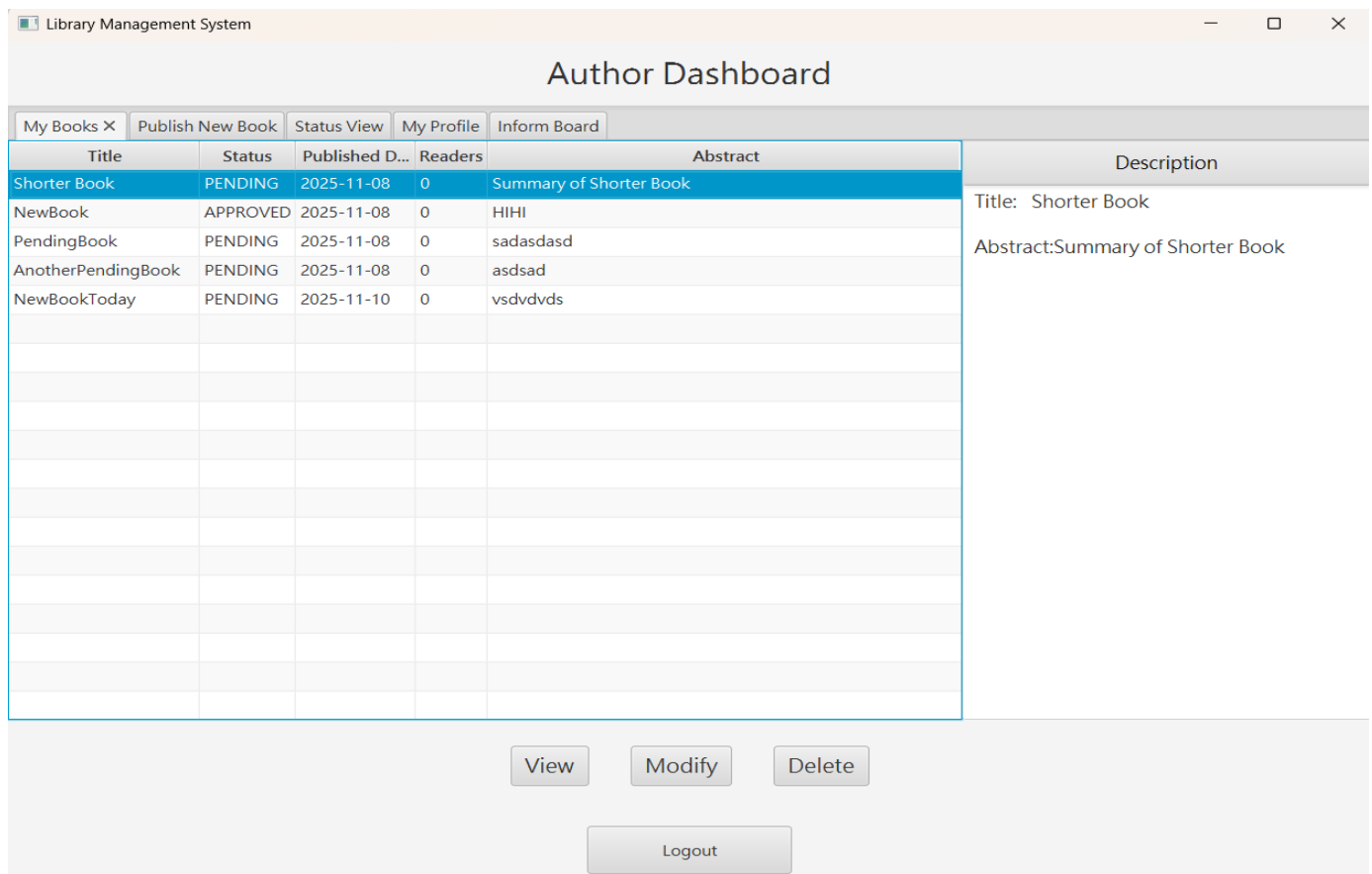
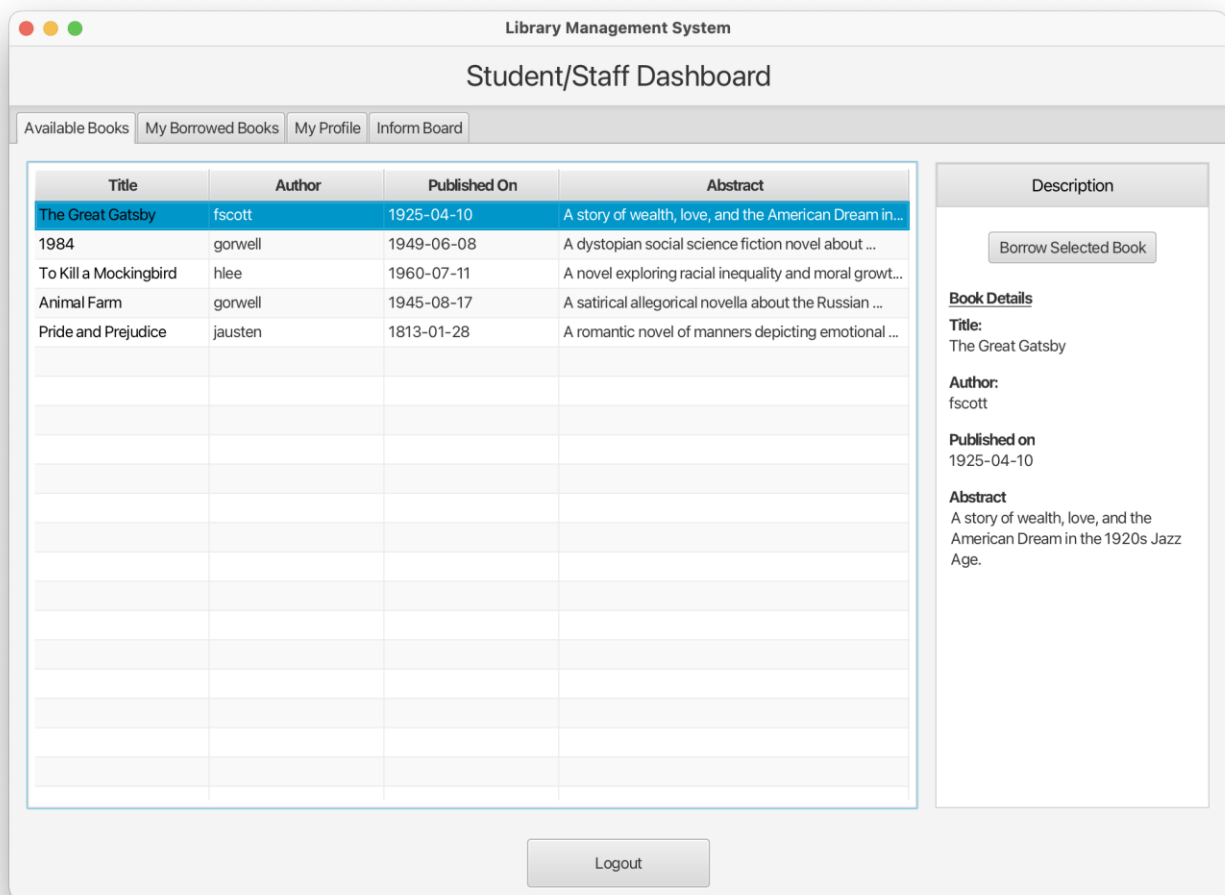
[Username and Password Requirements](#)

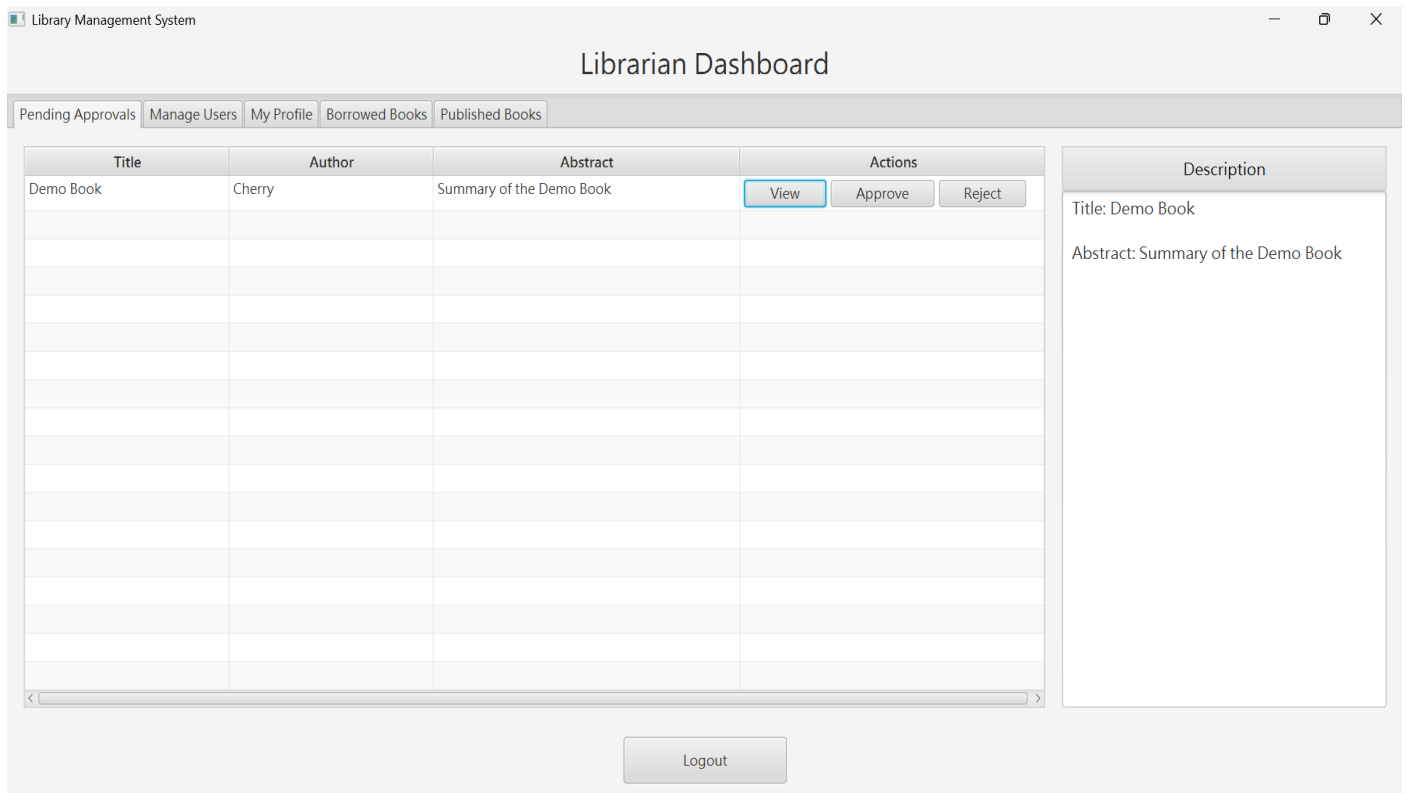
[Have an account? Login](#)



5.4 Description Box

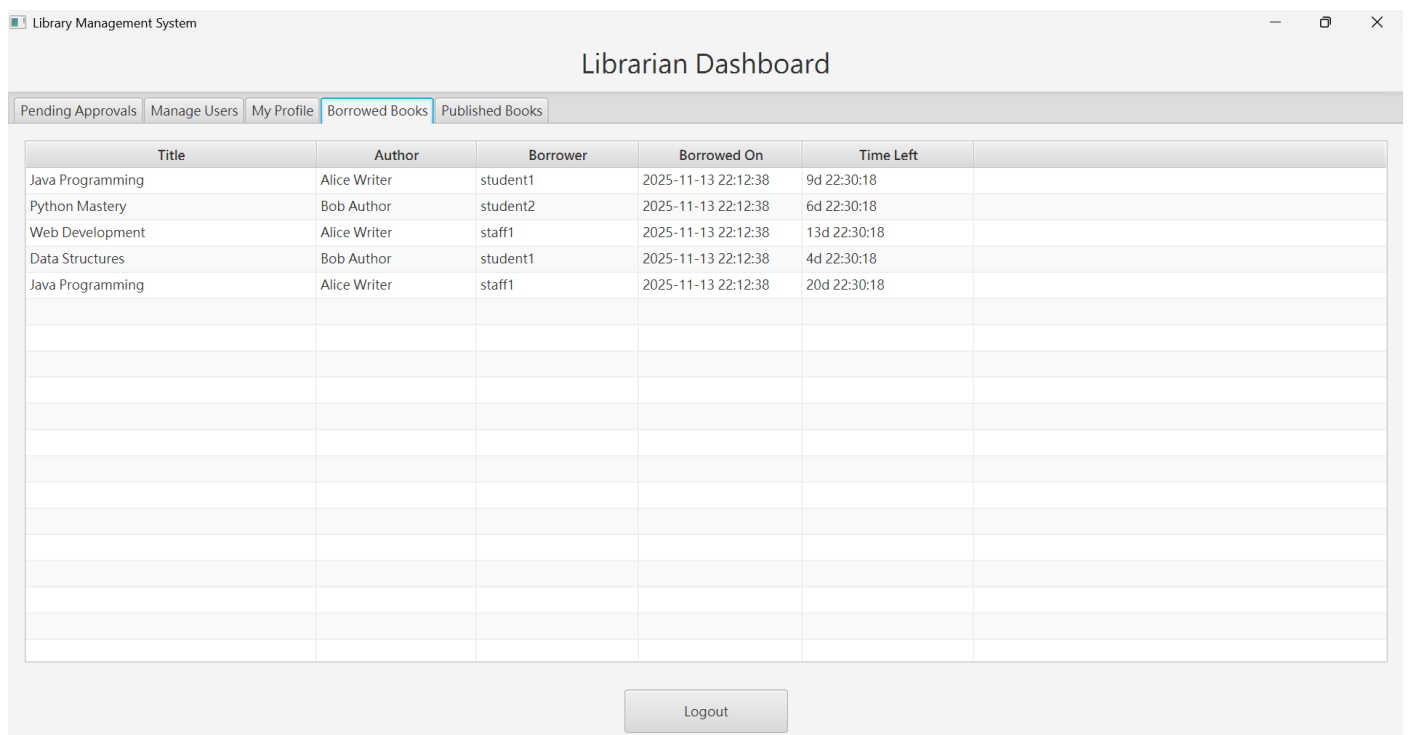
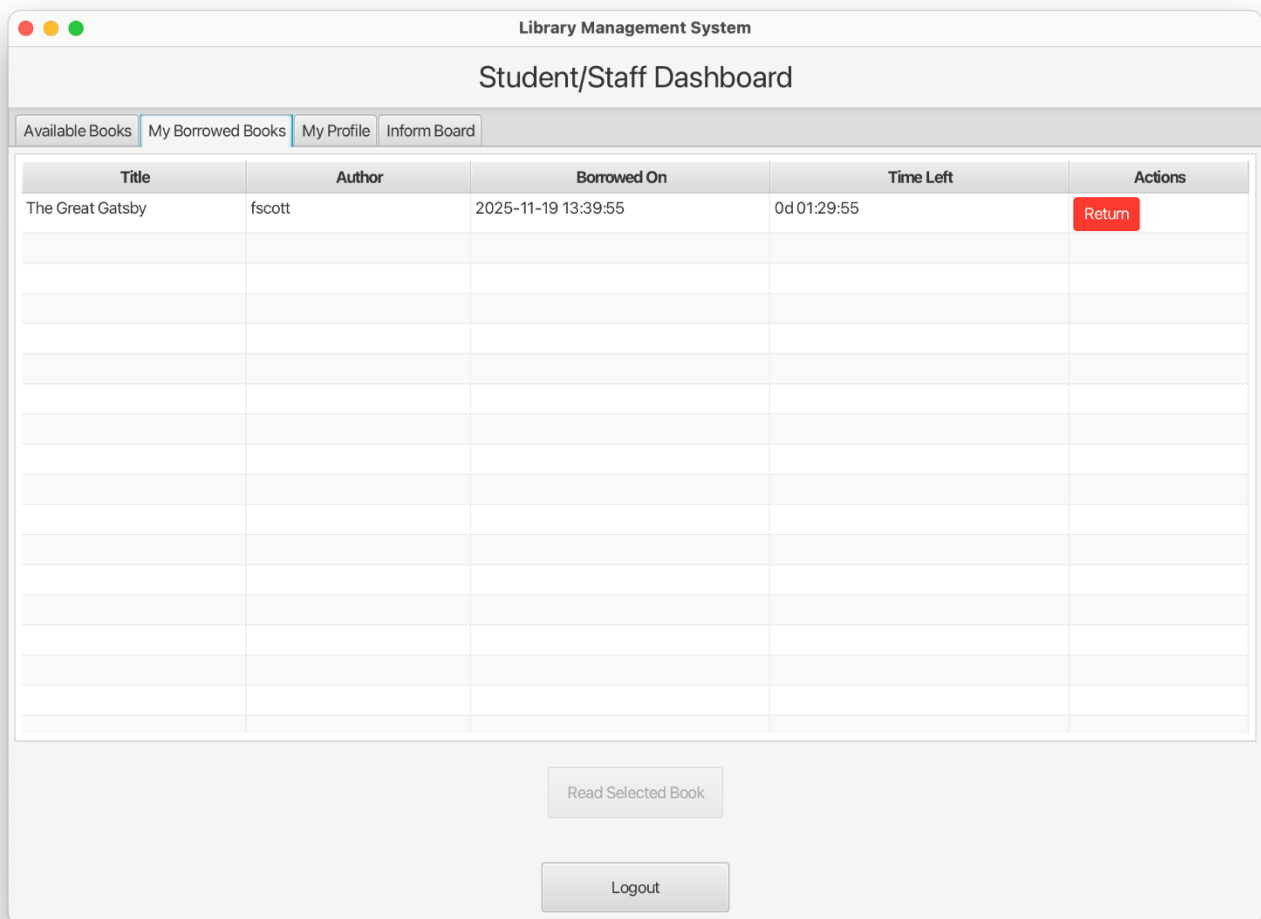
In the “Available Books” tab of the Student/Staff Dashboard, the “My Books” tab of the Author Dashboard, and the “Pending Approval” tab of the Librarian Dashboard, a Description Box is provided for each table to enable the user to view the book details, particularly when the title or abstract length exceeds the visible space within the table, preventing full readability.



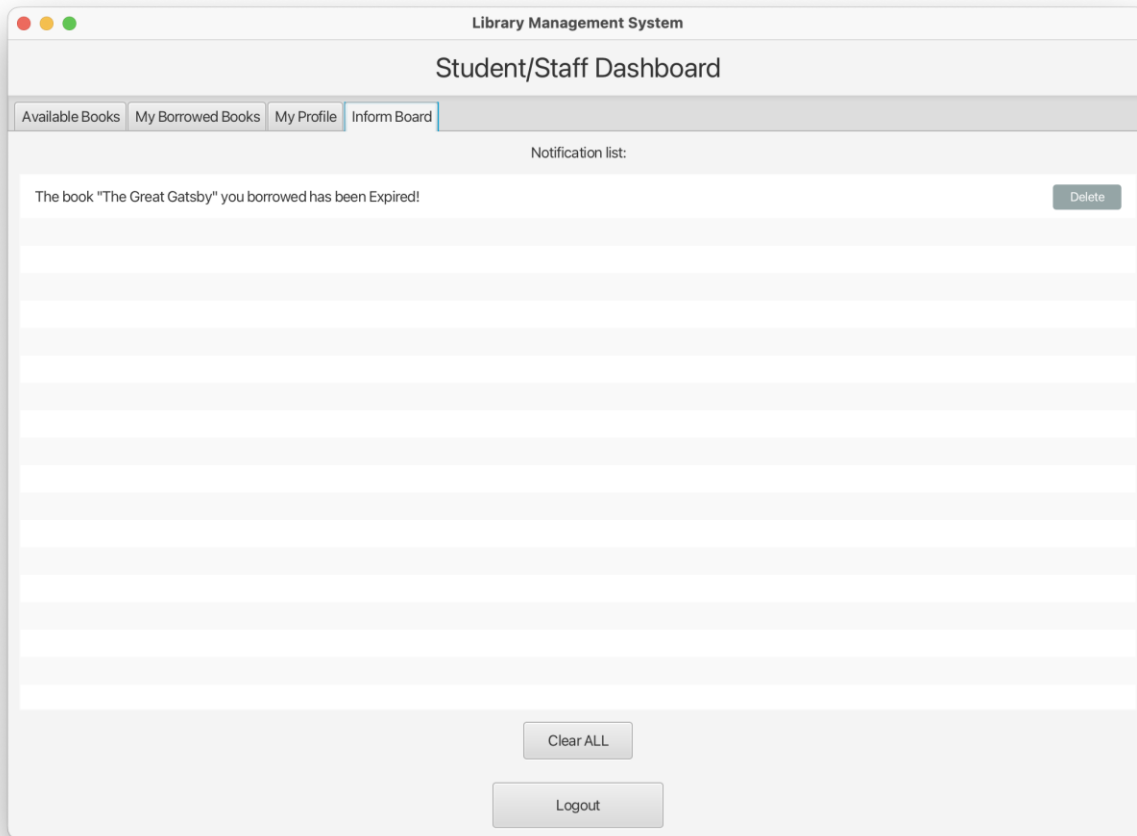


5.5 Real-time counting Time-Left

In the Borrow Book tab (located within the Student/Staff and Librarian dashboards), the “Time Left” column displays the remaining borrowing period in a countdown format that updates in real time. This allows users to clearly see the remaining time for their borrowed items.

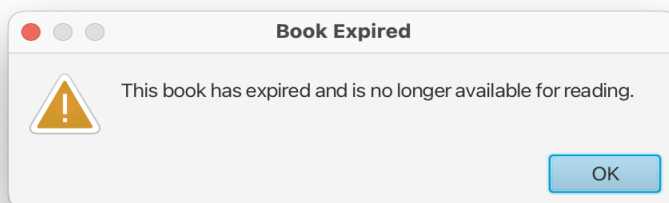
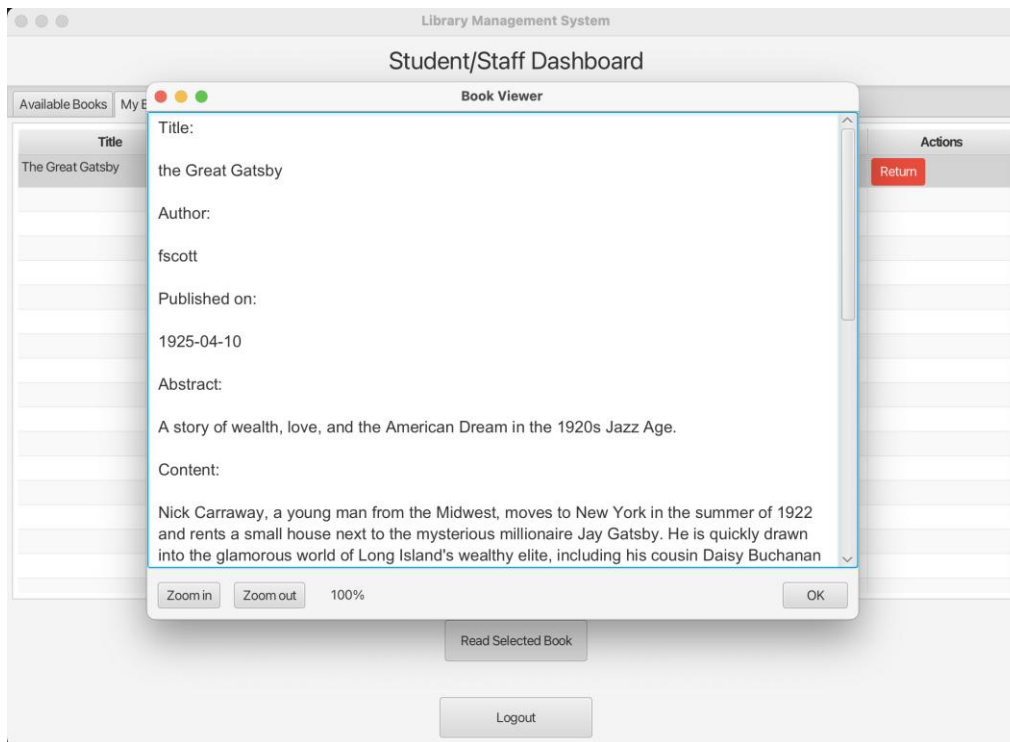


If a book has expired, it will be automatically removed from the table and marked as returned. A notification will be sent to the borrower to inform them that the borrowing period has expired.



5.6 Automatic Content Window Closure

The Student/Staff Dashboard features an automatic window closure function for the book content reader, which activates when a borrowed book reaches its expiration date. This ensures that access to the digital content is promptly revoked in accordance with the borrowing policy, maintaining system integrity. Prior to closure, the system displays a message to inform the user of the expiration.



5.7 Borrowing Duration Validation

The system validates duration input by filtering non-numeric characters and automatically limiting seconds to 59. It allows one field to be empty, prevents zero/negative values, and enforces a 14-day maximum. Clear error messages guide correction, while successful borrowing triggers a confirmation showing the book title and duration.

- a. Filter non-numeric characters automatically (e.g. inputting "a" in the fields)

Borrow Book

Please enter the borrowing duration. (Minutes and seconds)

Minutes:

Seconds:

Max: 20160 minutes (14 days)

Invalid input, please enter again

Cancel OK

- b. Restriction and Automatic Correction to “59” in second field (e.g. inputting “80” in the second field)

Borrow Book

Please enter the borrowing duration. (Minutes and seconds)

Minutes:

Seconds:

Max: 20160 minutes (14 days)

Invalid input, please enter again

Cancel OK

- c. Checking: Empty inputs in both fields

Borrow Book

Please enter the borrowing duration. (Minutes and seconds)

Minutes:

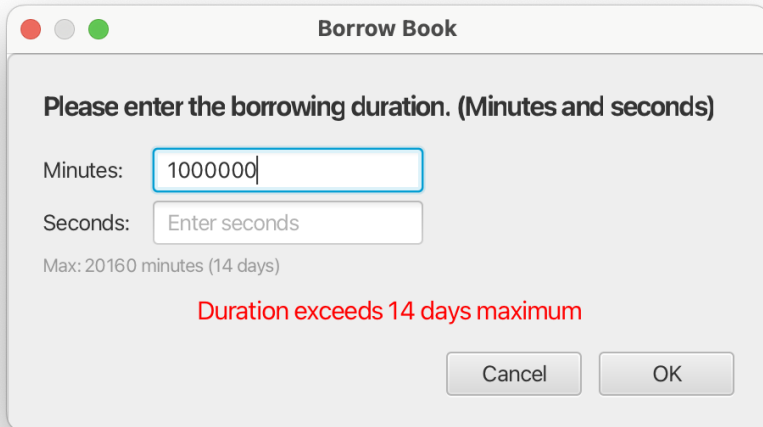
Seconds:

Max: 20160 minutes (14 days)

Please enter duration

Cancel OK

- d. Checking: Duration exceeding maximum limit



Borrow Book

Please enter the borrowing duration. (Minutes and seconds)

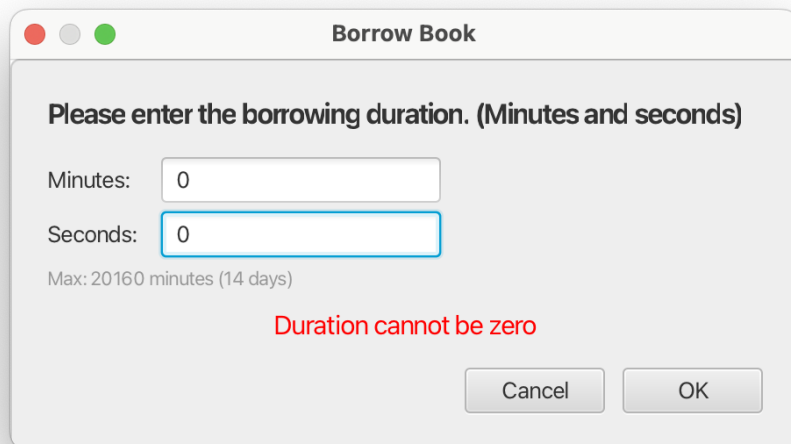
Minutes:

Seconds:

Max: 20160 minutes (14 days)

Duration exceeds 14 days maximum

- e. Checking: Non-positive input



Borrow Book

Please enter the borrowing duration. (Minutes and seconds)

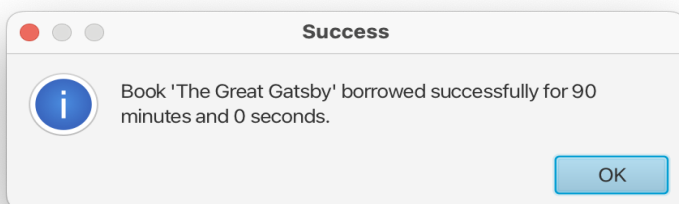
Minutes:

Seconds:

Max: 20160 minutes (14 days)

Duration cannot be zero

- f. Confirmation message (with corresponding book title and borrowing duration)



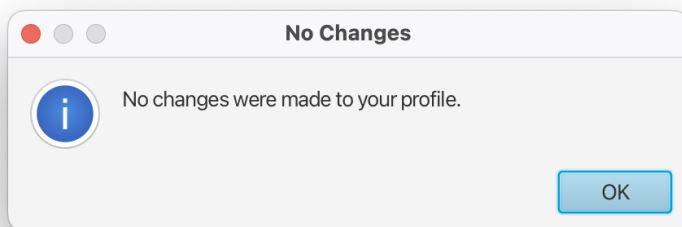
Success

 Book 'The Great Gatsby' borrowed successfully for 90 minutes and 0 seconds.

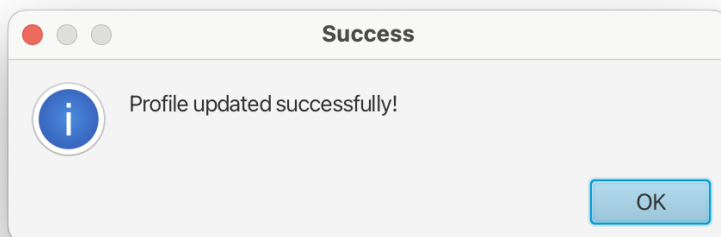
5.8 Profile Update functionality

The profile update (in all dashboards) functionality implements validation checks similar to the registration process to ensure data integrity. If the submitted form contains no modifications from the existing profile data, the system displays the message "No changes were made to your profile". Upon successful validation and update, the system displays a success message "Profile updated successfully!". Following a successful update, the input fields are automatically cleared and the interface buttons toggle back to their default state.

- a. Display message: No Modification



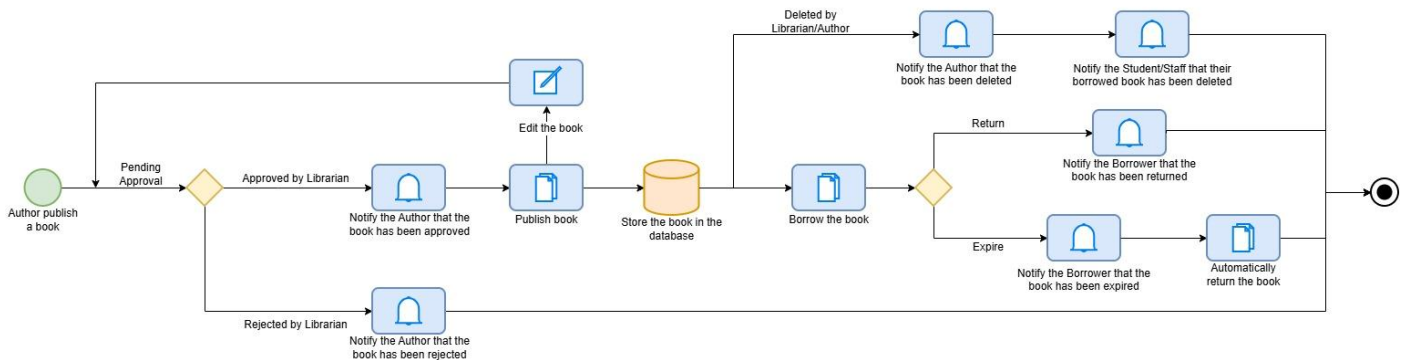
- a. Display message: Successful Update



- b. Reset the interface after updating

6. Workflow

The diagram below shows the workflow of the system.



7. Test Report

Tests were created for the data classes and controller classes in the implemented tasks (Student/Staff Dashboard, Author Dashboard, and Librarian Dashboard).

7.1 Report on the Unit Testing for the Implemented Tasks

The following screenshots show the result of the unit test for the implemented tasks. All tests were passed.

[Collapse](#) | [Expand](#)

```
C:\Users\laich\jdk\openjdk-24.0.2+12-54\bin\java.exe -javaagent:C:\Users\laich\AppData\Local\JetBrains\Idea\2025.2\captureAgent\debugger-agent.jar=file:///C:/Users/laich\AppData/Local/Temp/capture3805876430415833336.props -Djunit.jupiter.extensions.autodetection.enabled=true --add-opens
javafx.controls/javafx.scene.control=library --add-opens javafx.controls/javafx.scene.control.cell=library -
javaagent:C:\Users\laich\AppData\Local\JetBrains\Idea\2025.2\testAgent\intellij.platform.coverage.agent.jar=C:\Users\laich\AppData\Local\Temp\coverage329args -
Didea.coverage.calculate.hits=true -Didea.test.cyclic.buffer.size=1048576 "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=60771" -
Dkotlinx.coroutines.debug.enable.creation.stack.trace=false -Ddebugger.agent.enable.coroutines=true -Dkotlinx.coroutines.debug.enable.flows.stack.trace=true -
Dkotlinx.coroutines.debug.enable.mutable.state.flows.stack.trace=true --patch-module library=C:\LibraryManagementSystem\Skeleton\target\test-classes --add-reads library=ALL-
UNNAMED --add-opens library/library=ALL-UNNAMED --add-opens library/library.book=ALL-UNNAMED --add-opens library/library.controllers=ALL-UNNAMED --add-opens
library/library.user=ALL-UNNAMED --add-modules library -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8 -classpath
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base\21\javafx-base-21-win.jar;C:\Users\laich\m2\repository\org\mindrot\jbcrypt\0.3m\jbcrypt-0.3m.jar;C:\Users\laich\m2\repository\com\github\librepdf\openpdf\1.3.39\openpdf-
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com.intellij.rt.junit.JUnitStarter -ideVersion5 -junit5 @w@C:\Users\laich\AppData\Local\Temp\idea_working_dirs_junit.tmp @C:\Users\laich\AppData\Local\Temp\idea_junit.tmp -
socket60770 Process finished with exit code 0
```

UserTest

2.92 s

testStaticUpdateUser()	passed	276 ms
testSetPassword()	passed	295 ms
testSaveDuplicateUser()	passed	150 ms
testSetFullName()	passed	73 ms
testStaticUpdateNonExistentUser()	passed	2 ms
testUserConstructorWithInvalidStatus()	passed	1 ms
testUpdateNonExistentUser()	passed	77 ms
testRoleFromString()	passed	2 ms
testCheckUsernameLength_AllBranches()	passed	1 ms
testSaveUser()	passed	76 ms
testStaticSaveUser()	passed	74 ms
testUserConstructorWithInvalidRole()	passed	
testCheckFullNameLength_AllBranches()	passed	
testGetAllUsers()	passed	153 ms
testToString()	passed	74 ms
testUpdateUser()	passed	82 ms
testSetInvalidStatus()	passed	79 ms
testValidatePassword()	passed	1 ms
testStatusFromString()	passed	1 ms
testUserConstructorWithEnums()	passed	150 ms
testCheckUsernameLength()	passed	
testFileInitialization()	passed	3 ms
testUserConstructorWithStrings()	passed	75 ms
testRoleDisplayNames()	passed	
testUserPersistenceAcrossInstances()	passed	74 ms
testSelectUserByUsername()	passed	73 ms
testUsernameExists()	passed	72 ms
testEmptyDatabase()	passed	3 ms
testingcheckFullNameLength()	passed	1 ms
testCheckFullNameLength()	passed	1 ms
testCheckPassword()	passed	375 ms
testMultipleRoles()	passed	232 ms
testSetStatus()	passed	76 ms
testEquals()	passed	222 ms
testGetUserCount()	passed	145 ms
testStatusDisplayNames()	passed	
testCaseInsensitiveRoleAndStatus()	passed	
testValidateUsername()	passed	1 ms

AllTest	20.30 s
StudentDashboardControllerUITest	7.01 s
branch_HandleProfileUpdate_PasswordMismatch()	passed 294 ms
branch_LoadMyBorrowedBooksData_AllExpired()	passed 98 ms
branch_ClearBookDetails()	passed 78 ms
branch_LoadBooksData_ExceptionInGetApprovedBooks()	passed 84 ms
branch_CalculateTimeLeft_Expired()	passed 78 ms
branch_CalculateTimeLeft_ZeroDuration()	passed 78 ms
branch_HandleTabSelection_WithUser()	passed 77 ms
branch_LoadBooksData_EmptyApprovedBooks()	passed 78 ms
branch_SetUser_NullUser()	passed 76 ms
branch_UpdateBookDetails_ValidBook()	passed 77 ms
branch_SetupMyBorrowedBooksTable()	passed 89 ms
branch_RefreshNotifications_AfterClear()	passed 83 ms
branch_HandleProfileUpdate_InvalidPasswordFormat()	passed 76 ms
branch_LoadBooksData_WithUser()	passed 79 ms
branch_SetupNotificationList()	passed 77 ms
branch_LoadNotifications_NullUser()	passed 76 ms
branch_EnsureDataFiles()	passed 78 ms
branch_HandleProfileUpdate_InvalidFullNameLength()	passed 80 ms
branch_HandleProfileUpdate_ValidUpdate()	passed 79 ms
branch_InitializeData_NullUser()	passed 79 ms
branch_CalculateTimeLeft_LessThanOneSecond()	passed 87 ms
branch_HandleViewConfirmPassword_ToggleHide()	passed 73 ms
branch_HandleViewConfirmPassword_ToggleShow()	passed 77 ms
branch_HandleProfileUpdate_NoChanges()	passed 77 ms
branch_CalculateTimeLeft_VeryShortDuration()	passed 73 ms
branch_CalculateTimeLeft_ValidWithDays()	passed 76 ms
branch_HandleTabSelection_MultipleCalls()	passed 81 ms
branch_HandleViewPassword_ToggleHide()	passed 72 ms
branch_HandleViewPassword_ToggleShow()	passed 73 ms
branch_HandleProfileUpdate_ShortPassword()	passed 76 ms
branch_CalculateTimeLeft_ValidWithMinutesOnly()	passed 78 ms
branch_StartBorrowedBooksTimer()	passed 75 ms
branch_HandleReadBook_WithSelection()	passed 206 ms
branch_LoadNotifications_WithUser()	passed 74 ms
branch_StartBorrowedBooksTimer_MultipleInvocations()	passed 77 ms
branch_SetupBorrowedTableSelection_SelectionChanged()	passed 75 ms
branch_CalculateTimeLeft_NullDateTime()	passed 75 ms
branch_HandleReturnBook_NullBorrow()	passed 75 ms
branch_LoadMyBorrowedBooksData_WithUser()	passed 76 ms
branch_Initialize_AllFieldsNull()	passed 75 ms
branch_SetupBorrowedTableSelection_SelectionCleared()	passed 73 ms
branch_UpdateBookDetails_SpecialCharacters()	passed 76 ms
branch_ClearProfileFields()	passed 76 ms
branch_InitializeData_WithUser()	passed 75 ms
branch_LoadBooksData_Checking_DifferentAuthors()	passed 80 ms
branch_HandleViewConfirmPassword_InitialState()	passed 74 ms
branch_SetupTableColumns()	passed 75 ms
checking	
not yet expired	
branch_HandleProfileUpdate_UpdateSuccess()	passed 227 ms
branch_SetupInformBoardTab()	passed 74 ms
branch_FormatDateTime_Valid()	passed 79 ms
branch_LoadBooksData_Checking_DifferentTitles()	passed 74 ms
branch_CalculateTimeLeft_NullDuration()	passed 76 ms
branch_FormatDateTime_VariousNullScenarios()	passed 72 ms
branch_SetupTableSelectionListener()	passed 74 ms
branch_HandleViewPassword_TextFieldToPasswordField()	passed 73 ms
branch_HandleViewConfirmPassword_TextFieldToPasswordField()	passed 81 ms
branch_HandleClearNotification_NullNotification()	passed 77 ms
branch_SetUser_ValidUser()	passed 166 ms

branch_LoadMyBorrowedBooksData_ExceptionHandling()	passed	78 ms
branch_MethodExistence()	passed	78 ms
branch_ShowAlert_ValidInput()	passed	77 ms
branch_CalculateTimeLeft_ExactlyExpired()	passed	79 ms
branch_UpdateBookDetails_NullBook()	passed	74 ms
branch_HandleProfileTab_NullUser()	passed	74 ms
branch_SetupTableSelectionListener_SelectionChanged()	passed	78 ms
branch_LoadBooksData_Checking_NullBooks()	passed	78 ms
branch_CalculateTimeLeft_ValidWithHours()	passed	76 ms
branch_LoadBooksData_Checking_EqualBooks()	passed	78 ms
branch_RefreshNotifications()	passed	78 ms
checking not yet expired		
branch_SetupTableSelectionListener_SelectionCleared()	passed	74 ms
branch_HandleBorrowSelectedBook_WithSelection()	passed	97 ms
branch_SetupBorrowedTableCellFactories()	passed	73 ms
branch_HandleTabSelection_NullUser()	passed	76 ms
branch_ClearProfileFields_AllFieldsPopulated()	passed	76 ms
branch_HandleViewPassword_InitialState()	passed	75 ms
branch_SetupTableSelectionListener_NullTable()	passed	73 ms
branch_FormatDateTime_Null()	passed	77 ms
branch_HandleLogout()	passed	318 ms
branch_CalculateTimeLeft_NullBorrow()	passed	79 ms
checking not yet expired		
branch_Initialize_WithException()	passed	76 ms
BorrowDurationPopupControllerTest		34 ms
branch_ShowInvalidLabel()	passed	2 ms
branch_MaxDurationExceeded_MinutesOnly()	passed	2 ms
branch_ValidDuration_MinutesOnly()	passed	2 ms
branch_InvalidNumberFormat()	passed	2 ms
branch_HideInvalidLabel()	passed	2 ms
branch_NegativeMinutes()	passed	2 ms
branch_MaxDurationExceeded_MinutesAndSeconds()	passed	2 ms
branch_ZeroDuration()	passed	2 ms
branch_EmptyFields()	passed	1 ms
branch_BoundaryValues_OneMinute()	passed	2 ms
branch_OnlyMinutesField()	passed	3 ms
branch_BoundaryValues_OneSecond()	passed	3 ms
branch_SecondsExceed59()	passed	1 ms
branch_InputValidation_NonNumericCharacters()	passed	2 ms
branch_ValidDuration_BothFields()	passed	1 ms
branch_MaxValidDuration()	passed	1 ms
branch_OnlySecondsField()	passed	2 ms
branch_ValidDuration_SecondsOnly()	passed	2 ms
AuthorDashboardControllerLogicTest		10.82 s
branch_HandleProfileUpdate_PasswordMismatch()	passed	134 ms
branch_UpdatePieChart_OnlyApproved()	passed	127 ms
branch_BookSelection_NullSelection()	passed	98 ms
branch_HandleMyBooksDelete_ValidSelectionConfirmed()	passed	2.03 s
branch_HandleEmptyNotificationBoard()	passed	80 ms
branch_SetUser_NullUser()	passed	78 ms
branch_LoadUserBooks_WithUserAndBooks()	passed	77 ms
branch_HandleClearAll_WithNotificationsCancelled()	passed	620 ms
checking not yet expired		
branch_HandleNewBookPublish_NoFileSelected()	passed	79 ms
branch_HandleProfileUpdate_InvalidPasswordFormat()	passed	93 ms
branch_TabSelection_InformBoardTab()	passed	89 ms
branch_LoadNotifications_NullUser()	passed	76 ms
branch_HandleProfileUpdate_InvalidFullnameLength()	passed	70 ms
branch_HandleProfileUpdate_ValidUpdate()	passed	243 ms

branch_HandleProfileUpdate_NoChanges()	passed	97 ms
branch_HandleMyBooksModify_NullSelection()	passed	93 ms
branch_HandleMyBooksDelete_NullSelection()	passed	90 ms
branch_HandleNewPWHS_ToggleShowHide()	passed	93 ms
branch_HandleMyBooksDelete_ValidSelectionCancelled()	passed	548 ms
checking not yet expired		
branch_InitDescriptionBox()	passed	77 ms
branch_BookSelection_ValidSelection()	passed	70 ms
branch_Initialize_FirstTimeFlag()	passed	72 ms
branch_TabSelection_StatusViewTab()	passed	71 ms
branch_LoadNotifications_WithNotifications()	passed	70 ms
branch_DescriptionBox_SelectionAndClear()	passed	72 ms
branch_HandleProfileUpdate_ShortPassword()	passed	74 ms
branch_UpdateBarChart_SingleBook()	passed	96 ms
branch_HandleNewBookPublish_SuccessfulPublish()	passed	119 ms
branch_LoadNotifications_WithUser()	passed	125 ms
branch_PasswordToggle_InitialState()	passed	96 ms
branch_HandleMyBooksDelete_ApprovedAndBorrowed()	passed	80 ms
checking not yet expired		
branch_UpdateBarChart_EmptyList()	passed	87 ms
branch_ShowSimpleAlert_ValidInput()	passed	77 ms
branch_TableColumnWidth_ResizeListener()	passed	77 ms
branch_HandleProfileUpdate_UpdateSuccess()	passed	282 ms
checking not yet expired		
branch_SetProfileUsername_NullUser()	passed	85 ms
branch_HandleClearAll_WithNotifications()	passed	73 ms
branch_UpdateBarChart_FiveBooks()	passed	91 ms
branch_HandleNewBookPublish_EmptyTitle()	passed	77 ms
branch_FirstTimeFlag()	passed	89 ms
checking not yet expired		
branch_UpdatePieChart_OnlyRejected()	passed	76 ms
branch_UpdateBarChart_MoreThanFiveBooks()	passed	74 ms
branch_HandleMyBooksModify_ApprovedAndBorrowed()	passed	73 ms
branch_HandleMyBooksModify_ValidSelection()	passed	271 ms
checking not yet expired		
branch_HandleStatusRefresh()	passed	90 ms
branch_LoadUserBooks_NullUser()	passed	78 ms
branch_SetUser_ValidUser()	passed	149 ms
branch_UpdatePieChart_AllZero()	passed	71 ms
branch_HandleProfileUpdate_UpdateFailure()	passed	285 ms
checking not yet expired		
branch_MethodExistence()	passed	95 ms
branch_SetProfileFullname_NullUser()	passed	72 ms
branch_HandleClearAll_EmptyNotifications()	passed	71 ms
branch_HandleMyBooksView_NullSelection()	passed	92 ms
branch_SetProfileUsername_WithUser()	passed	90 ms
branch_ShowSimpleInformation_ValidInput()	passed	88 ms
branch_HandleNewBookPublish_AllFieldsValid()	passed	88 ms
branch_HandleProfileUpdate_EmptyFields()	passed	91 ms
branch_UpdatePieChart_MixedValues()	passed	96 ms
checking not yet expired		
branch_TabSelection_ProfileTab()	passed	77 ms
branch_EnsureStorageStructure()	passed	73 ms
branch_HandleClearAll_WithNotificationsConfirmed()	passed	1.36 s
checking not yet expired		
branch_HandleNewBookPublish_EmptyAbstract()	passed	80 ms
branch_HandleNewBookPublish_DuplicateBook()	passed	92 ms
branch_UpdatePieChart_OnlyPending()	passed	88 ms
branch_SetProfileFullname_WithUser()	passed	79 ms
branch_HandleLogout()	passed	158 ms
checking not yet expired		
branch_HandleConfirmationPWHS_ToggleShowHide()	passed	87 ms

StudentReadBookPopupTest		22 ms
branch_ZoomIn_MaxLimit()	passed	4 ms
branch_ZoomOut_MinLimit()	passed	1 ms
branch_UpdateButtonStates_MaxSize()	passed	2 ms
branch_UpdateButtonStates_MinSize()	passed	1 ms
branch_CloseWindow_WithTimer()	passed	1 ms
branch_Initialize()	passed	1 ms
branch_IsBookExpired_Expired()	passed	1 ms
branch_UpdateButtonStates_MiddleRange()	passed	1 ms
branch_ZoomIn_Normal()	passed	1 ms
branch_FontSizePercentageCalculation()	passed	2 ms
branch_SetBookContent()	passed	2 ms
branch_CheckAndHandleExpiration_NotExpired()	passed	1 ms
not yet expired		
branch_IsBookExpired_NotExpired()	passed	1 ms
branch_SetBorrowInfo_StartTimer()	passed	1 ms
branch_IsBookExpired_NullBorrowInfo()		passed
branch_CloseWindow_WithoutTimer()		passed
branch_HandleOKButton()	passed	1 ms
branch_ZoomOut_Normal()	passed	1 ms
RegisterControllerTest		417 ms
branch_UsernameTooShort()	passed	9 ms
branch_FullNameTooLong()	passed	2 ms
branch_FullNameTooShort()	passed	4 ms
branch_PasswordMismatch()	passed	3 ms
branch_PasswordTooShort()	passed	3 ms
branch_UsernameExists()	passed	87 ms
branch_InvalidUsername_PureNumber()	passed	3 ms
branch_HandleViewPassword_Hide()	passed	4 ms
branch_HandleViewPassword_Show()	passed	2 ms
branch_SuccessfulRegistration_PasswordField()	passed	77 ms
branch_PasswordNoLowercase()	passed	3 ms
branch_EmptyFields()	passed	3 ms
branch_SuccessfulRegistration_TextField()	passed	78 ms
branch_PasswordNoNumber()	passed	16 ms
branch_PasswordNoUppercase()	passed	8 ms
branch_WhitespaceTrimming()	passed	83 ms
branch_UsernameTooLong()	passed	23 ms
branch_InvalidUsername_SpecialChar()	passed	9 ms
LoginControllerTest		1.99 s
branch_UserDeactivated()	passed	158 ms
branch_SetRole_Author()	passed	17 ms
branch_SetRole_Staff()	passed	5 ms
branch_ShowSimpleToast()	passed	3 ms
branch_EmptyPassword()	passed	150 ms
branch_StaffLogin_StudentRole()	passed	241 ms
checking		
not yet expired		
branch_HandleViewPassword_Hide()	passed	21 ms
branch_HandleViewPassword_Show()	passed	5 ms
branch_SetRole_Librarian()	passed	3 ms
branch_UsernameNotExists()	passed	3 ms
branch_RoleCaseSensitivity()	passed	4 ms
branch_CorrectLogin_Librarian()	passed	220 ms
11月 22, 2025 1:55:49 上午 javafx.fxml.FXMLLoader\$ValueElement processValue WARNING: Loading FXML document with JavaFX API of version 25 by JavaFX runtime of version 21		
branch_CorrectLogin_Author()	passed	235 ms
11月 22, 2025 1:55:49 上午 javafx.fxml.FXMLLoader\$ValueElement processValue WARNING: Loading FXML document with JavaFX API of version 25 by JavaFX runtime of version 21		
branch_CorrectLogin_Student_PasswordField()	passed	202 ms
branch_SetRole_Invalid()	passed	5 ms
branch_UsernameTrimming()	passed	203 ms
checking		
not yet expired		
branch_SetRole_Student()	passed	3 ms
branch_WrongRoleAccess()	passed	156 ms
branch_CorrectLogin_Student_TextField()	passed	194 ms
branch_EmptyUsername()	passed	3 ms
branch_UsernameExists_WrongPassword()	passed	162 ms

LibrarianDashboardControllerUITest		17.64 s
branch_ShowAlert_NullParameters()	passed	235 ms
branch_FormatDuration_Hours()	passed	229 ms
checking not yet expired		
branch_HandleViewConfirmPassword_HidePassword()	passed	225 ms
branch_HandleProfileUpdate_PasswordMismatch()	passed	229 ms
branch_HandleProfileTab_WithNullUser()	passed	232 ms
branch_StartCountdownTimer()	passed	214 ms
branch_SetupPublish_TableConfiguration()	passed	217 ms
checking not yet expired		
branch_HandleViewPassword_ShowPassword()	passed	211 ms
branch_ShowAlert()	passed	203 ms
branch_FormatDuration_AllTimeRanges()	passed	287 ms
branch_SetUser_NullUser()	passed	288 ms
checking not yet expired		
branch_BorrowTable_ColumnWidthListener()	passed	347 ms
branch_IsTimeExpired_Expired()	passed	416 ms
branch_HandleProfileTab_WithValidUser()	passed	223 ms
checking not yet expired		
branch_HandlePublishTab_ValidInitialization()	passed	225 ms
branch_PublishActionsCell_EmptyCell()	passed	215 ms
PublishActionsCell empty cell test completed (possible structural limitation): javafx.scene.control.TableCell.getGraphic()		
branch_ShowSimpleToast()	passed	212 ms
branch_ApprovalActionsCell_ApproveAction()	passed	209 ms
checking not yet expired		
branch_PublishActionsCell_DeleteAction_Confirm()	passed	206 ms
branch_SetupBorrow_TableConfiguration()	passed	201 ms
branch_ApprovalActionsCell_Instantiation()	passed	201 ms
branch_CalculateTimeLeft_ValidBorrow()	passed	215 ms
branch_ApprovalActionsCell_ViewAbstractAction()	passed	206 ms
checking not yet expired		
branch_HandleProfileUpdate_NoChanges()	passed	209 ms
branch_LoadTableData_UserType()	passed	205 ms
branch_IsTimeExpired_NotExpired()	passed	209 ms
branch_PublishActionsCell_Instantiation()	passed	206 ms
branch_UsersActionsCell_DeactivateAction()	passed	203 ms
checking not yet expired		
branch_HandleProfileUpdate_DifferentNameAndPassword()	passed	345 ms
branch_FormatDuration_ZeroDaysNonZeroHours()	passed	205 ms
branch_UsersActionsCell_CurrentUser_NoButtons()	passed	205 ms
branch_HandleViewPassword_HidePassword()	passed	222 ms
checking not yet expired		
branch_LoadTableData_PublishType()	passed	218 ms
branch_HandleProfileUpdate_ShortPassword()	passed	210 ms
branch_FormatDuration_Minutes()	passed	210 ms
branch_ApprovalActionsCell_RejectAction_Cancel()	passed	211 ms
branch_UsersActionsCell_EmptyCell()	passed	208 ms
checking not yet expired		
UsersActionsCell empty cell test completed (possible structural limitation): javafx.scene.control.TableCell.getGraphic()		
branch_PasswordToggle_InitialState()	passed	214 ms
branch_ApprovalActionsCell_EmptyCell()	passed	204 ms
ApprovalActionsCell empty cell test completed (possible structural limitation): javafx.scene.control.TableCell.getGraphic()		
branch_CalculateTimeLeft_NullDateTime()	passed	207 ms
branch_SetUser_ValidLibrarianUser()	passed	275 ms
checking not yet expired		
branch_HandleProfileUpdate_PasswordSameAsCurrent()	passed	286 ms

branch_HandleProfileUpdate_AllUpdateBranches()	passed	278 ms
Profile update change branches: null		
branch_LoadTableData_BorrowType()	passed	208 ms
branch_ApprovalTable_ColumnWidthListener()	passed	207 ms
checking not yet expired		
branch_CalculateTimeLeft_NullDuration()	passed	205 ms
branch_PasswordToggle_AllStates()	passed	221 ms
branch_FormatDuration_Days()	passed	218 ms
branch_SetupApprovals_TableConfiguration()	passed	213 ms
branch_HandleUserTab_ValidInitialization()	passed	211 ms
checking not yet expired		
branch_SetupMethods_WidthListeners()	passed	209 ms
branch_PublishTable_ColumnWidthListener()	passed	214 ms
branch_SetupUsers_TableConfiguration()	passed	220 ms
branch_LoadTableData_ApprovalType()	passed	212 ms
branch_LoadTableData_NullBooks()	passed	546 ms
checking not yet expired LoadTableData null books test: The used MockMaker SubclassByteBuddyMockMaker does not support the creation of static mocks Mockito's inline mock maker supports static mocks based on the Instrumentation API. You can simply enable this mock mode, by placing the 'mockito-inline' artifact where you are currently using 'mockito-core'. Note that Mockito's inline mock maker is not supported on Android.		
branch_ApprovalActionsCell_RejectAction_Confirm()	passed	223 ms
branch_HandleViewConfirmPassword_ShowPassword()	passed	209 ms
branch_UsersActionsCell_Instantiation()	passed	205 ms
checking not yet expired		
branch_HandleProfileUpdate_AllValidationBranches()	passed	212 ms
Profile update validation branches: null		
branch_UsersActionsCell_UpdateItem_Branches()	passed	361 ms
branch_PublishActionsCell_DeleteAction_Cancel()	passed	207 ms
branch_TimeCalculation_AllBranches()	passed	208 ms
checking not yet expired		
branch_Initialize_WithoutFXML()	passed	214 ms
branch_UsersActionsCell_ActivateAction()	passed	213 ms
branch_HandleProfileUpdate_EmptyFields()	passed	218 ms
branch_LoadTableData_AllTableTypes()	passed	214 ms
LoadTableData branches: null		
branch_UsersActionsCell_AllUpdateItemBranches()	passed	357 ms
branch_TableType_EnumValues()	passed	210 ms
branch_CalculateTimeLeft_ExpiredBorrow()	passed	211 ms
branch_PublishActionsCell_ViewAction()	passed	211 ms
branch_LoadTableData_NullType()	passed	210 ms
checking not yet expired		
branch_HandleApprovalTab_ValidInitialization()	passed	210 ms
branch_HandleLogout()	passed	248 ms
branch_CalculateTimeLeft_NullBorrow()	passed	216 ms
branch_HandleBorrowTab_ValidInitialization()	passed	209 ms
checking not yet expired		
branch_UsersTable_ColumnWidthListener()	passed	213 ms
StudentDashboardControllerTest		2.81 s
branch_CalculateTimeLeft_Expired()	passed	71 ms
branch_ControllerInitialization()	passed	67 ms
branch_ProfileValidation_InvalidPasswordLength()	passed	70 ms

branch_BorrowReturn_ConfirmationLogic()	passed	68 ms
branch_BookStatus_Transitions()	passed	69 ms
branch_CalculateTimeLeft_ValidWithDays()	passed	72 ms
branch_ProfileValidation_EmptyFields()	passed	70 ms
branch_DataLoading_WithNullUser()	passed	73 ms
branch_ProfileValidation_PasswordMismatch()	passed	75 ms
branch_RefreshScenarios()	passed	71 ms
checking not yet expired		
branch_ErrorHandling_DataOperations()	passed	68 ms
branch_CalculateTimeLeft_ValidWithMinutesOnly()	passed	75 ms
branch_AutoReturn_ExpiredBooks()	passed	89 ms
branch_CalculateTimeLeft_NullDateTime()	passed	78 ms
branch_TimeCalculations_ExactlyOneDay()	passed	79 ms
branch_BookSelection_NoSelection()	passed	77 ms
branch_UserRole_Permissions()	passed	227 ms
branch_FormatDateTime_Valid()	passed	78 ms
branch_CalculateTimeLeft_NullDuration()	passed	75 ms
branch_ProfileValidation_InvalidFullNameLength()	passed	77 ms
branch_NotificationHandling_NonEmptyList()	passed	75 ms
branch_TimeCalculations_ExactlyOneHour()	passed	74 ms
branch_DurationCalculations_VariousScenarios()	passed	73 ms
branch_TimeCalculations_ExactlyOneMinute()	passed	78 ms
branch_PasswordToggle_ShowHideLogic()	passed	77 ms
branch_ErrorHandling_FileOperations()	passed	73 ms
branch_MethodExistence()	passed	73 ms
branch_TimeCalculations_EdgeCases()	passed	74 ms
branch_SuccessOperations()	passed	73 ms
branch_CalculateTimeLeft_ValidWithHours()	passed	79 ms
branch_StringValidation_ValidInputs()	passed	73 ms
branch_BookSelection_WithSelection()	passed	74 ms
branch_AutoReturn_NonExpiredBooks()	passed	74 ms
branch_FormatDateTime_Null()	passed	70 ms
branch_NotificationHandling_EmptyList()	passed	70 ms
branch_CalculateTimeLeft_NullBorrow()	passed	70 ms
NotificationTest		95 ms
testToStringWithBothNull()	passed	1 ms
testEqualsBothDifferent()		passed
testConstructorWithNullUsernameString()		passed
testFindByUserEdgeCases()	passed	1 ms
testLoadAllNotificationsEmptyFile()	passed	1 ms
testVeryLongMessage()	passed	2 ms
testPrivateConstructorIndirectly()	passed	1 ms
testNotificationOrderingConsistency()	passed	5 ms
testSetReceiverWithSameUser()	passed	1 ms
testAllNotificationTypes()		passed
testDeleteNonExistentNotification()	passed	1 ms
testConstructorWithNullBookTitle()	passed	1 ms
testMultipleNotificationsSameUser()	passed	8 ms
testNotificationWithUnicode()	passed	7 ms
testSetMessage()	passed	1 ms
testSelectByNotificationPrivateMethod()	passed	8 ms
testStaticSaveEdgeCases()	passed	2 ms
testConstructorWithUsernameString()	passed	1 ms
testNotificationTypeCombinations()		passed
testStaticSaveMethod()	passed	3 ms
testLoadAllNotificationsWithCorruptFile()	passed	4 ms

testCacheLoadedFlag()	passed	1 ms
testGetReceiverWithNullUsernameButReceiverSet()	passed	
testToStringWithNullMessage()	passed	
testDeleteFunctionality()	passed	2 ms
testInitializeFileWhenExists()	passed	1 ms
testFileReadError()	passed	3 ms
testSaveDuplicateNotification()	passed	3 ms
testSaveAllNotificationWithEmptyList()	passed	1 ms
testSaveWithEmptyMessage()	passed	2 ms
testFileOperationsWithDirectoryCreationIssues()	passed	2 ms
testConstructorWithEmptyUsernameString()	passed	1 ms
testGetReceiverUsernameDirectly()	passed	
testEqualsSameUserDifferentMessage()	passed	
testToStringWithNullReceiverUsername()	passed	
testEqualsDifferentUserSameMessage()	passed	
testEmptyNotifications()	passed	1 ms
testSelectByUserWithNull()	passed	1 ms
testDeleteNullNotification()	passed	
testConstructorWithEmptyBookTitle()	passed	1 ms
testBasicSaveAndRetrieve()	passed	2 ms
testGetReceiverWithBothNull()	passed	
testSaveWithMessageContainingTabs()	passed	2 ms
testSelectByUserWithEmptyString()	passed	1 ms
testEqualsWithDifferentClass()	passed	1 ms
testPrivateConstructorFormatWithNonExistentUser()	passed	2 ms
Fail to extract notification: library.user.UserDoesNotExistException: User nonExistentUser123 does not exist!		
testConstructorWithNonExistentUsername()	passed	1 ms
testSaveWithLongMessage()	passed	1 ms
testCacheBehavior()	passed	2 ms
testSaveAllNotificationWithNull()	passed	
Notifications list cannot be null!		
testNotificationPersistence()	passed	2 ms
testEqualsConsistency()	passed	
testSelectByUserWithWhitespace()	passed	2 ms
testConstructorWithNullType()	passed	1 ms
testConstructorWithNullUser()	passed	
testSaveWithMessageContainingNewlines()	passed	2 ms
testMultipleOperationsSequence()	passed	4 ms
testNotificationOrdering()	passed	3 ms
testAllNotificationTypeMessages()	passed	
testEqualsSameReference()	passed	1 ms
testEqualsWithNull()	passed	
testSetReceiverUsernameNullAfterReceiverSet()	passed	1 ms
BookTest		153 ms
testGetBookByAuthor_AllBranches()	passed	13 ms
testGetAllBooks_AllBranches()	passed	8 ms
testViewBook_DirectoryInsteadOfFile()	passed	2 ms
testEditBook_AllBranches()	passed	12 ms
testUpdateBooks_AllBranches()	passed	3 ms
testEnsureStorageStructure_AllBranches()	passed	2 ms
testEditBook_RenameIOException()	passed	5 ms

testBookStatusEnum_AllBranches()	passed	2 ms
testToString_AllBranches()	passed	2 ms
testUpdateBookStatus_CaseInsensitive()	passed	3 ms
testSetIsBorrowed_AllBranches()	passed	6 ms
testEnsureStorageStructure_IOException()	passed	3 ms
testViewBook_AllBranches()	passed	5 ms
testBookStatusFromString_AllBranches()	passed	4 ms
testPublishBook_FileCopyIOException()	passed	3 ms
testPublishBook_AllBranches()	passed	14 ms
testGettersAndSetters_AllBranches()	passed	1 ms
testUpdateBookStatus_AllBranches()	passed	6 ms
Error invoking method updateBookStatus: null		
testGetStatusNumber_AllBranches()	passed	6 ms
testFileHandlingEdgeCases_AllBranches()	passed	4 ms
testGetPendingBook_AllBranches()	passed	4 ms
testEqualsAndHashCode_AllBranches()	passed	2 ms
testQuoteField_AllBranches()	passed	3 ms
testGetCurrentDate_AllBranches()	passed	2 ms
testGetApprovedBook_AllBranches()	passed	4 ms
testGetBestFiveBooks_AllBranches()	passed	8 ms
testIncrementNoOfTimeBorrowed_AllBranches()	passed	4 ms
Error invoking method incrementNoOfTimeBorrowed: null		
testSetIsBorrowed_NullBook()	passed	2 ms
testMakeUniqueContentDirectory_AllBranches()	passed	3 ms
testDeleteBook_AllBranches()	passed	5 ms
Error invoking method deleteBook: null		
testUpdateBookStatus_InvalidStatus()	passed	3 ms
testPublishBook_DeletedBookAllowed()	passed	4 ms
testConstructor_AllBranches()	passed	1 ms
testGetParticularBook_AllBranches()	passed	4 ms
BorrowTest		21.93 s
testReturnBook_NullBookValidUser_Failure()	passed	297 ms
testBorrowBook_NegativeMinutesParameter_Failure()	passed	314 ms
testBorrowBook_NullBook_Failure()	passed	295 ms
testBorrowBook_MaxDuration_Success()	passed	307 ms
testNotificationConstructor_NullNotificationType_ThrowsException()	passed	302 ms
testReturnExpiredBook_CreatesNotifications()	passed	303 ms
testNotification_Equals_Success()	passed	302 ms
testSave_EmptyMessage_ThrowsException()	passed	295 ms
testReturnBook_DifferentUser_Failure()	passed	368 ms
testNotificationConstructor_NullBookTitle_ThrowsException()	passed	295 ms
testCountBorrowed_EmptyTitle_Zero()	passed	286 ms
testCountBorrowed_NullTitle_Zero()	passed	301 ms
testNotification_AllTypes_GenerateCorrectMessages()	passed	306 ms
testReturnBook_OthersStillBorrow_Success()	passed	390 ms
testGetBorrowByUser_TriggersExpirationCheck()	passed	297 ms
testGetBorrowByUser_NonExistentUser_EmptyList()	passed	286 ms
testGetBorrowByUser_NullUsername_EmptyList()	passed	307 ms
testNotification_ToString_CorrectFormat()	passed	293 ms
testBorrowBook_ZeroSecondsDuration_Success()	passed	303 ms
testBorrowBook_StaffUser_Failure()	passed	311 ms
testReturnExpiredBook_MixedBorrows_Separated()	passed	384 ms
testCountBorrowed_NonExistentBook_Zero()	passed	304 ms
testBorrowBook_DuplicateBorrow_Failure()	passed	299 ms

testCountBorrowed_MultipleActiveBorrows_CorrectCount()	passed	444 ms
testGetBorrowByUser_MultipleBorrows_Returned()	passed	303 ms
testReturnExpiredBook_MultipleBorrows_Success()	passed	377 ms
testBorrowConstructor_WithMinutesSeconds_SetsDurationCorrectly()	passed	300 ms
testReturnBook_MultipleBorrowersOneReturns_Success()	passed	383 ms
testBorrowBook_NegativeSecondsParameter_Failure()	passed	316 ms
testNotificationConstructor_NullReceiverUsername_ThrowsException()	passed	302 ms
testCountBorrowed_ExpiredBorrows_Zero()	passed	306 ms
testEnsureDataDirectory_CreatesDirectory()	passed	295 ms
testDeleteNotification_NullNotification_ReturnsFalse()	passed	298 ms
testCountBorrowed_EmptyAuthor_Zero()	passed	306 ms
testCountBorrowed_NullBook_Zero()	passed	307 ms
testCountBorrowed_NullAuthor_Zero()	passed	293 ms
testNotification_NotEquals_DifferentMessage()	passed	298 ms
testReturnBook_NonExistentUser_Failure()	passed	305 ms
testBorrowBook_ValidBookInvalidUserRole_Failure()	passed	304 ms
testBorrowBook_ZeroDuration_Success()	passed	306 ms
testGetReceiver_NonExistentUser_ThrowsException()	passed	288 ms
testBorrowBook_LibrarianUser_Failure()	passed	298 ms
testCountBorrowed_MixedActiveExpired_CountsOnlyActive()	passed	367 ms
testNotificationConstructor_EmptyBookTitle_ThrowsException()	passed	300 ms
testReturnBook_NullBook_Failure()	passed	298 ms
testNotificationConstructor_EmptyReceiverUsername_ThrowsException()	passed	297 ms
testBorrowBook_ValidUserInvalidBookStatus_Failure()	passed	298 ms
testCountBorrowed_TriggersExpirationCheck()	passed	292 ms
testSelectByUser_ReturnsNotifications()	passed	297 ms
testGetBorrowByUser_EmptyUsername_EmptyList()	passed	290 ms
testNotificationConstructor_NonExistentUser_ThrowsException()	passed	293 ms
testBorrowBook_RejectedBook_Failure()	passed	290 ms
testBorrowBook_InactiveUser_Failure()	passed	289 ms
testBorrowConstructor_WithDuration_SetsFieldsCorrectly()	passed	298 ms
testBorrowBook_NegativeDuration_Failure()	passed	305 ms
testBorrowBook_NonExistentUser_Failure()	passed	299 ms
testNotification_NotEquals_DifferentReceiver()	passed	383 ms
testGetBorrowByUser_OnlyExpiredBorrows_EmptyList()	passed	287 ms
testGetReceiver_LazyLoading_Success()	passed	297 ms
testReturnBook_EmptyUsername_Failure()	passed	304 ms
testBorrowBook_EmptyUsername_Failure()	passed	302 ms
testBorrowBook_MinutesSecondsParameters_Success()	passed	304 ms
testLoadAllBorrows_InvalidLines_SkipsInvalid()	passed	297 ms
testBorrowBook_PendingBook_Failure()	passed	298 ms
testReturnBook_NoActiveBorrow_Failure()	passed	298 ms
testReturnExpiredBook_NonExpiredBorrow_Preserved()	passed	302 ms
testReturnBook_LastBorrow_Success()	passed	300 ms
testGetAllBorrows_EmptyFile_EmptyList()	passed	300 ms
testBorrowBook_NullUsername_Failure()	passed	297 ms
testReturnBook_NullUsername_Failure()	passed	289 ms
testReturnBook_BookNotBorrowedByUser_Failure()	passed	284 ms
LibrarianDashboardControllerTest		1.96 s
branch_IsTimeExpired_NullDuration()	passed	74 ms
branch_CalculateTimeLeft_Expired()	passed	72 ms
branch_ControllerInitialization()	passed	77 ms
branch_TableTypeEnumValues()	passed	71 ms
branch_SetUser()	passed	137 ms
branch_IsTimeExpired_NullBorrow()	passed	72 ms
branch_CalculateTimeLeft_ValidWithDays()	passed	75 ms
branch_IsTimeExpired_True()	passed	71 ms
branch_FormatDuration_WithSecondsOnly()	passed	74 ms

branch_IsTimeExpired_False()	passed	73 ms
branch_CalculateTimeLeft_ValidWithMinutesOnly()	passed	71 ms
branch_FormatDuration_EdgeCases()	passed	71 ms
branch_CalculateTimeLeft_NullDateTime()	passed	70 ms
branch_IsTimeExpired_ExactlyZero()	passed	69 ms
branch_FormatDuration_WithHoursOnly()	passed	71 ms
branch_CalculateTimeLeft_NullDuration()	passed	74 ms
branch_FormatDuration_ZeroDuration()	passed	75 ms
branch_MethodExistence()	passed	76 ms
branch_TimeCalculations_EdgeCases()	passed	75 ms
branch_FormatDuration_WithDaysHoursMinutesSeconds()	passed	73 ms
branch_FormatDuration_WithMinutesOnly()	passed	75 ms
branch_CalculateTimeLeft_ValidWithHours()	passed	73 ms
branch_LoadTableData_AllTableTypes()	passed	74 ms
branch_IsTimeExpired_NullDateTime()	passed	74 ms
branch_DataHandlingScenarios()	passed	72 ms
branch_CalculateTimeLeft_NullBorrow()	passed	70 ms

7.2 Report on the Coverage Test

The following screenshots show the coverage summary of the unit testing for the implemented tasks. Eventually, 65.4% of branch coverage was attained in the library package, which includes the data classes (library.book and library.user) and controller classes (library.controllers).

Current scope: all classes

Overall Coverage Summary

Package	Class, %	Method, %	Branch, %	Line, %
all classes	75.8% (25/33)	76.5% (270/353)	65.4% (513/785)	68% (1279/1881)

Coverage Breakdown

Package	Class, %	Method, %	Branch, %	Line, %
library	100% (1/1)	25% (1/4)		12.5% (1/8)
library.book	100% (3/3)	100% (60/60)	83.1% (143/172)	96.6% (310/321)
library.controllers	61.9% (13/21)	63.3% (138/218)	52.1% (259/497)	55.3% (714/1290)
library.user	100% (8/8)	100% (71/71)	95.7% (111/116)	96.9% (254/262)

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Current scope: [all classes](#) | [library.book](#)

Coverage Summary for Package: library.book

Package	Class, %	Method, %	Branch, %	Line, %
library.book	100% (3/3)	100% (60/60)	83.1% (143/172)	96.6% (310/321)

Class 	Class, %	Method, %	Branch, %	Line, %
Book	100% (1/1)	100% (36/36)	82.4% (84/102)	97% (195/201)
BookStatus	100% (1/1)	100% (5/5)	83.3% (5/6)	100% (12/12)
Borrow	100% (1/1)	100% (19/19)	84.4% (54/64)	95.4% (103/108)

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Current scope: [all classes](#) | [library.controllers](#)

Coverage Summary for Package: library.controllers

Package	Class, %	Method, %	Branch, %	Line, %
library.controllers	61.9% (13/21)	63.3% (138/218)	52.1% (259/497)	55.3% (714/1290)

Class 	Class, %	Method, %	Branch, %	Line, %
AuthorDashboardController	50% (1/2)	75.7% (28/37)	54.4% (62/114)	60.5% (161/266)
BorrowDurationPopupController	100% (1/1)	53.3% (8/15)	57.9% (22/38)	50.7% (36/71)
EditWindowController	100% (1/1)	60% (3/5)	0% (0/6)	18.8% (3/16)
HomeController	100% (1/1)	20% (1/5)		9.1% (1/11)
LibrarianDashboardController	83.3% (5/6)	61.5% (32/52)	47.5% (58/122)	51.1% (182/356)
LoginController	100% (1/1)	70% (7/10)	95.2% (20/21)	85.1% (57/67)
RegisterController	100% (1/1)	58.3% (7/12)	81.2% (26/32)	62.9% (44/70)
StudentDashboardController	16.7% (1/6)	63.3% (38/60)	39.3% (55/140)	52.6% (192/365)
StudentReadBookPopup	100% (1/1)	87.5% (14/16)	66.7% (16/24)	74.5% (38/51)
ViewWindowController	0% (0/1)	0% (0/6)		0% (0/17)

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Current scope: [all classes](#) | [library.user](#)

Coverage Summary for Package: library.user

Package	Class, %	Method, %	Branch, %	Line, %
library.user	100% (8/8)	100% (71/71)	95.7% (111/116)	96.9% (254/262)

Class 	Class, %	Method, %	Branch, %	Line, %
Notification	100% (2/2)	100% (26/26)	96.1% (49/51)	97.2% (105/108)
Role	100% (1/1)	100% (5/5)	100% (5/5)	100% (12/12)
Status	100% (1/1)	100% (5/5)	100% (2/2)	100% (10/10)
User	100% (2/2)	100% (33/33)	94.8% (55/58)	96.2% (125/130)
UserAlreadyExistsException	100% (1/1)	100% (1/1)		100% (1/1)
UserDoesNotExistException	100% (1/1)	100% (1/1)		100% (1/1)

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